CELT

0.5.0

Generated by Doxygen 1.5.5

Thu Oct 9 07:19:40 2008

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Chapter 4

Module Documentation

4.1 Encoding and decoding

Functions

- EXPORT CELTMode * celt_mode_create (celt_int32_t Fs, int channels, int frame_size, int *error)
- EXPORT void celt_mode_destroy (CELTMode *mode)
- EXPORT int celt_mode_info (const CELTMode *mode, int request, celt_int32_t *value)
- EXPORT CELTEncoder * celt_encoder_create (const CELTMode *mode)
- EXPORT void celt_encoder_destroy (CELTEncoder *st)
- EXPORT int celt_encode_float (CELTEncoder *st, const float *pcm, float *optional_synthesis, unsigned char *compressed, int nbCompressedBytes)
- EXPORT int celt_encode (CELTEncoder *st, const celt_int16_t *pcm, celt_int16_t *optional_synthesis, unsigned char *compressed, int nbCompressedBytes)
- EXPORT int celt_encoder_ctl (CELTEncoder *st, int request,...)
- EXPORT CELTDecoder * celt_decoder_create (const CELTMode *mode)
- EXPORT void celt_decoder_destroy (CELTDecoder *st)
- EXPORT int celt_decode_float (CELTDecoder *st, unsigned char *data, int len, float *pcm)
- EXPORT int celt_decode (CELTDecoder *st, unsigned char *data, int len, celt_int16_t *pcm)

4.1.1 Detailed Description

4.1.2 Function Documentation

4.1.2.1 EXPORT int celt_decode (CELTDecoder * *st*, unsigned char * *data*, int *len*, celt_int16_t * *pcm*)

Decodes a frame of audio.

Parameters:

st Decoder state

data Compressed data produced by an encoder

len Number of bytes to read from "data". This MUST be exactly the number of bytes returned by the encoder. Using a larger value WILL NOT WORK.

pcm One frame (frame_size samples per channel) of decoded PCM will be returned here in 16-bit PCM format (native endian).

Returns:

Error code.

4.1.2.2 EXPORT int celt_decode_float (CELTDecoder * *st*, unsigned char * *data*, int *len*, float * *pcm*)

Decodes a frame of audio.

Parameters:

st Decoder state

data Compressed data produced by an encoder

len Number of bytes to read from "data". This MUST be exactly the number of bytes returned by the encoder. Using a larger value WILL NOT WORK.

pcm One frame (frame_size samples per channel) of decoded PCM will be returned here in float format.

Returns:

Error code.

4.1.2.3 EXPORT CELTDecoder* celt_decoder_create (const CELTMode * *mode*)

Creates a new decoder state. Each stream needs its own decoder state (can't be shared across simultaneous streams).

Parameters:

mode Contains all the information about the characteristics of the stream (must be the same characteristics as used for the encoder)

Returns:

Newly created decoder state.

Definition at line 780 of file celt.c.

References CELTDecoder::block_size, CELT_OK, CHANNELS, check_mode(), CELTDecoder::frame_size, CELTDecoder::last_pitch_index, MAX_PERIOD, CELTMode::mdctSize, CELTDecoder::mode, CELTMode::nbEBands, CELTDecoder::oldBandE, CELTDecoder::out_mem, CELTMode::overlap, CELTDecoder::overlap, and CELTDecoder::preemph_memD.

Referenced by main().

4.1.2.4 EXPORT void celt_decoder_destroy (CELTDecoder * *st*)

Destroys a a decoder state.

Parameters:

st Decoder state to be destroyed

Definition at line 807 of file celt.c.

References CELT_OK, check_mode(), CELTDecoder::mode, CELTDecoder::oldBandE, CELTDecoder::out_mem, and CELTDecoder::preemph_memD.

Referenced by main().

4.1.2.5 EXPORT int celt_encode (CELTEncoder * *st*, const celt_int16_t * *pcm*, celt_int16_t * *optional_synthesis*, unsigned char * *compressed*, int *nbCompressedBytes*)

Encodes a frame of audio.

Parameters:

st Encoder state

pcm PCM audio in signed 16-bit format (native endian). There must be exactly frame_size samples per channel. The input data is overwritten by a copy of what the remote decoder would decode.

optional_synthesis If not NULL, the encoder copies the audio signal that the decoder would decode. It is the same as calling the decoder on the compressed data, just faster.

compressed The compressed data is written here

nbCompressedBytes Number of bytes to use for compressing the frame (can change from one frame to another)

Returns:

Number of bytes written to "compressed". Should be the same as "nbCompressedBytes" unless the stream is VBR. If negative, an error has occured (see error codes). It is IMPORTANT that the length returned be somehow transmitted to the decoder. Otherwise, no decoding is possible.

4.1.2.6 EXPORT int celt_encode_float (CELTEncoder * *st*, const float * *pcm*, float * *optional_synthesis*, unsigned char * *compressed*, int *nbCompressedBytes*)

Encodes a frame of audio.

Parameters:

st Encoder state

pcm PCM audio in signed float format. There must be exactly frame_size samples per channel. The input data is overwritten by a copy of what the remote decoder would decode.

optional_synthesis If not NULL, the encoder copies the audio signal that the decoder would decode. It is the same as calling the decoder on the compressed data, just faster.

compressed The compressed data is written here

nbCompressedBytes Number of bytes to use for compressing the frame (can change from one frame to another)

Returns:

Number of bytes written to "compressed". Should be the same as "nbCompressedBytes" unless the stream is VBR. If negative, an error has occured (see error codes). It is IMPORTANT that the length returned be somehow transmitted to the decoder. Otherwise, no decoding is possible.

4.1.2.7 EXPORT CELTEncoder* celt_encoder_create (const CELTMode * *mode*)

Creates a new encoder state. Each stream needs its own encoder state (can't be shared across simultaneous streams).

Parameters:

mode Contains all the information about the characteristics of the stream (must be the same characteristics as used for the decoder)

Returns:

Newly created encoder state.

Definition at line 97 of file celt.c.

References CELTEncoder::block_size, CELTEncoder::buf, CELT_OK, check_mode(), ec_byte_writeinit(), ec_enc_init(), CELTEncoder::enc, CELTEncoder::frame_size, CELTMode::Fs, CELTEncoder::in_mem, MAX_PERIOD, CELTMode::mdctSize, CELTEncoder::mode, CELTMode::nbChannels, CELTMode::nbEBands, CELTEncoder::oldBandE, CELTEncoder::out_mem, CELTMode::overlap, CELTEncoder::overlap, CELTEncoder::pitch_enabled, CELTEncoder::preemph_memD, CELTEncoder::preemph_memE, and psydecay_init().

Referenced by main().

4.1.2.8 EXPORT int celt_encoder_ctl (CELTEncoder * *st*, int *request*, ...)

Query and set encoder parameters

Parameters:

st Encoder state

request Parameter to change or query

value Pointer to a 32-bit int value

Returns:

Error code

4.1.2.9 EXPORT void celt_encoder_destroy (CELTEncoder * *st*)

Destroys a an encoder state.

Parameters:

st Encoder state to be destroyed

Definition at line 135 of file celt.c.

References CELTEncoder::buf, CELT_OK, check_mode(), ec_byte_writeclear(), CELTEncoder::in_mem, CELTEncoder::mode, CELTEncoder::oldBandE, CELTEncoder::out_mem, CELTEncoder::preemph_memD, CELTEncoder::preemph_memE, and psydecay_clear().

Referenced by main().

4.1.2.10 EXPORT CELTMode* celt_mode_create (celt_int32_t *Fs*, int *channels*, int *frame_size*, int * *error*)

Creates a new mode struct. This will be passed to an encoder or decoder. The mode MUST NOT BE DESTROYED until the encoders and decoders that use it are destroyed as well.

Parameters:

Fs Sampling rate (32000 to 64000 Hz)

channels Number of channels

frame_size Number of samples (per channel) to encode in each packet (64 - 256)

lookahead Extra latency (in samples per channel) in addition to the frame size (between 32 and frame_size). The larger that value, the better the quality (at the expense of latency)

error Returned error code (if NULL, no error will be returned)

Returns:

A newly created mode

Definition at line 277 of file modes.c.

References ALLOC_STACK, CELTMode::bits, CELTMode::bits_stereo, CELT_BAD_ARG, CELT_COPY, CELT_OK, compute_alloc_cache(), CELTMode::eBands, CELTMode::ePredCoef, CELTMode::fft, CELTMode::Fs, M_PI, CELTMode::marker_end, CELTMode::marker_start, MAX_PERIOD, CELTMode::mdct, mdct_init(), CELTMode::mdctSize, MIN32, MIN_BINS, MODEVALID, CELTMode::nbChannels, CELTMode::nbEBands, CELTMode::nbShortMdcts, CELTMode::overlap, pitch_state_alloc(), CELTMode::prob, CELTMode::psy, psydecay_init(), Q15ONE, QCONST16, quant_prob_alloc(), CELTMode::shortMdct, CELTMode::shortMdctSize, CELTMode::shortWindow, TOTAL_MODES, and CELTMode::window.

Referenced by main().

4.1.2.11 EXPORT void celt_mode_destroy (CELTMode * *mode*)

Destroys a mode struct. Only call this after all encoders and decoders using this mode are destroyed as well.

Parameters:

mode Mode to be destroyed

Definition at line 418 of file modes.c.

References CELTMode::allocVectors, CELTMode::bits, CELTMode::bits_stereo, CELT_OK, check_mode(), CELTMode::eBands, CELTMode::energy_alloc, CELTMode::fft, CELTMode::marker_end, CELTMode::marker_start, CELTMode::mdct, mdct_clear(), MODEFREED, CELTMode::nbEBands, CELTMode::pBands, pitch_state_free(), CELTMode::prob, CELTMode::psy, psydecay_clear(), quant_prob_free(), CELTMode::shortMdct, and CELTMode::window.

Referenced by main().

4.1.2.12 EXPORT int celt_mode_info (const CELTMode * *mode*, int *request*, celt_int32_t * *value*)

Query information from a mode

Definition at line 55 of file modes.c.

References CELT_BITSTREAM_VERSION, CELT_GET_BITSTREAM_VERSION, CELT_GET_-FRAME_SIZE, CELT_GET_LOOKAHEAD, CELT_GET_NB_CHANNELS, CELT_OK, CELT_-UNIMPLEMENTED, CELTMode::mdctSize, CELTMode::nbChannels, and CELTMode::overlap.

Referenced by celt_header_init(), and main().

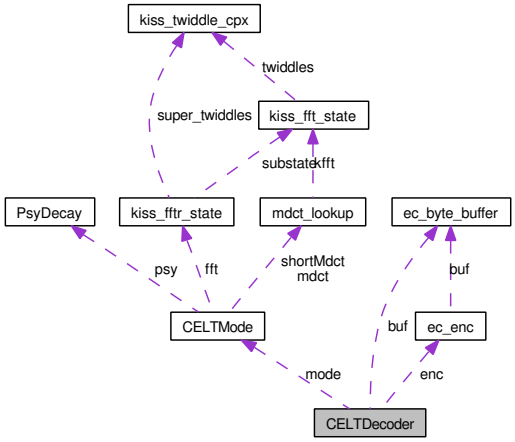
Chapter 5

Data Structure Documentation

5.1 CELTDecoder Struct Reference

Decoder state.

Collaboration diagram for CELTDecoder:



Data Fields

- const CELTMode * mode
- int frame_size
- int block_size
- int overlap
- ec_byte_buffer buf
- ec_enc enc
- celt_sig_t *restrict preemph_memD
- celt_sig_t * out_mem
- celt_word16_t * oldBandE
- int last_pitch_index
- const CELTMode * mode
- celt_sig_t *restrict preemph_memD

- celt_sig_t * out_mem
- celt_word16_t * oldBandE

5.1.1 Detailed Description

Decoder state.

Decoder state

Definition at line 762 of file celt.c.

5.1.2 Field Documentation

5.1.2.1 const CELTMode* CELTDecoder::mode

Definition at line 763 of file celt.c.

Referenced by celt_decoder_create(), and celt_decoder_destroy().

5.1.2.2 int CELTDecoder::frame_size

Definition at line 764 of file celt.c.

Referenced by celt_decoder_create().

5.1.2.3 int CELTDecoder::block_size

Definition at line 765 of file celt.c.

Referenced by celt_decoder_create().

5.1.2.4 int CELTDecoder::overlap

Definition at line 766 of file celt.c.

Referenced by celt_decoder_create().

5.1.2.5 ec_byte_buffer CELTDecoder::buf

Definition at line 768 of file celt.c.

5.1.2.6 ec_enc CELTDecoder::enc

Definition at line 769 of file celt.c.

5.1.2.7 celt_sig_t* restrict CELTDecoder::preemph_memD

Definition at line 771 of file celt.c.

Referenced by celt_decoder_create(), and celt_decoder_destroy().

5.1.2.8 celt_sig_t* CELTDecoder::out_mem

Definition at line 773 of file celt.c.

Referenced by celt_decoder_create(), and celt_decoder_destroy().

5.1.2.9 celt_word16_t* CELTDecoder::oldBandE

Definition at line 775 of file celt.c.

Referenced by celt_decoder_create(), and celt_decoder_destroy().

5.1.2.10 int CELTDecoder::last_pitch_index

Definition at line 777 of file celt.c.

Referenced by celt_decoder_create().

5.1.2.11 const CELTMode* CELTDecoder::mode

Definition at line 518 of file celt_shaping1.c.

5.1.2.12 celt_sig_t* restrict CELTDecoder::preemph_memD

Definition at line 526 of file celt_shaping1.c.

5.1.2.13 celt_sig_t* CELTDecoder::out_mem

Definition at line 528 of file celt_shaping1.c.

5.1.2.14 celt_word16_t* CELTDecoder::oldBandE

Definition at line 530 of file celt_shaping1.c.

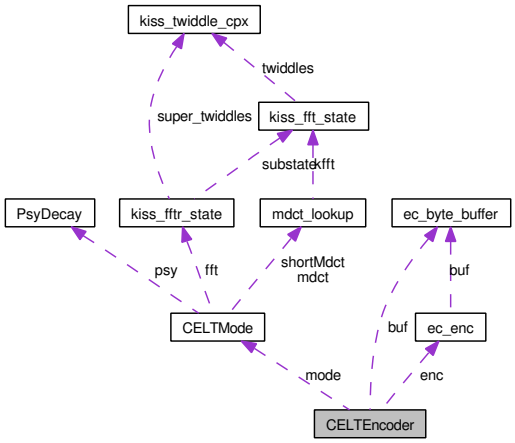
The documentation for this struct was generated from the following files:

- libcelt/celt.c
- libcelt/celt_shaping1.c

5.2 CELTEncoder Struct Reference

Encoder state.

Collaboration diagram for CELTEncoder:



Data Fields

- const CELTMode * mode
- int frame_size
- int block_size
- int overlap
- int channels
- int pitch_enabled
- ec_byte_buffer buf
- ec_enc enc
- celt_word16_t *restrict preemph_memE
- celt_sig_t *restrict preemph_memD
- celt_sig_t * in_mem
- celt_sig_t * out_mem
- celt_word16_t * oldBandE
- const CELTMode * mode
- celt_word16_t *restrict preemph_memE
- celt_sig_t *restrict preemph_memD
- celt_sig_t * in_mem
- celt_sig_t * out_mem
- celt_word16_t * oldBandE

5.2.1 Detailed Description

Encoder state.

Encoder state

Definition at line 72 of file celt.c.

5.2.2 Field Documentation

5.2.2.1 const CELTMode* CELTEncoder::mode

Mode used by the encoder

Definition at line 73 of file celt.c.

Referenced by celt_encoder_create(), and celt_encoder_destroy().

5.2.2.2 int CELTEncoder::frame_size

Definition at line 74 of file celt.c.

Referenced by celt_encoder_create().

5.2.2.3 int CELTEncoder::block_size

Definition at line 75 of file celt.c.

Referenced by celt_encoder_create().

5.2.2.4 int CELTEncoder::overlap

Definition at line 76 of file celt.c.

Referenced by celt_encoder_create().

5.2.2.5 int CELTEncoder::channels

Definition at line 77 of file celt.c.

5.2.2.6 int CELTEncoder::pitch_enabled

Definition at line 79 of file celt.c.

Referenced by celt_encoder_create().

5.2.2.7 ec_byte_buffer CELTEncoder::buf

Definition at line 81 of file celt.c.

Referenced by celt_encoder_create(), and celt_encoder_destroy().

5.2.2.8 ec_enc CELTEncoder::enc

Definition at line 82 of file celt.c.

Referenced by celt_encoder_create().

5.2.2.9 celt_word16_t* restrict CELTEncoder::preemph_memE

Definition at line 84 of file celt.c.

Referenced by celt_encoder_create(), and celt_encoder_destroy().

5.2.2.10 celt_sig_t* restrict CELTEncoder::preemph_memD

Definition at line 85 of file celt.c.

Referenced by celt_encoder_create(), and celt_encoder_destroy().

5.2.2.11 celt_sig_t* CELTEncoder::in_mem

Definition at line 87 of file celt.c.

Referenced by celt_encoder_create(), and celt_encoder_destroy().

5.2.2.12 celt_sig_t* CELTEncoder::out_mem

Definition at line 88 of file celt.c.

Referenced by celt_encoder_create(), and celt_encoder_destroy().

5.2.2.13 celt_word16_t* CELTEncoder::oldBandE

Definition at line 90 of file celt.c.

Referenced by celt_encoder_create(), and celt_encoder_destroy().

5.2.2.14 const CELTMode* CELTEncoder::mode

Mode used by the encoder

Definition at line 60 of file celt_shaping1.c.

5.2.2.15 celt_word16_t* restrict CELTEncoder::preemph_memE

Definition at line 69 of file celt_shaping1.c.

5.2.2.16 celt_sig_t* restrict CELTEncoder::preemph_memD

Definition at line 70 of file celt_shaping1.c.

5.2.2.17 celt_sig_t* CELTEncoder::in_mem

Definition at line 72 of file celt_shaping1.c.

5.2.2.18 celt_sig_t* CELTEncoder::out_mem

Definition at line 73 of file celt_shaping1.c.

5.2.2.19 celt_word16_t* CELTEncoder::oldBandE

Definition at line 75 of file celt_shaping1.c.

The documentation for this struct was generated from the following files:

- libcelt/celt.c
- libcelt/celt_shaping1.c

5.3 CELTHeader Struct Reference

Header data.

`#include <celt_header.h>`

Data Fields

- char codec_id [8]
- char codec_version [20]
- celt_int32_t version_id
- celt_int32_t header_size
- celt_int32_t sample_rate
- celt_int32_t nb_channels
- celt_int32_t frame_size
- celt_int32_t overlap
- celt_int32_t bytes_per_packet
- celt_int32_t extra_headers

5.3.1 Detailed Description

Header data.

Header data to be used for Ogg files (or possibly other encapsulation)

Definition at line 45 of file celt_header.h.

5.3.2 Field Documentation

5.3.2.1 char CELTHeader::codec_id[8]

MUST be "CELT " (four spaces)

Definition at line 46 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.2 char CELTHeader::codec_version[20]

Version used (as string)

Definition at line 47 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.3 celt_int32_t CELTHeader::version_id

Version id (negative for until stream is frozen)

Definition at line 48 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.4 celt_int32_t CELTHeader::header_size

Size of this header

Definition at line 49 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.5 celt_int32_t CELTHeader::sample_rate

Sampling rate of the original audio

Definition at line 50 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.6 celt_int32_t CELTHeader::nb_channels

Number of channels

Definition at line 51 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.7 celt_int32_t CELTHeader::frame_size

Samples per frame (per channel)

Definition at line 52 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.8 celt_int32_t CELTHeader::overlap

Overlapping samples (per channel)

Definition at line 53 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.9 celt_int32_t CELTHeader::bytes_per_packet

Number of bytes per compressed packet (0 if unknown)

Definition at line 54 of file celt_header.h.

Referenced by celt_header_init().

5.3.2.10 celt_int32_t CELTHeader::extra_headers

Number of additional headers that follow this header

Definition at line 55 of file celt_header.h.

Referenced by celt_header_init().

The documentation for this struct was generated from the following file:

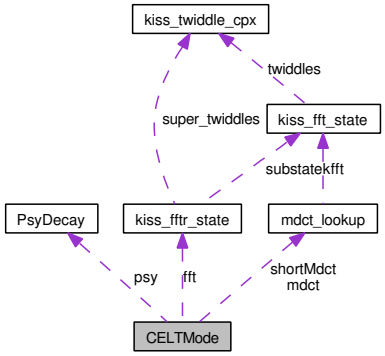
- libcelt/celt_header.h

5.4 CELTMode Struct Reference

Mode definition.

`#include <modes.h>`

Collaboration diagram for CELTMode:



Data Fields

- celt_uint32_t marker_start
- celt_int32_t Fs
- int overlap
- int mdctSize
- int nbChannels
- int nbEBands
- int nbPBands
- int pitchEnd
- const celt_int16_t * eBands
- const celt_int16_t * pBands
- celt_word16_t ePredCoef
- int nbAllocVectors
- const celt_int16_t * allocVectors
- const celt_int16_t *const * bits
- const celt_int16_t *const * bits_stereo
- mdct_lookup mdct
- kiss_fftr_cfg fft
- const celt_word16_t * window
- int nbShortMdcts
- int shortMdctSize
- mdct_lookup shortMdct
- const celt_word16_t * shortWindow
- struct PsyDecay psy
- int * prob
- const celt_int16_t * energy_alloc
- celt_uint32_t marker_end

5.4.1 Detailed Description

Mode definition.

Mode definition (opaque)

Definition at line 71 of file modes.h.

5.4.2 Field Documentation

5.4.2.1 celt_uint32_t CELTMode::marker_start

Definition at line 72 of file modes.h.

Referenced by celt_mode_create(), celt_mode_destroy(), and check_mode().

5.4.2.2 celt_int32_t CELTMode::Fs

Definition at line 73 of file modes.h.

Referenced by celt_encoder_create(), celt_header_init(), celt_mode_create(), and dump_modes().

5.4.2.3 int CELTMode::overlap

Definition at line 74 of file modes.h.

Referenced by celt_decoder_create(), celt_encoder_create(), celt_header_init(), celt_mode_create(), celt_mode_info(), do_plc(), dump_header(), and dump_modes().

5.4.2.4 int CELTMode::mdctSize

Definition at line 75 of file modes.h.

Referenced by celt_decoder_create(), celt_encoder_create(), celt_header_init(), celt_mode_create(), celt_mode_info(), dump_header(), and dump_modes().

5.4.2.5 int CELTMode::nbChannels

Definition at line 76 of file modes.h.

Referenced by celt_encoder_create(), celt_header_init(), celt_mode_create(), celt_mode_info(), compute_allocation(), dump_header(), dump_modes(), quant_coarse_energy(), quant_fine_energy(), renormalise_bands(), unquant_coarse_energy(), and unquant_fine_energy().

5.4.2.6 int CELTMode::nbEBands

Definition at line 78 of file modes.h.

Referenced by celt_decoder_create(), celt_encoder_create(), celt_mode_create(), celt_mode_destroy(), compute_alloc_cache(), compute_allocation(), compute_band_energies(), denormalise_bands(), dump_modes(), normalise_bands(), quant_bands(), quant_coarse_energy(), quant_fine_energy(), quant_prob_alloc(), renormalise_bands(), stereo_mix(), unquant_bands(), unquant_coarse_energy(), and unquant_fine_energy().

5.4.2.7 int CELTMode::nbPBands

Definition at line 79 of file modes.h.

Referenced by compute_pitch_gain(), dump_modes(), and pitch_quant_bands().

5.4.2.8 int CELTMode::pitchEnd

Definition at line 80 of file modes.h.

Referenced by compute_alloc_cache(), dump_modes(), quant_bands(), and unquant_bands().

5.4.2.9 const celt_int16_t* CELTMode::eBands

Definition for each "pseudo-critical band"

Definition at line 82 of file modes.h.

Referenced by celt_mode_create(), celt_mode_destroy(), compute_alloc_cache(), compute_band_energies(), denormalise_bands(), dump_modes(), normalise_bands(), quant_bands(), renormalise_bands(), stereo_mix(), and unquant_bands().

5.4.2.10 const celt_int16_t* CELTMode::pBands

Definition of the bands used for the pitch

Definition at line 83 of file modes.h.

Referenced by celt_mode_destroy(), compute_pitch_gain(), dump_modes(), and pitch_quant_bands().

5.4.2.11 celt_word16_t CELTMode::ePredCoef

Prediction coefficient for the energy encoding

Definition at line 85 of file modes.h.

Referenced by celt_mode_create(), and dump_modes().

5.4.2.12 int CELTMode::nbAllocVectors

Number of lines in the matrix below

Definition at line 87 of file modes.h.

Referenced by compute_allocation(), and dump_modes().

5.4.2.13 const celt_int16_t* CELTMode::allocVectors

Number of bits in each band for several rates

Definition at line 88 of file modes.h.

Referenced by celt_mode_destroy(), compute_allocation(), and dump_modes().

5.4.2.14 const celt_int16_t* const* CELTMode::bits

Cache for pulses->bits mapping in each band

Definition at line 90 of file modes.h.

Referenced by celt_mode_create(), celt_mode_destroy(), compute_allocation(), dump_modes(), quant_bands(), and unquant_bands().

5.4.2.15 const celt_int16_t* const* CELTMode::bits_stereo

Cache for pulses->bits mapping in each band

Definition at line 92 of file modes.h.

Referenced by celt_mode_create(), celt_mode_destroy(), compute_allocation(), dump_modes(), quant_bands(), and unquant_bands().

5.4.2.16 mdct_lookup CELTMode::mdct

Definition at line 95 of file modes.h.

Referenced by celt_mode_create(), and celt_mode_destroy().

5.4.2.17 kiss_fftr_cfg CELTMode::fft

Definition at line 96 of file modes.h.

Referenced by celt_mode_create(), and celt_mode_destroy().

5.4.2.18 const celt_word16_t* CELTMode::window

Definition at line 98 of file modes.h.

Referenced by celt_mode_create(), celt_mode_destroy(), do_plc(), and dump_modes().

5.4.2.19 int CELTMode::nbShortMdcts

Definition at line 100 of file modes.h.

Referenced by celt_mode_create(), dump_modes(), quant_bands(), and unquant_bands().

5.4.2.20 int CELTMode::shortMdctSize

Definition at line 101 of file modes.h.

Referenced by celt_mode_create(), and dump_modes().

5.4.2.21 mdct_lookup CELTMode::shortMdct

Definition at line 102 of file modes.h.

Referenced by celt_mode_create(), and celt_mode_destroy().

5.4.2.22 const celt_word16_t* CELTMode::shortWindow

Definition at line 103 of file modes.h.

Referenced by celt_mode_create().

5.4.2.23 struct PsyDecay CELTMode::psy [read]

Definition at line 105 of file modes.h.

Referenced by celt_mode_create(), celt_mode_destroy(), and dump_modes().

5.4.2.24 int* CELTMode::prob

Definition at line 107 of file modes.h.

Referenced by celt_mode_create(), and celt_mode_destroy().

5.4.2.25 const celt_int16_t* CELTMode::energy_alloc

Definition at line 108 of file modes.h.

Referenced by celt_mode_destroy(), compute_allocation(), and dump_modes().

5.4.2.26 celt_uint32_t CELTMode::marker_end

Definition at line 110 of file modes.h.

Referenced by celt_mode_create(), celt_mode_destroy(), and check_mode().

The documentation for this struct was generated from the following file:

- libcelt/modes.h

5.5 ec_byte_buffer Struct Reference

`#include <entcode.h>`

Data Fields

- unsigned char * buf
- unsigned char * ptr
- long storage

5.5.1 Detailed Description

Definition at line 25 of file entcode.h.

5.5.2 Field Documentation

5.5.2.1 unsigned char* ec_byte_buffer::buf

Definition at line 26 of file entcode.h.

Referenced by ec_byte_look1(), ec_byte_look4(), ec_byte_read1(), ec_byte_read4(), ec_byte_readinit(), ec_byte_write1(), ec_byte_write4(), ec_byte_writeclear(), ec_byte_writecopy(), ec_byte_writeinit(), and ec_byte_writetrunc().

5.5.2.2 unsigned char* ec_byte_buffer::ptr

Definition at line 27 of file entcode.h.

Referenced by ec_byte_adv1(), ec_byte_adv4(), ec_byte_look1(), ec_byte_look4(), ec_byte_read1(), ec_byte_read4(), ec_byte_readinit(), ec_byte_write1(), ec_byte_write4(), ec_byte_writecopy(), ec_byte_writeinit(), and ec_byte_writetrunc().

5.5.2.3 long ec_byte_buffer::storage

Definition at line 28 of file entcode.h.

Referenced by ec_byte_look1(), ec_byte_look4(), ec_byte_read1(), ec_byte_read4(), ec_byte_readinit(), ec_byte_write1(), ec_byte_write4(), ec_byte_writecopy(), and ec_byte_writeinit().

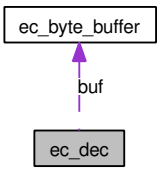
The documentation for this struct was generated from the following file:

- libcelt/entcode.h

5.6 ec_dec Struct Reference

`#include <entdec.h>`

Collaboration diagram for ec_dec:



Data Fields

- ec_byte_buffer * buf
- int rem
- ec_uint32 rng
- ec_uint32 dif
- ec_uint32 nrm

5.6.1 Detailed Description

Definition at line 12 of file entdec.h.

5.6.2 Field Documentation

5.6.2.1 ec_byte_buffer* ec_dec::buf

Definition at line 14 of file entdec.h.

Referenced by ec_dec_init(), and ec_dec_tell().

5.6.2.2 int ec_dec::rem

Definition at line 16 of file entdec.h.

Referenced by ec_dec_init().

5.6.2.3 ec_uint32 ec_dec::rng

Definition at line 18 of file entdec.h.

Referenced by ec_dec_init(), ec_dec_tell(), ec_dec_update(), ec_decode(), and ec_decode_bin().

5.6.2.4 ec_uint32 ec_dec::dif

Definition at line 21 of file entdec.h.

Referenced by ec_dec_init(), ec_dec_update(), ec_decode(), and ec_decode_bin().

5.6.2.5 ec_uint32 ec_dec::nrm

Definition at line 23 of file entdec.h.

Referenced by ec_dec_update(), ec_decode(), and ec_decode_bin().

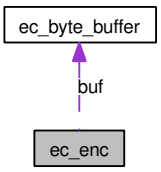
The documentation for this struct was generated from the following file:

- libcelt/entdec.h

5.7 ec_enc Struct Reference

`#include <entenc.h>`

Collaboration diagram for ec_enc:



Data Fields

- ec_byte_buffer * buf
- int rem
- size_t ext
- ec_uint32 rng
- ec_uint32 low

5.7.1 Detailed Description

Definition at line 13 of file entenc.h.

5.7.2 Field Documentation

5.7.2.1 ec_byte_buffer* ec_enc::buf

Definition at line 15 of file entenc.h.

Referenced by ec_enc_init(), and ec_enc_tell().

5.7.2.2 int ec_enc::rem

Definition at line 17 of file entenc.h.

Referenced by ec_enc_done(), ec_enc_init(), and ec_enc_tell().

5.7.2.3 size_t ec_enc::ext

Definition at line 19 of file entenc.h.

Referenced by ec_enc_done(), ec_enc_init(), and ec_enc_tell().

5.7.2.4 ec_uint32 ec_enc::rng

Definition at line 21 of file entenc.h.

Referenced by ec_enc_done(), ec_enc_init(), ec_enc_tell(), ec_encode(), and ec_encode_bin().

5.7.2.5 ec_uint32 ec_enc::low

Definition at line 23 of file entenc.h.

Referenced by ec_enc_done(), ec_enc_init(), ec_encode(), and ec_encode_bin().

The documentation for this struct was generated from the following file:

- libcelt/entenc.h

5.8 kiss_fft_cpx Struct Reference

`#include <kiss_fft.h>`

Data Fields

- kiss_fft_scalar r
- kiss_fft_scalar i

5.8.1 Detailed Description

Definition at line 69 of file kiss_fft.h.

5.8.2 Field Documentation

5.8.2.1 kiss_fft_scalar kiss_fft_cpx::r

Definition at line 70 of file kiss_fft.h.

Referenced by kiss_fft_stride(), kiss_fftr_twiddles(), and kiss_fftri().

5.8.2.2 kiss_fft_scalar kiss_fft_cpx::i

Definition at line 71 of file kiss_fft.h.

Referenced by kiss_fft_stride(), kiss_fftr_twiddles(), and kiss_fftri().

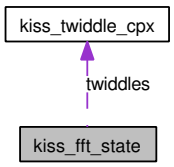
The documentation for this struct was generated from the following file:

- libcelt/kiss_fft.h

5.9 kiss_fft_state Struct Reference

`#include <_kiss_fft_guts.h>`

Collaboration diagram for kiss_fft_state:



Data Fields

- int nfft
- kiss_fft_scalar scale
- int factors [2 ∗MAXFACTORS]
- int ∗ bitrev
- kiss_twiddle_cpx twiddles [1]

5.9.1 Detailed Description

Definition at line 33 of file _kiss_fft_guts.h.

5.9.2 Field Documentation

5.9.2.1 int kiss_fft_state::nfft

Definition at line 34 of file _kiss_fft_guts.h.

Referenced by kiss_fft_alloc(), kiss_fft_stride(), kiss_fftr_twiddles(), kiss_fftri(), and kiss_ifft_stride().

5.9.2.2 kiss_fft_scalar kiss_fft_state::scale

Definition at line 36 of file _kiss_fft_guts.h.

Referenced by kiss_fft_alloc(), kiss_fft_stride(), and kiss_fftr_alloc().

5.9.2.3 int kiss_fft_state::factors[2 ∗MAXFACTORS]

Definition at line 38 of file _kiss_fft_guts.h.

Referenced by kiss_fft_alloc(), kiss_fft_stride(), kiss_fftr_inplace(), kiss_fftri(), and kiss_ifft_stride().

5.9.2.4 int∗ kiss_fft_state::bitrev

Definition at line 39 of file _kiss_fft_guts.h.

Referenced by kiss_fft_alloc(), kiss_fft_stride(), kiss_fftri(), and kiss_ifft_stride().

5.9.2.5 kiss_twiddle_cpx kiss_fft_state::twiddles[1]

Definition at line 40 of file _kiss_fft_guts.h.

Referenced by kiss_fft_alloc().

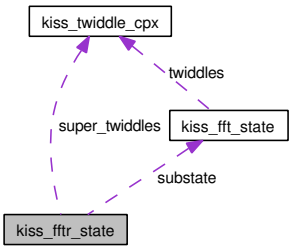
The documentation for this struct was generated from the following file:

- libcelt/_kiss_fft_guts.h

5.10 kiss_fftr_state Struct Reference

`#include <kiss_fftr.h>`

Collaboration diagram for kiss_fftr_state:



Data Fields

- kiss_fft_cfg substate
- kiss_twiddle_cpx * super_twiddles

5.10.1 Detailed Description

Definition at line 24 of file kiss_fftr.h.

5.10.2 Field Documentation

5.10.2.1 kiss_fft_cfg kiss_fftr_state::substate

Definition at line 25 of file kiss_fftr.h.

Referenced by kiss_fftr(), kiss_fftr_alloc(), kiss_fftr_inplace(), kiss_fftr_twiddles(), and kiss_fftri().

5.10.2.2 kiss_twiddle_cpx* kiss_fftr_state::super_twiddles

Definition at line 26 of file kiss_fftr.h.

Referenced by kiss_fftr_alloc(), kiss_fftr_twiddles(), and kiss_fftri().

The documentation for this struct was generated from the following file:

- libcelt/kiss_fftr.h

5.11 kiss_twiddle_cpx Struct Reference

`#include <kiss_fft.h>`

Data Fields

- kiss_twiddle_scalar r
- kiss_twiddle_scalar i

5.11.1 Detailed Description

Definition at line 74 of file kiss_fft.h.

5.11.2 Field Documentation

5.11.2.1 kiss_twiddle_scalar kiss_twiddle_cpx::r

Definition at line 75 of file kiss_fft.h.

5.11.2.2 kiss_twiddle_scalar kiss_twiddle_cpx::i

Definition at line 76 of file kiss_fft.h.

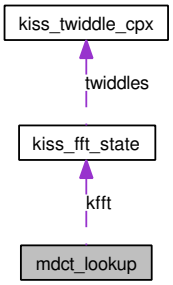
The documentation for this struct was generated from the following file:

- libcelt/kiss_fft.h

5.12 mdct_lookup Struct Reference

`#include <mdct.h>`

Collaboration diagram for mdct_lookup:



Data Fields

- int n
- kiss_fft_cfg kfft
- kiss_twiddle_scalar *restrict trig

5.12.1 Detailed Description

Definition at line 51 of file mdct.h.

5.12.2 Field Documentation

5.12.2.1 int mdct_lookup::n

Definition at line 52 of file mdct.h.

Referenced by mdct_backward(), mdct_forward(), and mdct_init().

5.12.2.2 kiss_fft_cfg mdct_lookup::kfft

Definition at line 53 of file mdct.h.

Referenced by mdct_backward(), mdct_clear(), mdct_forward(), and mdct_init().

5.12.2.3 kiss_twiddle_scalar* restrict mdct_lookup::trig

Definition at line 54 of file mdct.h.

Referenced by mdct_backward(), mdct_clear(), mdct_forward(), and mdct_init().

The documentation for this struct was generated from the following file:

- libcelt/mdct.h

5.13 PsyDecay Struct Reference

`#include <psy.h>`

Data Fields

- const celt_word16_t *restrict decayR

5.13.1 Detailed Description

Definition at line 36 of file psy.h.

5.13.2 Field Documentation

5.13.2.1 const celt_word16_t* restrict PsyDecay::decayR

Definition at line 38 of file psy.h.

Referenced by dump_modes(), psydecay_clear(), and psydecay_init().

The documentation for this struct was generated from the following file:

- libcelt/psy.h

Chapter 6

File Documentation

6.1 libcelt/_kiss_fft_guts.h File Reference

```
#include "kiss_fft.h"
```

Data Structures

- struct kiss_fft_state

Defines

- #define MIN(a, b) ((a)<(b) ? (a):(b))
- #define MAX(a, b) ((a)>(b) ? (a):(b))
- #define MAXFACTORS 32
- #define EXT32(a) (a)
- #define S_MUL(a, b) ((a)*(b))
- #define C_MUL(m, a, b)
- #define C_MULC(m, a, b)
- #define C_MUL4(m, a, b) C_MUL(m,a,b)
- #define C_FIXDIV(c, div)
- #define C_MULBYSCALAR(c, s)
- #define CHECK_OVERFLOW_OP(a, op, b)
- #define C_ADD(res, a, b)
- #define C_SUB(res, a, b)
- #define C_ADDTO(res, a)
- #define C_SUBFROM(res, a)
- #define KISS_FFT_COS(phase) (kiss_fft_scalar) cos(phase)
- #define KISS_FFT_SIN(phase) (kiss_fft_scalar) sin(phase)
- #define HALF_OF(x) ((x)*.5)
- #define kf_cexp(x, phase)
- #define kf_cexp2(x, phase)

6.1.1 Define Documentation

6.1.1.1 #define C_ADD(res, a, b)

Value:

```
do { \
            CHECK_OVERFLOW_OP((a).r,+,(b).r)\
            CHECK_OVERFLOW_OP((a).i,+,(b).i)\
            (res).r=(a).r+(b).r;  (res).i=(a).i+(b).i; \
    }while(0)
```

Definition at line 189 of file _kiss_fft_guts.h.

Referenced by kiss_fftri().

6.1.1.2 #define C_ADDTO(res, a)

Value:

```
do { \
            CHECK_OVERFLOW_OP((res).r,+,(a).r)\
            CHECK_OVERFLOW_OP((res).i,+,(a).i)\
            (res).r += (a).r;  (res).i += (a).i;\
    }while(0)
```

Definition at line 201 of file _kiss_fft_guts.h.

6.1.1.3 #define C_FIXDIV(c, div)

Definition at line 176 of file _kiss_fft_guts.h.

Referenced by kiss_fftr_twiddles().

6.1.1.4 #define C_MUL(m, a, b)

Value:

```
do{ (m).r = (a).r*(b).r - (a).i*(b).i;\
        (m).i = (a).r*(b).i + (a).i*(b).r; }while(0)
```

Definition at line 167 of file _kiss_fft_guts.h.

Referenced by kiss_fftri().

6.1.1.5 #define C_MUL4(m, a, b) C_MUL(m,a,b)

Definition at line 174 of file _kiss_fft_guts.h.

6.1.1.6 #define C_MULBYSCALAR(c, s)

Value:

```
do{ (c).r *= (s);\
        (c).i *= (s); }while(0)
```

Definition at line 177 of file _kiss_fft_guts.h.

6.1.1.7 #define C_MULC(m, a, b)

Value:

```
do{ (m).r = (a).r*(b).r + (a).i*(b).i;\
        (m).i = (a).i*(b).r - (a).r*(b).i; }while(0)
```

Definition at line 170 of file _kiss_fft_guts.h.

Referenced by kiss_fftr_twiddles().

6.1.1.8 #define C_SUB(res, a, b)

Value:

```
do { \
            CHECK_OVERFLOW_OP((a).r,-,(b).r)\
            CHECK_OVERFLOW_OP((a).i,-,(b).i)\
            (res).r=(a).r-(b).r;  (res).i=(a).i-(b).i; \
    }while(0)
```

Definition at line 195 of file _kiss_fft_guts.h.

Referenced by kiss_fftri().

6.1.1.9 #define C_SUBFROM(res, a)

Value:

```
do {\
            CHECK_OVERFLOW_OP((res).r,-,(a).r)\
            CHECK_OVERFLOW_OP((res).i,-,(a).i)\
            (res).r -= (a).r;  (res).i -= (a).i; \
    }while(0)
```

Definition at line 208 of file _kiss_fft_guts.h.

6.1.1.10 #define CHECK_OVERFLOW_OP(a, op, b)

Definition at line 185 of file _kiss_fft_guts.h.

Referenced by kiss_fftr_twiddles().

6.1.1.11 #define EXT32(a) (a)

Definition at line 164 of file _kiss_fft_guts.h.

Referenced by kiss_fftr_twiddles().

6.1.1.12 #define HALF_OF(x) ((x)*.5)

Definition at line 229 of file _kiss_fft_guts.h.

Referenced by kiss_fftr_twiddles().

6.1.1.13 #define kf_cexp(x, phase)

Value:

```
do{ \
                (x)->r = KISS_FFT_COS(phase);\
                (x)->i = KISS_FFT_SIN(phase);\
        }while(0)
```

Definition at line 232 of file _kiss_fft_guts.h.

Referenced by kiss_fft_alloc(), and kiss_fftr_alloc().

6.1.1.14 #define kf_cexp2(x, phase)

Value:

```
do{ \
      (x)->r = TRIG_UPSCALE*celt_cos_norm((phase));\
      (x)->i = TRIG_UPSCALE*celt_cos_norm((phase)-32768);\
}while(0)
```

Definition at line 238 of file _kiss_fft_guts.h.

Referenced by kiss_fft_alloc(), and kiss_fftr_alloc().

6.1.1.15 #define KISS_FFT_COS(phase) (kiss_fft_scalar) cos(phase)

Definition at line 227 of file _kiss_fft_guts.h.

6.1.1.16 #define KISS_FFT_SIN(phase) (kiss_fft_scalar) sin(phase)

Definition at line 228 of file _kiss_fft_guts.h.

6.1.1.17 #define MAX(a, b) ((a)>(b) ? (a):(b))

Definition at line 19 of file _kiss_fft_guts.h.

6.1.1.18 #define MAXFACTORS 32

Definition at line 27 of file _kiss_fft_guts.h.

6.1.1.19 #define MIN(a, b) ((a)<(b) ? (a):(b))

Definition at line 18 of file _kiss_fft_guts.h.

6.1.1.20 #define S_MUL(a, b) ((a)*(b))

Definition at line 166 of file _kiss_fft_guts.h.

Referenced by mdct_backward(), and mdct_forward().

6.2 libcelt/arch.h File Reference

Various architecture definitions for CELT.

```
#include "celt_types.h"
```

Defines

- #define celt_fatal(str) _celt_fatal(str, __FILE__, __LINE__);
- #define celt_assert(cond)
- #define celt_assert2(cond, message)
- #define IMUL32(a, b) ((a)*(b))
- #define ABS(x) ((x) < 0 ? (-(x)) : (x))
- #define ABS16(x) ((x) < 0 ? (-(x)) : (x))
- #define MIN16(a, b) ((a) < (b) ? (a) : (b))
- #define MAX16(a, b) ((a) > (b) ? (a) : (b))
- #define ABS32(x) ((x) < 0 ? (-(x)) : (x))
- #define MIN32(a, b) ((a) < (b) ? (a) : (b))
- #define MAX32(a, b) ((a) > (b) ? (a) : (b))
- #define IMAX(a, b) ((a) > (b) ? (a) : (b))
- #define UADD32(a, b) ((a)+(b))
- #define USUB32(a, b) ((a)-(b))
- #define PRINT_MIPS(file)
- #define Q15ONE 1.0f
- #define Q30ONE 1.0f
- #define NORM_SCALING 1.f
- #define NORM_SCALING_1 1.f
- #define ENER_SCALING 1.f
- #define ENER_SCALING_1 1.f
- #define PGAIN_SCALING 1.f
- #define PGAIN_SCALING_1 1.f
- #define DB_SCALING 1.f
- #define DB_SCALING_1 1.f
- #define EPSILON 1e-15f
- #define VERY_SMALL 1e-15f
- #define VERY_LARGE32 1e15f
- #define VERY_LARGE16 1e15f
- #define Q15_ONE ((celt_word16_t)1.f)
- #define Q15_ONE_1 ((celt_word16_t)1.f)
- #define QCONST16(x, bits) (x)
- #define QCONST32(x, bits) (x)
- #define NEG16(x) (-(x))
- #define NEG32(x) (-(x))
- #define EXTRACT16(x) (x)
- #define EXTEND32(x) (x)
- #define SHR16(a, shift) (a)
- #define SHL16(a, shift) (a)
- #define SHR32(a, shift) (a)
- #define SHL32(a, shift) (a)
- #define PSHR16(a, shift) (a)

- #define PSHR32(a, shift) (a)
- #define VSHR32(a, shift) (a)
- #define SATURATE16(x, a) (x)
- #define SATURATE32(x, a) (x)
- #define PSHR(a, shift) (a)
- #define SHR(a, shift) (a)
- #define SHL(a, shift) (a)
- #define SATURATE(x, a) (x)
- #define ROUND16(a, shift) (a)
- #define HALF32(x) (.5f*(x))
- #define ADD16(a, b) ((a)+(b))
- #define SUB16(a, b) ((a)-(b))
- #define ADD32(a, b) ((a)+(b))
- #define SUB32(a, b) ((a)-(b))
- #define MULT16_16_16(a, b) ((a)*(b))
- #define MULT16_16(a, b) ((celt_word32_t)(a)*(celt_word32_t)(b))
- #define MAC16_16(c, a, b) ((c)+(celt_word32_t)(a)*(celt_word32_t)(b))
- #define MULT16_32_Q11(a, b) ((a)*(b))
- #define MULT16_32_Q13(a, b) ((a)*(b))
- #define MULT16_32_Q14(a, b) ((a)*(b))
- #define MULT16_32_Q15(a, b) ((a)*(b))
- #define MULT16_32_Q16(a, b) ((a)*(b))
- #define MULT16_32_P15(a, b) ((a)*(b))
- #define MULT32_32_Q31(a, b) ((a)*(b))
- #define MAC16_32_Q11(c, a, b) ((c)+(a)*(b))
- #define MAC16_32_Q15(c, a, b) ((c)+(a)*(b))
- #define MAC16_16_Q11(c, a, b) ((c)+(a)*(b))
- #define MAC16_16_Q13(c, a, b) ((c)+(a)*(b))
- #define MAC16_16_P13(c, a, b) ((c)+(a)*(b))
- #define MULT16_16_Q11_32(a, b) ((a)*(b))
- #define MULT16_16_Q13(a, b) ((a)*(b))
- #define MULT16_16_Q14(a, b) ((a)*(b))
- #define MULT16_16_Q15(a, b) ((a)*(b))
- #define MULT16_16_P15(a, b) ((a)*(b))
- #define MULT16_16_P13(a, b) ((a)*(b))
- #define MULT16_16_P14(a, b) ((a)*(b))
- #define DIV32_16(a, b) (((celt_word32_t)(a))/(celt_word16_t)(b))
- #define PDIV32_16(a, b) (((celt_word32_t)(a))/(celt_word16_t)(b))
- #define DIV32(a, b) (((celt_word32_t)(a))/(celt_word32_t)(b))
- #define PDIV32(a, b) (((celt_word32_t)(a))/(celt_word32_t)(b))
- #define SCALEIN(a) ((a)*32768.)
- #define SCALEOUT(a) ((a)*(1/32768.))
- #define BYTES_PER_CHAR 1
- #define BITS_PER_CHAR 8
- #define LOG2_BITS_PER_CHAR 3

Typedefs

- typedef float celt_word16_t
- typedef float celt_word32_t
- typedef float celt_sig_t
- typedef float celt_norm_t
- typedef float celt_ener_t
- typedef float celt_pgain_t
- typedef float celt_mask_t

6.2.1 Detailed Description

Various architecture definitions for CELT.

Definition in file arch.h.

6.2.2 Define Documentation

6.2.2.1 #define ABS(x) ((x) < 0 ? (-(x)) : (x))

Absolute integer value.

Definition at line 52 of file arch.h.

6.2.2.2 #define ABS16(x) ((x) < 0 ? (-(x)) : (x))

Absolute 16-bit value.

Definition at line 53 of file arch.h.

6.2.2.3 #define ABS32(x) ((x) < 0 ? (-(x)) : (x))

Absolute 32-bit value.

Definition at line 56 of file arch.h.

6.2.2.4 #define ADD16(a, b) ((a)+(b))

Definition at line 183 of file arch.h.

Referenced by _celt_lpc(), alg_quant(), and find_spectral_pitch().

6.2.2.5 #define ADD32(a, b) ((a)+(b))

Definition at line 185 of file arch.h.

Referenced by celt_decode(), celt_encode(), compute_masking(), compute_pitch_gain(), fir_mem16(), iir_mem16(), kiss_fftr_twiddles(), mdct_init(), and quant_pitch().

6.2.2.6 #define BITS_PER_CHAR 8

Definition at line 235 of file arch.h.

6.2.2.7 #define BYTES_PER_CHAR 1

Definition at line 234 of file arch.h.

6.2.2.8 #define celt_assert(cond)

Definition at line 46 of file arch.h.

Referenced by cwrsi32(), cwrsi64(), icwrs32(), icwrs64(), ncwrs_u32(), ncwrs_unext32(), and ncwrs_unext64().

6.2.2.9 #define celt_assert2(cond, message)

Definition at line 47 of file arch.h.

Referenced by alg_quant().

6.2.2.10 #define celt_fatal(str) _celt_fatal(str, __FILE__, __LINE__);

Definition at line 41 of file arch.h.

Referenced by denormalise_bands(), kf_work(), ki_work(), kiss_fft_stride(), and kiss_ifft_stride().

6.2.2.11 #define DB_SCALING 1.f

Definition at line 148 of file arch.h.

6.2.2.12 #define DB_SCALING_1 1.f

Definition at line 149 of file arch.h.

6.2.2.13 #define DIV32(a, b) (((celt_word32_t)(a))/(celt_word32_t)(b))

Definition at line 216 of file arch.h.

Referenced by kiss_fft_alloc(), kiss_fftr_alloc(), and mdct_init().

6.2.2.14 #define DIV32_16(a, b) (((celt_word32_t)(a))/(celt_word16_t)(b))

Definition at line 214 of file arch.h.

Referenced by _celt_lpc(), and stereo_mix().

6.2.2.15 #define ENER_SCALING 1.f

Definition at line 143 of file arch.h.

6.2.2.16 #define ENER_SCALING_1 1.f

Definition at line 144 of file arch.h.

6.2.2.17 #define EPSILON 1e-15f

Definition at line 151 of file arch.h.

Referenced by compute_pitch_gain(), find_spectral_pitch(), renormalise_vector(), and stereo_mix().

6.2.2.18 #define EXTEND32(x) (x)

Definition at line 164 of file arch.h.

Referenced by _celt_lpc(), celt_encode(), celt_encode_float(), fir_mem16(), iir_mem16(), mdct_init(), quant_pitch(), and stereo_mix().

6.2.2.19 #define EXTRACT16(x) (x)

Definition at line 163 of file arch.h.

Referenced by alg_quant(), celt_encode_float(), compute_pitch_gain(), find_spectral_pitch(), fir_mem16(), and iir_mem16().

6.2.2.20 #define HALF32(x) (.5f*(x))

Definition at line 181 of file arch.h.

Referenced by mdct_forward().

6.2.2.21 #define IMAX(a, b) ((a) > (b) ? (a) : (b))

Maximum int value.

Definition at line 59 of file arch.h.

6.2.2.22 #define IMUL32(a, b) ((a)*(b))

Definition at line 50 of file arch.h.

Referenced by ec_dec_update(), ec_encode(), and ec_encode_bin().

6.2.2.23 #define LOG2_BITS_PER_CHAR 3

Definition at line 236 of file arch.h.

6.2.2.24 #define MAC16_16(c, a, b) ((c)+(celt_word32_t)(a)*(celt_word32_t)(b))

Definition at line 189 of file arch.h.

Referenced by alg_quant(), compute_pitch_gain(), fir_mem16(), iir_mem16(), intra_prediction(), renormalise_vector(), and vq_index().

6.2.2.25 #define MAC16_16_P13(c, a, b) ((c)+(a)*(b))

Definition at line 205 of file arch.h.

Referenced by _celt_lpc().

6.2.2.26 #define MAC16_16_Q11(c, a, b) ((c)+(a)*(b))

Definition at line 203 of file arch.h.

6.2.2.27 #define MAC16_16_Q13(c, a, b) ((c)+(a)*(b))

Definition at line 204 of file arch.h.

6.2.2.28 #define MAC16_32_Q11(c, a, b) ((c)+(a)*(b))

Definition at line 200 of file arch.h.

6.2.2.29 #define MAC16_32_Q15(c, a, b) ((c)+(a)*(b))

Definition at line 201 of file arch.h.

Referenced by celt_decode_float(), and celt_encode_float().

6.2.2.30 #define MAX16(a, b) ((a) > (b) ? (a) : (b))

Maximum 16-bit value.

Definition at line 55 of file arch.h.

6.2.2.31 #define MAX32(a, b) ((a) > (b) ? (a) : (b))

Maximum 32-bit value.

Definition at line 58 of file arch.h.

Referenced by stereo_mix().

6.2.2.32 #define MIN16(a, b) ((a) < (b) ? (a) : (b))

Maximum 16-bit value.

Definition at line 54 of file arch.h.

6.2.2.33 #define MIN32(a, b) ((a) < (b) ? (a) : (b))

Maximum 32-bit value.

Definition at line 57 of file arch.h.

Referenced by celt_mode_create().

6.2.2.34 #define MULT16_16(a, b) ((celt_word32_t)(a)∗(celt_word32_t)(b))

Definition at line 188 of file arch.h.

Referenced by _celt_lpc(), alg_quant(), celt_encode(), celt_encode_float(), compute_masking(), find_spectral_pitch(), fir_mem16(), iir_mem16(), intra_prediction(), intra_unquant(), quant_pitch(), renormalise_vector(), and stereo_mix().

6.2.2.35 #define MULT16_16_16(a, b) ((a)∗(b))

Definition at line 187 of file arch.h.

Referenced by find_spectral_pitch().

6.2.2.36 #define MULT16_16_P13(a, b) ((a)∗(b))

Definition at line 211 of file arch.h.

6.2.2.37 #define MULT16_16_P14(a, b) ((a)∗(b))

Definition at line 212 of file arch.h.

6.2.2.38 #define MULT16_16_P15(a, b) ((a)∗(b))

Definition at line 210 of file arch.h.

6.2.2.39 #define MULT16_16_Q11_32(a, b) ((a)∗(b))

Definition at line 206 of file arch.h.

6.2.2.40 #define MULT16_16_Q13(a, b) ((a)∗(b))

Definition at line 207 of file arch.h.

Referenced by _celt_lpc().

6.2.2.41 #define MULT16_16_Q14(a, b) ((a)∗(b))

Definition at line 208 of file arch.h.

Referenced by stereo_mix().

6.2.2.42 #define MULT16_16_Q15(a, b) ((a)∗(b))

Definition at line 209 of file arch.h.

Referenced by alg_quant(), exp_rotation(), find_spectral_pitch(), pitch_quant_bands(), quant_bands(), renormalise_vector(), and unquant_bands().

6.2.2.43 #define MULT16_32_P15(a, b) ((a)*(b))

Definition at line 196 of file arch.h.

6.2.2.44 #define MULT16_32_Q11(a, b) ((a)*(b))

Definition at line 191 of file arch.h.

6.2.2.45 #define MULT16_32_Q13(a, b) ((a)*(b))

Definition at line 192 of file arch.h.

Referenced by denormalise_bands().

6.2.2.46 #define MULT16_32_Q14(a, b) ((a)*(b))

Definition at line 193 of file arch.h.

6.2.2.47 #define MULT16_32_Q15(a, b) ((a)*(b))

Definition at line 194 of file arch.h.

Referenced by celt_decode(), celt_encode(), celt_encode_float(), compute_pitch_gain(), denormalise_bands(), mdct_backward(), and mdct_forward().

6.2.2.48 #define MULT16_32_Q16(a, b) ((a)*(b))

Definition at line 195 of file arch.h.

6.2.2.49 #define MULT32_32_Q31(a, b) ((a)*(b))

Definition at line 198 of file arch.h.

6.2.2.50 #define NEG16(x) (-(x))

Definition at line 161 of file arch.h.

Referenced by iir_mem16().

6.2.2.51 #define NEG32(x) (-(x))

Definition at line 162 of file arch.h.

Referenced by _celt_lpc().

6.2.2.52 #define NORM_SCALING 1.f

Definition at line 141 of file arch.h.

Referenced by alg_quant().

6.2.2.53 #define NORM_SCALING_1 1.f

Definition at line 142 of file arch.h.

6.2.2.54 #define PDIV32(a, b) (((celt_word32_t)(a))/(celt_word32_t)(b))

Definition at line 217 of file arch.h.

6.2.2.55 #define PDIV32_16(a, b) (((celt_word32_t)(a))/(celt_word16_t)(b))

Definition at line 215 of file arch.h.

6.2.2.56 #define PGAIN_SCALING 1.f

Definition at line 145 of file arch.h.

6.2.2.57 #define PGAIN_SCALING_1 1.f

Definition at line 146 of file arch.h.

6.2.2.58 #define PRINT_MIPS(file)

Definition at line 63 of file arch.h.
Referenced by main().

6.2.2.59 #define PSHR(a, shift) (a)

Definition at line 175 of file arch.h.

6.2.2.60 #define PSHR16(a, shift) (a)

Definition at line 169 of file arch.h.

6.2.2.61 #define PSHR32(a, shift) (a)

Definition at line 170 of file arch.h.
Referenced by _celt_lpc(), fir_mem16(), iir_mem16(), kiss_fftr_twiddles(), and renormalise_vector().

6.2.2.62 #define Q15_ONE ((celt_word16_t)1.f)

Definition at line 155 of file arch.h.
Referenced by intra_prediction(), and intra_unquant().

6.2.2.63 #define Q15_ONE_1 ((celt_word16_t)1.f)

Definition at line 156 of file arch.h.

6.2.2.64 #define Q15ONE 1.0f

Definition at line 138 of file arch.h.

Referenced by celt_encode_float(), celt_mode_create(), intra_fold(), psydecay_init(), and quant_pitch().

6.2.2.65 #define Q30ONE 1.0f

Definition at line 139 of file arch.h.

6.2.2.66 #define QCONST16(x, bits) (x)

Definition at line 158 of file arch.h.

Referenced by alg_quant(), celt_encode(), celt_encode_float(), celt_mode_create(), compute_pitch_gain(), exp_rotation(), and renormalise_bands().

6.2.2.67 #define QCONST32(x, bits) (x)

Definition at line 159 of file arch.h.

Referenced by celt_encode(), celt_encode_float(), and quant_pitch().

6.2.2.68 #define ROUND16(a, shift) (a)

Definition at line 180 of file arch.h.

Referenced by alg_quant().

6.2.2.69 #define SATURATE(x, a) (x)

Definition at line 178 of file arch.h.

Referenced by fir_mem16(), and iir_mem16().

6.2.2.70 #define SATURATE16(x, a) (x)

Definition at line 172 of file arch.h.

6.2.2.71 #define SATURATE32(x, a) (x)

Definition at line 173 of file arch.h.

6.2.2.72 #define SCALEIN(a) ((a)∗32768.)

Definition at line 219 of file arch.h.

Referenced by celt_encode_float().

6.2.2.73 #define SCALEOUT(a) ((a)∗(1/32768.))

Definition at line 220 of file arch.h.

Referenced by celt_decode_float(), celt_encode(), and celt_encode_float().

6.2.2.74 #define SHL(a, shift) (a)

Definition at line 177 of file arch.h.

6.2.2.75 #define SHL16(a, shift) (a)

Definition at line 166 of file arch.h.

Referenced by alg_quant(), quant_bands(), and unquant_bands().

6.2.2.76 #define SHL32(a, shift) (a)

Definition at line 168 of file arch.h.

Referenced by _celt_lpc(), celt_encode(), celt_encode_float(), denormalise_bands(), kiss_fft_alloc(), kiss_fftr_alloc(), mdct_init(), renormalise_vector(), and stereo_mix().

6.2.2.77 #define SHR(a, shift) (a)

Definition at line 176 of file arch.h.

6.2.2.78 #define SHR16(a, shift) (a)

Definition at line 165 of file arch.h.

Referenced by quant_pitch().

6.2.2.79 #define SHR32(a, shift) (a)

Definition at line 167 of file arch.h.

Referenced by alg_quant(), celt_encode(), celt_encode_float(), compute_pitch_gain(), find_spectral_-pitch(), and kiss_fftr_twiddles().

6.2.2.80 #define SUB16(a, b) ((a)-(b))

Definition at line 184 of file arch.h.

Referenced by _celt_lpc(), alg_quant(), find_spectral_pitch(), and vq_index().

6.2.2.81 #define SUB32(a, b) ((a)-(b))

Definition at line 186 of file arch.h.

Referenced by _celt_lpc(), celt_encode(), celt_encode_float(), and kiss_fftr_twiddles().

6.2.2.82 #define UADD32(a, b) ((a)+(b))

Definition at line 60 of file arch.h.

6.2.2.83 #define USUB32(a, b) ((a)-(b))

Definition at line 61 of file arch.h.

6.2.2.84 #define VERY_LARGE16 1e15f

Definition at line 154 of file arch.h.

Referenced by alg_quant().

6.2.2.85 #define VERY_LARGE32 1e15f

Definition at line 153 of file arch.h.

Referenced by vq_index().

6.2.2.86 #define VERY_SMALL 1e-15f

Definition at line 152 of file arch.h.

6.2.2.87 #define VSHR32(a, shift) (a)

Definition at line 171 of file arch.h.

Referenced by stereo_mix().

6.2.3 Typedef Documentation

6.2.3.1 typedef float celt_ener_t

Definition at line 134 of file arch.h.

6.2.3.2 typedef float celt_mask_t

Definition at line 136 of file arch.h.

6.2.3.3 typedef float celt_norm_t

Definition at line 133 of file arch.h.

6.2.3.4 typedef float celt_pgain_t

Definition at line 135 of file arch.h.

6.2.3.5 typedef float celt_sig_t

Definition at line 132 of file arch.h.

6.2.3.6 typedef float celt_word16_t

Definition at line 129 of file arch.h.

6.2.3.7 typedef float celt_word32_t

Definition at line 130 of file arch.h.

6.3 libcelt/bands-pc.c File Reference

```
#include <math.h>
#include "bands.h"
#include "modes.h"
#include "vq.h"
#include "cwrs.h"
#include "stack_alloc.h"
#include "os_support.h"
#include "mathops.h"
```

Functions

- void exp_rotation (celt_norm_t *X, int len, int dir, int stride, int iter)
- void compute_band_energies (const CELTMode *m, const celt_sig_t *X, celt_ener_t *bank)
- void normalise_bands (const CELTMode *m, const celt_sig_t *restrict freq, celt_norm_t *restrict X, const celt_ener_t *bank)
- void renormalise_bands (const CELTMode *m, celt_norm_t *restrict X)
- void denormalise_bands (const CELTMode *m, const celt_norm_t *restrict X, celt_sig_t *restrict freq, const celt_ener_t *bank)
- void compute_pitch_gain (const CELTMode *m, const celt_norm_t *X, const celt_norm_t *P, celt_pgain_t *gains)
- void pitch_quant_bands (const CELTMode *m, celt_norm_t *restrict P, const celt_pgain_t *restrict gains)
- void quant_bands (const CELTMode *m, celt_norm_t *restrict X, celt_norm_t *P, celt_mask_t *W, int total_bits, ec_enc *enc)
- void unquant_bands (const CELTMode *m, celt_norm_t *restrict X, celt_norm_t *P, int total_bits, ec_dec *dec)
- void stereo_mix (const CELTMode *m, celt_norm_t *X, const celt_ener_t *bank, int dir)

Variables

- const celt_word16_t sqrtC_1 [2] = {QCONST16(1.f, 14), QCONST16(1.414214f, 14)}

6.3.1 Function Documentation

6.3.1.1 void compute_band_energies (const CELTMode * *m*, const celt_sig_t * *X*, celt_ener_t * *bands*)

Compute the amplitude (sqrt energy) in each of the bands

Parameters:

m Mode data

X Spectrum

bands Square root of the energy for each band (returned)

Definition at line 155 of file bands-pc.c.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), and renormalise_bands().

6.3.1.2 void compute_pitch_gain (const CELTMode * *m*, const celt_norm_t * *X*, const celt_norm_t * *P*, celt_pgain_t * *gains*)

Compute the pitch predictor gain for each pitch band

Parameters:

m Mode data

X Spectrum to predict

P Pitch vector (normalised)

gains Gain computed for each pitch band (returned)

bank Square root of the energy for each band

Definition at line 232 of file bands-pc.c.

Referenced by celt_encode(), and celt_encode_float().

6.3.1.3 void denormalise_bands (const CELTMode * *m*, const celt_norm_t *restrict *X*, celt_sig_t *restrict *freq*, const celt_ener_t * *bands*)

Denormalise each band of X to restore full amplitude

Parameters:

m Mode data

X Spectrum (returned de-normalised)

bands Square root of the energy for each band

Definition at line 209 of file bands-pc.c.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.3.1.4 void exp_rotation (celt_norm_t * *X*, int *len*, int *dir*, int *stride*, int *iter*)

Applies a series of rotations so that pulses are spread like a two-sided exponential. The effect of this is to reduce the tonal noise created by the sparse spectrum resulting from the pulse codebook

Definition at line 45 of file bands-pc.c.

References k, MULT16_16_Q15, and QCONST16.

Referenced by quant_bands(), and unquant_bands().

6.3.1.5 void normalise_bands (const CELTMode * *m*, const celt_sig_t *restrict *freq*, celt_norm_t *restrict *X*, const celt_ener_t * *bands*)

Normalise each band of X such that the energy in each band is equal to 1

Parameters:

m Mode data

X Spectrum (returned normalised)

bands Square root of the energy for each band

Definition at line 176 of file bands-pc.c.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), and renormalise_bands().

6.3.1.6 void pitch_quant_bands (const CELTMode * *m*, celt_norm_t *restrict *P*, const celt_pgain_t *restrict *gains*)

Definition at line 270 of file bands-pc.c.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.3.1.7 void quant_bands (const CELTMode * *m*, celt_norm_t *restrict *X*, celt_norm_t * *P*, celt_mask_t * *W*, int *total_bits*, ec_enc * *enc*)

Definition at line 288 of file bands-pc.c.

References alg_quant(), ALLOC, celt_sqrt, CHANNELS, compute_allocation(), CELTMode::eBands, ec_enc_tell(), exp_rotation(), intra_fold(), intra_prediction(), MULT16_16_Q15, n, CELTMode::nbEBands, CELTMode::pitchEnd, RESTORE_STACK, SAVE_STACK, SHL16, and VARDECL.

Referenced by celt_encode(), and celt_encode_float().

6.3.1.8 void renormalise_bands (const CELTMode * *m*, celt_norm_t *restrict *X*)

Definition at line 196 of file bands-pc.c.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.3.1.9 void stereo_mix (const CELTMode * *m*, celt_norm_t * *X*, const celt_ener_t * *bank*, int *dir*)

Definition at line 410 of file bands-pc.c.

References celt_sqrt, CHANNELS, DIV32_16, CELTMode::eBands, EPSILON, EXTEND32, MAX32, MULT16_16, MULT16_16_Q14, CELTMode::nbEBands, SHL32, and VSHR32.

6.3.1.10 void unquant_bands (const CELTMode * *m*, celt_norm_t *restrict *X*, celt_norm_t * *P*, int *total_bits*, ec_dec * *dec*)

Definition at line 353 of file bands-pc.c.

References alg_unquant(), ALLOC, celt_sqrt, CHANNELS, compute_allocation(), CELTMode::eBands, ec_dec_tell(), exp_rotation(), intra_fold(), intra_unquant(), MULT16_16_Q15, n, CELTMode::nbEBands, CELTMode::pitchEnd, RESTORE_STACK, SAVE_STACK, SHL16, and VARDECL.

Referenced by celt_decode(), and celt_decode_float().

6.3.2 Variable Documentation

6.3.2.1 const celt_word16_t sqrtC_1[2] = {QCONST16(1.f, 14), QCONST16(1.414214f, 14)}

Definition at line 79 of file bands-pc.c.

Referenced by denormalise_bands().

6.4 libcelt/bands.c File Reference

```
#include <math.h>
#include "bands.h"
#include "modes.h"
#include "vq.h"
#include "cwrs.h"
#include "stack_alloc.h"
#include "os_support.h"
#include "mathops.h"
#include "rate.h"
```

Functions

- void compute_band_energies (const CELTMode *m, const celt_sig_t *X, celt_ener_t *bank)
- void normalise_bands (const CELTMode *m, const celt_sig_t *restrict freq, celt_norm_t *restrict X, const celt_ener_t *bank)
- void renormalise_bands (const CELTMode *m, celt_norm_t *restrict X)
- void denormalise_bands (const CELTMode *m, const celt_norm_t *restrict X, celt_sig_t *restrict freq, const celt_ener_t *bank)
- void compute_pitch_gain (const CELTMode *m, const celt_norm_t *X, const celt_norm_t *P, celt_pgain_t *gains)
- void pitch_quant_bands (const CELTMode *m, celt_norm_t *restrict P, const celt_pgain_t *restrict gains)
- void stereo_decision (const CELTMode *m, celt_norm_t *restrict X, int *stereo_mode, int len)
- void quant_bands (const CELTMode *m, celt_norm_t *restrict X, celt_norm_t *P, celt_mask_t *W, const celt_ener_t *bandE, const int *stereo_mode, int *pulses, int shortBlocks, int total_bits, ec_enc *enc)
- void unquant_bands (const CELTMode *m, celt_norm_t *restrict X, celt_norm_t *P, const celt_ener_t *bandE, const int *stereo_mode, int *pulses, int shortBlocks, int total_bits, ec_dec *dec)

Variables

- const celt_word16_t sqrtC_1 [2] = {QCONST16(1.f, 14), QCONST16(1.414214f, 14)}

6.4.1 Function Documentation

6.4.1.1 void compute_band_energies (const CELTMode * *m*, const celt_sig_t * *X*, celt_ener_t * *bands*)

Compute the amplitude (sqrt energy) in each of the bands

Parameters:

m Mode data

X Spectrum

bands Square root of the energy for each band (returned)

Definition at line 111 of file bands.c.

References CHANNELS, CELTMode::eBands, and CELTMode::nbEBands.

6.4.1.2 void compute_pitch_gain (const CELTMode * *m*, const celt_norm_t * *X*, const celt_norm_t * *P*, celt_pgain_t * *gains*)

Compute the pitch predictor gain for each pitch band

Parameters:

m Mode data

X Spectrum to predict

P Pitch vector (normalised)

gains Gain computed for each pitch band (returned)

bank Square root of the energy for each band

Definition at line 191 of file bands.c.

References ADD32, celt_div, CHANNELS, EPSILON, EXTRACT16, MAC16_16, MULT16_32_Q15, CELTMode::nbPBands, CELTMode::pBands, QCONST16, and SHR32.

6.4.1.3 void denormalise_bands (const CELTMode * *m*, const celt_norm_t *restrict *X*, celt_sig_t *restrict *freq*, const celt_ener_t * *bands*)

Denormalise each band of X to restore full amplitude

Parameters:

m Mode data

X Spectrum (returned de-normalised)

bands Square root of the energy for each band

Definition at line 167 of file bands.c.

References celt_fatal, CHANNELS, CELTMode::eBands, MULT16_32_Q13, MULT16_32_Q15, CELTMode::nbEBands, SHL32, and sqrtC_1.

6.4.1.4 void normalise_bands (const CELTMode * *m*, const celt_sig_t *restrict *freq*, celt_norm_t *restrict *X*, const celt_ener_t * *bands*)

Normalise each band of X such that the energy in each band is equal to 1

Parameters:

m Mode data

X Spectrum (returned normalised)

bands Square root of the energy for each band

Definition at line 132 of file bands.c.

References CHANNELS, CELTMode::eBands, and CELTMode::nbEBands.

6.4.1.5 void pitch_quant_bands (const CELTMode * *m*, celt_norm_t *restrict *P*, const celt_pgain_t *restrict *gains*)

Definition at line 229 of file bands.c.

References CHANNELS, MULT16_16_Q15, CELTMode::nbPBands, and CELTMode::pBands.

6.4.1.6 void quant_bands (const CELTMode * *m*, celt_norm_t *restrict *X*, celt_norm_t * *P*, celt_mask_t * *W*, const celt_ener_t * *bandE*, const int * *stereo_mode*, int * *pulses*, int *time_domain*, int *total_bits*, ec_enc * *enc*)

Quantisation/encoding of the residual spectrum

Parameters:

m Mode data

X Residual (normalised)

P Pitch vector (normalised)

W Perceptual weighting

total_bits Total number of bits that can be used for the frame (including the ones already spent)

enc Entropy encoder

Definition at line 333 of file bands.c.

References alg_quant(), ALLOC, BITRES, CELTMode::bits, CELTMode::bits_stereo, celt_sqrt, CHANNELS, CELTMode::eBands, ec_enc_tell(), intra_fold(), intra_prediction(), MULT16_16_Q15, n, CELTMode::nbEBands, CELTMode::nbShortMdcts, CELTMode::pitchEnd, RESTORE_STACK, SAVE_STACK, SHL16, and VARDECL.

6.4.1.7 void renormalise_bands (const CELTMode * *m*, celt_norm_t *restrict *X*)

Definition at line 152 of file bands.c.

References ALLOC, CHANNELS, compute_band_energies(), CELTMode::eBands, CELTMode::nbChannels, CELTMode::nbEBands, normalise_bands(), QCONST16, renormalise_vector(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.4.1.8 void stereo_decision (const CELTMode * *m*, celt_norm_t *restrict *X*, int * *stereo_mode*, int *len*)

Definition at line 321 of file bands.c.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.4.1.9 void unquant_bands (const CELTMode * *m*, celt_norm_t *restrict *X*, celt_norm_t * *P*, const celt_ener_t * *bandE*, const int * *stereo_mode*, int * *pulses*, int *time_domain*, int *total_bits*, ec_dec * *dec*)

Decoding of the residual spectrum

Parameters:

m Mode data

X Residual (normalised)

P Pitch vector (normalised)

total_bits Total number of bits that can be used for the frame (including the ones already spent)

dec Entropy decoder

Definition at line 423 of file bands.c.

References alg_unquant(), ALLOC, BITRES, CELTMode::bits, CELTMode::bits_stereo, celt_sqrt, CHANNELS, CELTMode::eBands, ec_dec_tell(), intra_fold(), intra_unquant(), MULT16_16_Q15, n, CELTMode::nbEBands, CELTMode::nbShortMdcts, CELTMode::pitchEnd, RESTORE_STACK, SAVE_STACK, SHL16, and VARDECL.

6.4.2 Variable Documentation

6.4.2.1 const celt_word16_t sqrtC_1[2] = {QCONST16(1.f, 14), QCONST16(1.414214f, 14)}

Definition at line 46 of file bands.c.

6.5 libcelt/bands.h File Reference

```
#include "arch.h"
#include "modes.h"
#include "entenc.h"
#include "entdec.h"
#include "rate.h"
```

Functions

- void exp_rotation (celt_norm_t *X, int len, int dir, int stride, int iter)
- void compute_band_energies (const CELTMode *m, const celt_sig_t *X, celt_ener_t *bands)
- void normalise_bands (const CELTMode *m, const celt_sig_t *restrict freq, celt_norm_t *restrict X, const celt_ener_t *bands)
- void renormalise_bands (const CELTMode *m, celt_norm_t *restrict X)
- void denormalise_bands (const CELTMode *m, const celt_norm_t *restrict X, celt_sig_t *restrict freq, const celt_ener_t *bands)
- void compute_pitch_gain (const CELTMode *m, const celt_norm_t *X, const celt_norm_t *P, celt_pgain_t *gains)
- void pitch_quant_bands (const CELTMode *m, celt_norm_t *restrict P, const celt_pgain_t *restrict gains)
- void quant_bands (const CELTMode *m, celt_norm_t *restrict X, celt_norm_t *P, celt_mask_t *W, const celt_ener_t *bandE, const int *stereo_mode, int *pulses, int time_domain, int total_bits, ec_enc *enc)
- void unquant_bands (const CELTMode *m, celt_norm_t *restrict X, celt_norm_t *P, const celt_ener_t *bandE, const int *stereo_mode, int *pulses, int time_domain, int total_bits, ec_dec *dec)
- void stereo_decision (const CELTMode *m, celt_norm_t *restrict X, int *stereo_mode, int len)

6.5.1 Function Documentation

6.5.1.1 void compute_band_energies (const CELTMode * *m*, const celt_sig_t * *X*, celt_ener_t * *bands*)

Compute the amplitude (sqrt energy) in each of the bands

Parameters:

m Mode data

X Spectrum

bands Square root of the energy for each band (returned)

Definition at line 155 of file bands-pc.c.

References CHANNELS, CELTMode::eBands, and CELTMode::nbEBands.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), and renormalise_bands().

6.5.1.2 void compute_pitch_gain (const CELTMode * *m*, const celt_norm_t * *X*, const celt_norm_t * *P*, celt_pgain_t * *gains*)

Compute the pitch predictor gain for each pitch band

Parameters:

m Mode data

X Spectrum to predict

P Pitch vector (normalised)

gains Gain computed for each pitch band (returned)

bank Square root of the energy for each band

Definition at line 232 of file bands-pc.c.

References ADD32, celt_div, CHANNELS, EPSILON, EXTRACT16, MAC16_16, MULT16_32_Q15, CELTMode::nbPBands, CELTMode::pBands, QCONST16, and SHR32.

Referenced by celt_encode(), and celt_encode_float().

6.5.1.3 void denormalise_bands (const CELTMode * *m*, const celt_norm_t *restrict *X*, celt_sig_t *restrict *freq*, const celt_ener_t * *bands*)

Denormalise each band of X to restore full amplitude

Parameters:

m Mode data

X Spectrum (returned de-normalised)

bands Square root of the energy for each band

Definition at line 209 of file bands-pc.c.

References celt_fatal, CHANNELS, CELTMode::eBands, MULT16_32_Q13, MULT16_32_Q15, CELTMode::nbEBands, SHL32, and sqrtC_1.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.5.1.4 void exp_rotation (celt_norm_t * *X*, int *len*, int *dir*, int *stride*, int *iter*)

Applies a series of rotations so that pulses are spread like a two-sided exponential. The effect of this is to reduce the tonal noise created by the sparse spectrum resulting from the pulse codebook

Definition at line 45 of file bands-pc.c.

References k, MULT16_16_Q15, and QCONST16.

Referenced by quant_bands(), and unquant_bands().

6.5.1.5 void normalise_bands (const CELTMode * *m*, const celt_sig_t *restrict *freq*, celt_norm_t *restrict *X*, const celt_ener_t * *bands*)

Normalise each band of X such that the energy in each band is equal to 1

Parameters:

m Mode data

X Spectrum (returned normalised)

bands Square root of the energy for each band

Definition at line 176 of file bands-pc.c.

References CHANNELS, CELTMode::eBands, and CELTMode::nbEBands.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), and renormalise_bands().

6.5.1.6 void pitch_quant_bands (const CELTMode * *m*, celt_norm_t *restrict *P*, const celt_pgain_t *restrict *gains*)

Definition at line 270 of file bands-pc.c.

References CHANNELS, MULT16_16_Q15, CELTMode::nbPBands, and CELTMode::pBands.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.5.1.7 void quant_bands (const CELTMode * *m*, celt_norm_t *restrict *X*, celt_norm_t * *P*, celt_mask_t * *W*, const celt_ener_t * *bandE*, const int * *stereo_mode*, int * *pulses*, int *time_domain*, int *total_bits*, ec_enc * *enc*)

Quantisation/encoding of the residual spectrum

Parameters:

m Mode data

X Residual (normalised)

P Pitch vector (normalised)

W Perceptual weighting

total_bits Total number of bits that can be used for the frame (including the ones already spent)

enc Entropy encoder

Definition at line 333 of file bands.c.

References alg_quant(), ALLOC, BITRES, CELTMode::bits, CELTMode::bits_stereo, celt_sqrt, CHANNELS, CELTMode::eBands, ec_enc_tell(), intra_fold(), intra_prediction(), MULT16_16_Q15, n, CELTMode::nbEBands, CELTMode::nbShortMdcts, CELTMode::pitchEnd, RESTORE_STACK, SAVE_STACK, SHL16, and VARDECL.

6.5.1.8 void renormalise_bands (const CELTMode * *m*, celt_norm_t *restrict *X*)

Definition at line 196 of file bands-pc.c.

References ALLOC, CHANNELS, compute_band_energies(), CELTMode::eBands, CELTMode::nbChannels, CELTMode::nbEBands, normalise_bands(), QCONST16, renormalise_vector(), RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.5.1.9 void stereo_decision (const CELTMode * *m*, celt_norm_t *restrict *X*, int * *stereo_mode*, int *len*)

Definition at line 321 of file bands.c.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.5.1.10 void unquant_bands (const CELTMode * *m*, celt_norm_t *restrict *X*, celt_norm_t * *P*, const celt_ener_t * *bandE*, const int * *stereo_mode*, int * *pulses*, int *time_domain*, int *total_bits*, ec_dec * *dec*)

Decoding of the residual spectrum

Parameters:

m Mode data

X Residual (normalised)

P Pitch vector (normalised)

total_bits Total number of bits that can be used for the frame (including the ones already spent)

dec Entropy decoder

Definition at line 423 of file bands.c.

References alg_unquant(), ALLOC, BITRES, CELTMode::bits, CELTMode::bits_stereo, celt_sqrt, CHANNELS, CELTMode::eBands, ec_dec_tell(), intra_fold(), intra_unquant(), MULT16_16_Q15, n, CELTMode::nbEBands, CELTMode::nbShortMdcts, CELTMode::pitchEnd, RESTORE_STACK, SAVE_STACK, SHL16, and VARDECL.

6.6 libcelt/celt.c File Reference

```
#include "os_support.h"
#include "mdct.h"
#include <math.h>
#include "celt.h"
#include "pitch.h"
#include "kiss_fftr.h"
#include "bands.h"
#include "modes.h"
#include "entcode.h"
#include "quant_pitch.h"
#include "quant_bands.h"
#include "psy.h"
#include "rate.h"
#include "stack_alloc.h"
#include "mathops.h"
#include "float_cast.h"
#include <stdarg.h>
```

Data Structures

- struct CELTEncoder

  Encoder state.

- struct CELTDecoder

  Decoder state.

Defines

- #define CELT_C

Functions

- CELTEncoder * celt_encoder_create (const CELTMode *mode)
- void celt_encoder_destroy (CELTEncoder *st)
- int celt_encode_float (CELTEncoder *restrict st, const celt_sig_t *pcm, celt_sig_t *optional_synthesis, unsigned char *compressed, int nbCompressedBytes)
- int celt_encode (CELTEncoder *restrict st, const celt_int16_t *pcm, celt_int16_t *optional_synthesis, unsigned char *compressed, int nbCompressedBytes)
- int celt_encoder_ctl (CELTEncoder *restrict st, int request,...)
- CELTDecoder * celt_decoder_create (const CELTMode *mode)

- void celt_decoder_destroy (CELTDecoder *st)
- int celt_decode_float (CELTDecoder *restrict st, unsigned char *data, int len, celt_sig_t *restrict pcm)
- int celt_decode (CELTDecoder *restrict st, unsigned char *data, int len, celt_int16_t *restrict pcm)

6.6.1 Define Documentation

6.6.1.1 #define CELT_C

Definition at line 36 of file celt.c.

6.6.2 Function Documentation

6.6.2.1 int celt_decode (CELTDecoder *restrict *st*, unsigned char * *data*, int *len*, celt_int16_t *restrict *pcm*)

Definition at line 1076 of file celt.c.

References ALLOC, celt_decode_float(), CHANNELS, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by main().

6.6.2.2 int celt_decode_float (CELTDecoder *restrict *st*, unsigned char * *data*, int *len*, celt_sig_t *restrict *pcm*)

< Interleaved signal MDCTs

< Interleaved normalised MDCTs

< Interleaved normalised pitch MDCTs

Definition at line 883 of file celt.c.

References ALLOC, CELT_CORRUPTED_DATA, CELT_INVALID_MODE, CELT_MOVE, CELT_OK, CHANNELS, check_mode(), compute_allocation(), compute_band_energies(), denormalise_bands(), ec_byte_readinit(), ec_dec_bits(), ec_dec_init(), ec_dec_tell(), ec_dec_uint(), MAC16_32_Q15, MAX_PERIOD, normalise_bands(), pitch_quant_bands(), renormalise_bands(), RESTORE_STACK, SAVE_STACK, SCALEOUT, stereo_decision(), unquant_bands(), unquant_coarse_energy(), unquant_fine_energy(), unquant_pitch(), and VARDECL.

Referenced by celt_decode().

6.6.2.3 int celt_encode (CELTEncoder *restrict *st*, const celt_int16_t * *pcm*, celt_int16_t * *optional_synthesis*, unsigned char * *compressed*, int *nbCompressedBytes*)

Definition at line 698 of file celt.c.

References ALLOC, celt_encode_float(), CHANNELS, RESTORE_STACK, SAVE_STACK, SCALEOUT, and VARDECL.

Referenced by main().

6.6.2.4 int celt_encode_float (CELTEncoder *restrict *st*, const celt_sig_t * *pcm*, celt_sig_t * *optional_synthesis*, unsigned char * *compressed*, int *nbCompressedBytes*)

< Interleaved signal MDCTs

< Interleaved normalised MDCTs

< Interleaved normalised pitch MDCTs

Definition at line 378 of file celt.c.

References ALLOC, CELT_COPY, CELT_INTERNAL_ERROR, CELT_INVALID_MODE, CELT_MOVE, CELT_OK, celt_rcp, CHANNELS, check_mode(), compute_allocation(), compute_band_energies(), compute_mdct_masking(), compute_pitch_gain(), denormalise_bands(), ec_enc_bits(), ec_enc_done(), ec_enc_init(), ec_enc_tell(), ec_enc_uint(), EXTEND32, EXTRACT16, find_spectral_pitch(), MAC16_32_Q15, MAX_PERIOD, MULT16_16, MULT16_32_Q15, normalise_bands(), pitch_quant_bands(), Q15ONE, QCONST16, QCONST32, quant_bands(), quant_coarse_energy(), quant_fine_energy(), quant_pitch(), renormalise_bands(), RESTORE_STACK, SAVE_STACK, SCALEIN, SCALEOUT, SHL32, SHR32, stereo_decision(), SUB32, and VARDECL.

Referenced by celt_encode().

6.6.2.5 int celt_encoder_ctl (CELTEncoder *restrict *st*, int *request*, ...)

Definition at line 722 of file celt.c.

References CELT_BAD_ARG, CELT_OK, CELT_SET_COMPLEXITY_REQUEST, and CELT_UNIMPLEMENTED.

Referenced by main().

6.7 libcelt/celt.h File Reference

Contains all the functions for encoding and decoding audio streams.

```
#include "celt_types.h"
```

Defines

- #define EXPORT
- #define _celt_check_int(x) (((void)((x) == (int)0)), (int)(x))
- #define CELT_OK 0
- #define CELT_BAD_ARG -1
- #define CELT_INVALID_MODE -2
- #define CELT_INTERNAL_ERROR -3
- #define CELT_CORRUPTED_DATA -4
- #define CELT_UNIMPLEMENTED -5
- #define CELT_SET_COMPLEXITY_REQUEST 2
- #define CELT_SET_COMPLEXITY(x) CELT_SET_COMPLEXITY_REQUEST, _celt_check_int(x)
- #define CELT_GET_FRAME_SIZE 1000
- #define CELT_GET_LOOKAHEAD 1001
- #define CELT_GET_NB_CHANNELS 1002
- #define CELT_GET_BITSTREAM_VERSION 2000

Typedefs

- typedef struct CELTEncoder CELTEncoder

  Encoder state.

- typedef struct CELTDecoder CELTDecoder
- typedef struct CELTMode CELTMode

Functions

- EXPORT CELTMode * celt_mode_create (celt_int32_t Fs, int channels, int frame_size, int *error)
- EXPORT void celt_mode_destroy (CELTMode *mode)
- EXPORT int celt_mode_info (const CELTMode *mode, int request, celt_int32_t *value)
- EXPORT CELTEncoder * celt_encoder_create (const CELTMode *mode)
- EXPORT void celt_encoder_destroy (CELTEncoder *st)
- EXPORT int celt_encode_float (CELTEncoder *st, const float *pcm, float *optional_synthesis, unsigned char *compressed, int nbCompressedBytes)
- EXPORT int celt_encode (CELTEncoder *st, const celt_int16_t *pcm, celt_int16_t *optional_synthesis, unsigned char *compressed, int nbCompressedBytes)
- EXPORT int celt_encoder_ctl (CELTEncoder *st, int request,...)
- EXPORT CELTDecoder * celt_decoder_create (const CELTMode *mode)
- EXPORT void celt_decoder_destroy (CELTDecoder *st)
- EXPORT int celt_decode_float (CELTDecoder *st, unsigned char *data, int len, float *pcm)
- EXPORT int celt_decode (CELTDecoder *st, unsigned char *data, int len, celt_int16_t *pcm)

6.7.1 Detailed Description

Contains all the functions for encoding and decoding audio streams.

Definition in file celt.h.

6.7.2 Define Documentation

6.7.2.1 #define _celt_check_int(x) (((void)((x) == (int)0)), (int)(x))

Definition at line 54 of file celt.h.

6.7.2.2 #define CELT_BAD_ARG -1

An (or more) invalid argument (e.g. out of range)

Definition at line 60 of file celt.h.

Referenced by celt_encoder_ctl(), celt_header_to_packet(), and celt_mode_create().

6.7.2.3 #define CELT_CORRUPTED_DATA -4

The data passed (e.g. compressed data to decoder) is corrupted

Definition at line 66 of file celt.h.

Referenced by celt_decode(), and celt_decode_float().

6.7.2.4 #define CELT_GET_BITSTREAM_VERSION 2000

GET the bit-stream version for compatibility check

Definition at line 83 of file celt.h.

Referenced by celt_header_init(), and celt_mode_info().

6.7.2.5 #define CELT_GET_FRAME_SIZE 1000

GET the frame size used in the current mode

Definition at line 76 of file celt.h.

Referenced by celt_mode_info(), and main().

6.7.2.6 #define CELT_GET_LOOKAHEAD 1001

GET the lookahead used in the current mode

Definition at line 78 of file celt.h.

Referenced by celt_mode_info(), and main().

6.7.2.7 #define CELT_GET_NB_CHANNELS 1002

GET the number of channels used in the current mode

Definition at line 80 of file celt.h.

Referenced by celt_mode_info(), and main().

6.7.2.8 #define CELT_INTERNAL_ERROR -3

An internal error was detected

Definition at line 64 of file celt.h.

Referenced by celt_encode(), and celt_encode_float().

6.7.2.9 #define CELT_INVALID_MODE -2

The mode struct passed is invalid

Definition at line 62 of file celt.h.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), and check_mode().

6.7.2.10 #define CELT_OK 0

No error

Definition at line 58 of file celt.h.

Referenced by celt_decode(), celt_decode_float(), celt_decoder_create(), celt_decoder_destroy(), celt_encode(), celt_encode_float(), celt_encoder_create(), celt_encoder_ctl(), celt_encoder_destroy(), celt_mode_create(), celt_mode_destroy(), celt_mode_info(), and check_mode().

6.7.2.11 #define CELT_SET_COMPLEXITY(x) CELT_SET_COMPLEXITY_REQUEST, _celt_check_int(x)

Controls the complexity from 0-10 (int)

Definition at line 73 of file celt.h.

Referenced by main().

6.7.2.12 #define CELT_SET_COMPLEXITY_REQUEST 2

Definition at line 71 of file celt.h.

Referenced by celt_encoder_ctl().

6.7.2.13 #define CELT_UNIMPLEMENTED -5

Invalid/unsupported request number

Definition at line 68 of file celt.h.

Referenced by celt_encoder_ctl(), and celt_mode_info().

6.7.2.14 #define EXPORT

Definition at line 51 of file celt.h.

6.7.3 Typedef Documentation

6.7.3.1 typedef struct CELTDecoder CELTDecoder

State of the decoder. One decoder state is needed for each stream. It is initialised once at the beginning of the stream. Do ∗not∗ re-initialise the state for every frame

Definition at line 96 of file celt.h.

6.7.3.2 typedef struct CELTEncoder CELTEncoder

Encoder state.

Contains the state of an encoder. One encoder state is needed for each stream. It is initialised once at the beginning of the stream. Do ∗not∗ re-initialise the state for every frame.

Definition at line 91 of file celt.h.

6.7.3.3 typedef struct CELTMode CELTMode

The mode contains all the information necessary to create an encoder. Both the encoder and decoder need to be initialised with exactly the same mode, otherwise the quality will be very bad

Definition at line 101 of file celt.h.

6.8 libcelt/celt_header.h File Reference

```
#include "celt.h"
#include "celt_types.h"
```

Data Structures

- struct CELTHeader

  Header data.

Functions

- EXPORT void celt_header_init (CELTHeader *header, const CELTMode *m)
- EXPORT int celt_header_to_packet (const CELTHeader *header, unsigned char *packet, celt_uint32_t size)
- EXPORT int celt_header_from_packet (const unsigned char *packet, celt_uint32_t size, CELTHeader *header)

6.8.1 Function Documentation

6.8.1.1 EXPORT int celt_header_from_packet (const unsigned char * *packet*, celt_uint32_t *size*, CELTHeader * *header*)

Definition at line 106 of file header.c.

References CELT_COPY.

6.8.1.2 EXPORT void celt_header_init (CELTHeader * *header*, const CELTMode * *m*)

Creates a basic header struct

Definition at line 65 of file header.c.

References CELTHeader::bytes_per_packet, CELT_COPY, CELT_GET_BITSTREAM_VERSION, celt_mode_info(), CELTHeader::codec_id, CELTHeader::codec_version, CELTHeader::extra_headers, CELTHeader::frame_size, CELTMode::Fs, CELTHeader::header_size, CELTMode::mdctSize, CELTHeader::nb_channels, CELTMode::nbChannels, CELTMode::overlap, CELTHeader::overlap, CELTHeader::sample_rate, and CELTHeader::version_id.

6.8.1.3 EXPORT int celt_header_to_packet (const CELTHeader * *header*, unsigned char * *packet*, celt_uint32_t *size*)

Definition at line 80 of file header.c.

References CELT_BAD_ARG, CELT_COPY, and CELT_MEMSET.

6.9 libcelt/celt_shaping1.c File Reference

```
#include "os_support.h"
#include "mdct.h"
#include <math.h>
#include "celt.h"
#include "pitch.h"
#include "kiss_fftr.h"
#include "bands.h"
#include "modes.h"
#include "entcode.h"
#include "quant_pitch.h"
#include "quant_bands.h"
#include "psy.h"
#include "rate.h"
#include "stack_alloc.h"
```

Data Structures

- struct CELTEncoder

  Encoder state.

- struct CELTDecoder

  Decoder state.

Defines

- #define CELT_C

Functions

- CELTEncoder * celt_encoder_create (const CELTMode *mode)
- void celt_encoder_destroy (CELTEncoder *st)
- int celt_encode (CELTEncoder *restrict st, celt_int16_t *restrict pcm, unsigned char *compressed, int nbCompressedBytes)
- CELTDecoder * celt_decoder_create (const CELTMode *mode)
- void celt_decoder_destroy (CELTDecoder *st)
- int celt_decode (CELTDecoder *restrict st, unsigned char *data, int len, celt_int16_t *restrict pcm)

6.9.1 Define Documentation

6.9.1.1 #define CELT_C

Definition at line 36 of file celt_shaping1.c.

6.9.2 Function Documentation

6.9.2.1 int celt_decode (CELTDecoder *restrict *st*, unsigned char * *data*, int *len*, celt_int16_t *restrict *pcm*)

< Interleaved signal MDCTs

< Interleaved normalised MDCTs

< Interleaved normalised pitch MDCTs

Definition at line 634 of file celt_shaping1.c.

References ADD32, ALLOC, CELT_CORRUPTED_DATA, CELT_INVALID_MODE, CELT_MOVE, CELT_OK, CHANNELS, check_mode(), compute_band_energies(), denormalise_bands(), ec_byte_readinit(), ec_dec_init(), ec_dec_tell(), ec_dec_uint(), MAX_PERIOD, MULT16_32_Q15, normalise_bands(), pitch_quant_bands(), renormalise_bands(), RESTORE_STACK, SAVE_STACK, stereo_decision(), unquant_bands(), unquant_pitch(), and VARDECL.

6.9.2.2 int celt_encode (CELTEncoder *restrict *st*, celt_int16_t *restrict *pcm*, unsigned char * *compressed*, int *nbCompressedBytes*)

< Interleaved signal MDCTs

< Interleaved normalised MDCTs

< Interleaved normalised pitch MDCTs

Definition at line 228 of file celt_shaping1.c.

References ADD32, ALLOC, CELT_COPY, CELT_INTERNAL_ERROR, CELT_INVALID_MODE, CELT_MOVE, CELT_OK, CHANNELS, check_mode(), compute_band_energies(), compute_mdct_masking(), compute_pitch_gain(), denormalise_bands(), ec_enc_bits(), ec_enc_done(), ec_enc_init(), ec_enc_tell(), ec_enc_uint(), EXTEND32, find_spectral_pitch(), MAX_PERIOD, MULT16_16, MULT16_32_Q15, normalise_bands(), pitch_quant_bands(), QCONST16, QCONST32, quant_bands(), quant_pitch(), renormalise_bands(), RESTORE_STACK, SAVE_STACK, SHL32, SHR32, stereo_decision(), SUB32, and VARDECL.

6.10 libcelt/celt_types.h File Reference

CELT types.

Typedefs

- typedef short celt_int16_t
- typedef unsigned short celt_uint16_t
- typedef int celt_int32_t
- typedef unsigned int celt_uint32_t
- typedef long long celt_int64_t
- typedef unsigned long long celt_uint64_t

6.10.1 Detailed Description

CELT types.

Definition in file celt_types.h.

6.10.2 Typedef Documentation

6.10.2.1 typedef short celt_int16_t

Definition at line 159 of file celt_types.h.

6.10.2.2 typedef int celt_int32_t

Definition at line 161 of file celt_types.h.

6.10.2.3 typedef long long celt_int64_t

Definition at line 163 of file celt_types.h.

6.10.2.4 typedef unsigned short celt_uint16_t

Definition at line 160 of file celt_types.h.

6.10.2.5 typedef unsigned int celt_uint32_t

Definition at line 162 of file celt_types.h.

6.10.2.6 typedef unsigned long long celt_uint64_t

Definition at line 164 of file celt_types.h.

6.11 libcelt/check_ncwrs.m File Reference

Functions

- if (abs(D)< 1e6) y

Variables

- function y

6.11.1 Function Documentation

6.11.1.1 if ()

6.11.2 Variable Documentation

6.11.2.1 else y

Initial value:

```
 check_ncwrs(nc, N, K)
D = nc(N,K)-(nc(N,K-1)+nc(N-1,K)+nc(N-1,K-1))
```

Definition at line 1 of file check_ncwrs.m.

Referenced by alg_quant().

6.12 libcelt/cwrs-pc.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
```

Functions

- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_x, int *_s, celt_uint32_t *_u)
- void cwrsi64 (int _n, int _m, celt_uint64_t _i, int *_x, int *_s, celt_uint64_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, const int *_x, const int *_s, celt_uint32_t *_u)
- celt_uint64_t icwrs64 (int _n, int _m, const int *_x, const int *_s, celt_uint64_t *_u)
- void comb2pulse (int _n, int _m, int *restrict _y, const int *_x, const int *_s)
- void pulse2comb (int _n, int _m, int *_x, int *_s, const int *_y)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.12.1 Function Documentation

6.12.1.1 void comb2pulse (int _n, int _m, int *restrict _y, const int * _x, const int * _s)

Definition at line 274 of file cwrs-pc.c.

References CELT_MEMSET, and k.

Referenced by decode_pulses().

6.12.1.2 void cwrsi32 (int _n, int _m, celt_uint32_t _i, int * _x, int * _s, celt_uint32_t * _u)

Definition at line 170 of file cwrs-pc.c.

References k, and p.

6.12.1.3 void cwrsi64 (int _n, int _m, celt_uint64_t _i, int * _x, int * _s, celt_uint64_t * _u)

Definition at line 196 of file cwrs-pc.c.

References k, and p.

6.12.1.4 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 368 of file cwrs-pc.c.

Referenced by alg_unquant(), and decode_pulses().

6.12.1.5 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 332 of file cwrs-pc.c.

Referenced by alg_quant(), and encode_pulses().

6.12.1.6 celt_uint32_t icwrs32 (int _n, int _m, const int * _x, const int * _s, celt_uint32_t * _u)

Definition at line 226 of file cwrs-pc.c.

References k, and p.

6.12.1.7 celt_uint64_t icwrs64 (int _n, int _m, const int * _x, const int * _s, celt_uint64_t * _u)

Definition at line 248 of file cwrs-pc.c.

References k, and p.

6.12.1.8 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 146 of file cwrs-pc.c.

Referenced by get_required_bits().

6.12.1.9 celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t * _u)

Definition at line 155 of file cwrs-pc.c.

Referenced by get_required_bits().

6.12.1.10 celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t * _ui)

Definition at line 109 of file cwrs-pc.c.

References celt_assert, and ret.

Referenced by ncwrs_u32().

6.12.1.11 celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t * _ui)

Definition at line 126 of file cwrs-pc.c.

References celt_assert, and ret.

Referenced by ncwrs_u64().

6.12.1.12 void pulse2comb (int _n, int _m, int * _x, int * _s, const int * _y)

Definition at line 288 of file cwrs-pc.c.

References k, and n.

Referenced by encode_pulses().

6.13 libcelt/cwrs.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
#include "arch.h"
```

Defines

- #define MASK32 (0xFFFFFFFF)

Functions

- int log2_frac (ec_uint32 val, int frac)
- int fits_in32 (int _n, int _m)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_y, celt_uint32_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t *_nc, const int *_y, celt_uint32_t *_u)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.13.1 Define Documentation

6.13.1.1 #define MASK32 (0xFFFFFFFF)

Definition at line 95 of file cwrs.c.

6.13.2 Function Documentation

6.13.2.1 void cwrsi32 (int *_n*, int *_m*, celt_uint32_t *_i*, int * *_y*, celt_uint32_t * *_u*)

Definition at line 236 of file cwrs.c.

References celt_assert, k, p, and yj.

6.13.2.2 void decode_pulses (int * *_y*, int *N*, int *K*, ec_dec * *dec*)

Definition at line 350 of file cwrs.c.

References decode_pulses(), ec_dec_bits(), ec_dec_uint(), and fits_in32().

6.13.2.3 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 320 of file cwrs.c.

References ec_enc_bits(), ec_enc_uint(), encode_pulses(), and fits_in32().

6.13.2.4 int fits_in32 (int _n, int _m)

Definition at line 76 of file cwrs.c.

Referenced by decode_pulses(), encode_pulses(), and get_required_bits().

6.13.2.5 int get_required_bits (int N, int K, int frac)

Definition at line 301 of file cwrs.c.

Referenced by compute_alloc_cache(), and get_required_bits().

6.13.2.6 celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t * _nc, const int * _y, celt_uint32_t * _u)

Definition at line 265 of file cwrs.c.

6.13.2.7 int log2_frac (ec_uint32 val, int frac)

Definition at line 52 of file cwrs.c.

References EC_ILOG.

Referenced by get_required_bits().

6.13.2.8 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 197 of file cwrs.c.

References celt_assert, k, and um2.

6.14 libcelt/cwrs.c.BACKUP.9121.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
#include "arch.h"
```

Defines

- #define MASK32 (0xFFFFFFFF)

Functions

- int log2_frac (ec_uint32 val, int frac)
- int fits_in32 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- if (_m<=6)
- while (++k< _n)

Variables

- <<<<<<< HEAD:libcelt/cwrs.c=======celt_uint64_tncwrs_unext64(int_n, celt_uint64_t *_ui){celt_uint64_tret;celt_uint64_tui0;celt_uint64_tui1;intj;ret=ui0=1;celt_assert(_n >=2);j=1;do{ui1=_ui celt_assert [j]+_ui[j-1]+ui0;_ui[j-1]=ui0;ui0=ui1;ret+=ui0;}while(++j< _n);_ui[j-1]=ui0;returnret<<=1;}>>>>>>>ChangeCWRSindexingtousePyramidVQ'smagnitudeordering.:libcelt/cwrs.celt_uint32_tncwrs_u32(int_n, int_m, celt_uint32_t *_u){celt_uint32_tret;celt_uint32_tum2;intk;celt_assert(_m >(_n >=2)
- um2 = _u[0]=1
- k = 1
- do ret = _u[k]
- yj = k
- _y [j] = yj-(yj<<1&-s)
- <<<<<<< HEAD:libcelt/cwrs.c=======voidcwrsi64(int_n, int_m, celt_uint64_t_i, celt_uint64_tnc, int *_y, celt_uint64_t *_u){celt_uint64_tp;celt_uint64_tq;intj;intk;celt_assert(_n >0);p=_nc;q=0;j=0;k=_m;do{ints;intyj;p-=q;q=_u ordering [_n-j-1];p-=q;s=_i >=p;if(s) _i-=p;yj=k;while(q >_i){uprev64(_u, _n-j,–k >0);p=q;q=_u[_n-j-1];}_i-=q;yj-=k;_y[j]=yj-(yj<< 1 &-s);}while(++j< _n);}>>>>>> Change CWRS indexing to use Pyramid VQ s magnitude
- <<<<<<< HEAD:libcelt/cwrs.c=======voidcwrsi64(int_n, int_m, celt_uint64_t_i, celt_uint64_tnc, int *_y, celt_uint64_t *_u){celt_uint64_tp;celt_uint64_tq;intj;intk;celt_assert(_n >0);p=_nc;q=0;j=0;k=_m;do{ints;intyj;p-=q;q=_u _m [_n-j-1];p-=q;s=_i >=p;if(s) _i-=p;yj=k;while(q >_i){uprev64(_u, _n-j,–k >0);p=q;q=_u[_n-j-1];}_i-=q;yj-=k;_y[j]=yj-(yj<< 1 &-s);}while(++j< _n);}>>>>>> Change CWRS indexing to use Pyramid VQ s magnitude int

- <<<<<<< HEAD:libcelt/cwrs.c=======voidcwrsi64(int_n, int_m, celt_uint64_t_i, celt_-uint64_t_nc, int *_y, celt_uint64_t *_u){celt_uint64_tp;celt_uint64_tq;intj;intk;celt_assert(_-n >0);p=_nc;q=0;j=0;k=_m;do{ints;intyj;p-=q;q=_u _nc [_n-j-1];p-=q;s=_i >=p;if(s) _i-=p;yj=k;while(q >_i){uprev64(_u, _n-j,–k >0);p=q;q=_u[_n-j-1];}_i-=q;yj-=k;_y[j]=yj-(yj<< 1 &-s);}while(++j< _n);}>>>>>> Change CWRS indexing to use Pyramid VQ s magnitude int celt_uint32_t *

6.14.1 Define Documentation

6.14.1.1 #define MASK32 (0xFFFFFFFF)

Definition at line 95 of file cwrs.c.BACKUP.9121.c.

6.14.2 Function Documentation

6.14.2.1 int fits_in32 (int *_n*, int *_m*)

Definition at line 76 of file cwrs.c.BACKUP.9121.c.

6.14.2.2 if (_m<= *6*)

Definition at line 247 of file cwrs.c.BACKUP.9121.c.

References _m, and k.

6.14.2.3 int log2_frac (ec_uint32 *val*, int *frac*)

Definition at line 52 of file cwrs.c.BACKUP.9121.c.

References EC_ILOG.

6.14.2.4 celt_uint32_t ncwrs_unext32 (int *_n*, celt_uint32_t * *_ui*)

Definition at line 198 of file cwrs.c.BACKUP.9121.c.

References celt_assert, and ret.

6.14.2.5 while ()

6.14.3 Variable Documentation

6.14.3.1 <<<<<<<HEAD:libcelt/cwrs.c=======voidcwrsi64(int_n,int_m,celt_uint64_-t_i,celt_uint64_t_nc,int*_y,celt_uint64_t*_u){celt_uint64_tp;celt_uint64_-tq;intj;intk;celt_assert(_n>0);p=_nc;q=0;j=0;k=_m;do{ints;intyj;p-=q;q=_u _m[_n-j-1];p-=q;s=_i>=p;if(s)_i-=p;yj=k;while(q>_i){uprev64(_u,_n-j,–k>0);p=q;q=_-u[_n-j-1];}_i-=q;yj-=k;_y[j]=yj-(yj<<1&-s);}while(++j<_n);}>>>>>> Change CWRS indexing to use Pyramid VQ s magnitude int

Definition at line 372 of file cwrs.c.BACKUP.9121.c.

Referenced by if().

6.14.3.2 <<<<<<<HEAD:libcelt/cwrs.c=======voidcwrsi64(int_n,int_m,celt_uint64_-t_i,celt_uint64_t_nc,int*_y,celt_uint64_t*_u){celt_uint64_tp;celt_uint64_-tq;intj;intk;celt_assert(_n>0);p=_nc;q=0;j=0;k=_m;do{ints;intyj;p-=q;q=_u_nc[_n-j-1];p-=q;s=_i>=p;if(s)_i-=p;yj=k;while(q>_i){uprev64(_u,_n-j,–k>0);p=q;q=_-u[_n-j-1];}_i-=q;yj-=k;_y[j]=yj-(yj<<1&-s);}while(++j<_n);}>>>>>> Change CWRS indexing to use Pyramid VQ s magnitude int celt_uint32_t*

Definition at line 372 of file cwrs.c.BACKUP.9121.c.

6.14.3.3 <<<<<<<HEAD:libcelt/cwrs.c=======voidcwrsi64(int_n,int_m,celt_uint64_-t_i,celt_uint64_t_nc,int*_y,celt_uint64_t*_u){celt_uint64_tp;celt_uint64_-tq;intj;intk;celt_assert(_n>0);p=_nc;q=0;j=0;k=_m;do{ints;intyj;p-=q;q=_u_y[_n-j-1];p-=q;s=_i>=p;if(s)_i-=p;yj=k;while(q>_i){uprev64(_u,_n-j,–k>0);p=q;q=_-u[_n-j-1];}_i-=q;yj-=k;_y[j]=yj-(yj<<1&-s);}while(++j<_n);}>>>>>> Change CWRS indexing to use Pyramid VQ s magnitude int celt_uint32_t const int* () = yj-(yj<<1&-s)

Definition at line 323 of file cwrs.c.BACKUP.9121.c.

6.14.3.4 <<<<<<<HEAD:libcelt/cwrs.c=======celt_uint64_tncwrs_-unext64(int_n,celt_uint64_t*_ui){celt_uint64_tret;celt_uint64_-tui0;celt_uint64_tui1;intj;ret=ui0=1;celt_assert(_n>=2);j=1;do{ui1=_uicelt_assert[j]+_ui[j-1]+ui0;_ui[j-1]=ui0;ui0=ui1;ret+=ui0;}while(++j<_n);_ui[j-1]=ui0;returnret<<=1;}>>>>>>>ChangeCWRSindexingtousePyramidVQ'smagnitudeordering.:libcelt/cvuint32_tncwrs_u32(int_n,int_m,celt_uint32_t*_u){celt_uint32_tret;celt_uint32_-tum2;intk;celt_assert(_m>(_n >=2)

6.14.3.5 k = 1

Definition at line 272 of file cwrs.c.BACKUP.9121.c.

Referenced by comb2pulse(), cwrsi32(), cwrsi64(), dump_modes(), exp_rotation(), icwrs32(), icwrs64(), if(), kiss_fftr_twiddles(), kiss_fftri(), ncwrs_u32(), ncwrs_u64(), and pulse2comb().

6.14.3.6 <<<<<<<HEAD:libcelt/cwrs.c=======voidcwrsi64(int_n,int_m,celt_uint64_-t_i,celt_uint64_t_nc,int*_y,celt_uint64_t*_u){celt_uint64_tp;celt_uint64_-tq;intj;intk;celt_assert(_n>0);p=_nc;q=0;j=0;k=_m;do{ints;intyj;p-=q;q=_uordering[_n-j-1];p-=q;s=_i>=p;if(s)_i-=p;yj=k;while(q>_i){uprev64(_u,_n-j,–k>0);p=q;q=_u[_n-j-1];}_i-=q;yj-=k;_y[j]=yj-(yj<<1&-s);}while(++j<_n);}>>>>>> Change CWRS indexing to use Pyramid VQ s magnitude

Definition at line 372 of file cwrs.c.BACKUP.9121.c.

6.14.3.7 do ret = _u[k]

Definition at line 273 of file cwrs.c.BACKUP.9121.c.

Referenced by ec_ilog(), ncwrs_u32(), ncwrs_unext32(), and ncwrs_unext64().

6.14.3.8 um2 = _u[0]=1

Definition at line 246 of file cwrs.c.BACKUP.9121.c.

Referenced by ncwrs_u32().

6.14.3.9 yj = k

Definition at line 322 of file cwrs.c.BACKUP.9121.c.

Referenced by cwrsi32(), and cwrsi64().

6.15 libcelt/cwrs.c.BASE.9121.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
```

Functions

- int log2_frac64 (ec_uint64 val, int frac)
- int fits_in32 (int _n, int _m)
- int fits_in64 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_x, int *_s, celt_uint32_t *_u)
- void cwrsi64 (int _n, int _m, celt_uint64_t _i, int *_x, int *_s, celt_uint64_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, const int *_x, const int *_s, celt_uint32_t *_u)
- celt_uint64_t icwrs64 (int _n, int _m, const int *_x, const int *_s, celt_uint64_t *_u)
- void comb2pulse (int _n, int _m, int *restrict _y, const int *_x, const int *_s)
- void pulse2comb (int _n, int _m, int *_x, int *_s, const int *_y)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.15.1 Function Documentation

6.15.1.1 void comb2pulse (int _n, int _m, int *restrict _y, const int * _x, const int * _s)

Definition at line 362 of file cwrs.c.BASE.9121.c.

References CELT_MEMSET, and k.

6.15.1.2 void cwrsi32 (int _n, int _m, celt_uint32_t _i, int * _x, int * _s, celt_uint32_t * _u)

Definition at line 261 of file cwrs.c.BASE.9121.c.

References k, and p.

6.15.1.3 void cwrsi64 (int _n, int _m, celt_uint64_t _i, int * _x, int * _s, celt_uint64_t * _u)

Definition at line 285 of file cwrs.c.BASE.9121.c.

References k, and p.

6.15.1.4 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 488 of file cwrs.c.BASE.9121.c.

References ALLOC, comb2pulse(), decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), fits_in64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.15.1.5 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 439 of file cwrs.c.BASE.9121.c.

References ALLOC, ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), fits_in64(), pulse2comb(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.15.1.6 int fits_in32 (int _n, int _m)

Definition at line 101 of file cwrs.c.BASE.9121.c.

6.15.1.7 int fits_in64 (int _n, int _m)

Definition at line 119 of file cwrs.c.BASE.9121.c.

Referenced by decode_pulses(), encode_pulses(), and get_required_bits().

6.15.1.8 int get_required_bits (int N, int K, int frac)

Definition at line 420 of file cwrs.c.BASE.9121.c.

References ALLOC, fits_in64(), get_required_bits(), log2_frac64(), ncwrs_u64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.15.1.9 celt_uint32_t icwrs32 (int _n, int _m, const int * _x, const int * _s, celt_uint32_t * _u)

Definition at line 315 of file cwrs.c.BASE.9121.c.

References k, and p.

6.15.1.10 celt_uint64_t icwrs64 (int _n, int _m, const int * _x, const int * _s, celt_uint64_t * _u)

Definition at line 336 of file cwrs.c.BASE.9121.c.

References k, and p.

6.15.1.11 int log2_frac64 (ec_uint64 val, int frac)

Definition at line 77 of file cwrs.c.BASE.9121.c.

Referenced by get_required_bits().

6.15.1.12 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 237 of file cwrs.c.BASE.9121.c.

References CELT_MEMSET, k, and ncwrs_unext32().

6.15.1.13 celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t * _u)

Definition at line 246 of file cwrs.c.BASE.9121.c.

References CELT_MEMSET, k, and ncwrs_unext64().

6.15.1.14 celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t * _ui)

Definition at line 200 of file cwrs.c.BASE.9121.c.

References celt_assert, and ret.

6.15.1.15 celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t * _ui)

Definition at line 217 of file cwrs.c.BASE.9121.c.

References celt_assert, and ret.

6.15.1.16 void pulse2comb (int _n, int _m, int * _x, int * _s, const int * _y)

Definition at line 376 of file cwrs.c.BASE.9121.c.

References k, and n.

6.16 libcelt/cwrs.c.BASE.9303.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
```

Functions

- int log2_frac64 (ec_uint64 val, int frac)
- int fits_in32 (int _n, int _m)
- int fits_in64 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_x, int *_s, celt_uint32_t *_u)
- void cwrsi64 (int _n, int _m, celt_uint64_t _i, int *_x, int *_s, celt_uint64_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, const int *_x, const int *_s, celt_uint32_t *_u)
- celt_uint64_t icwrs64 (int _n, int _m, const int *_x, const int *_s, celt_uint64_t *_u)
- void comb2pulse (int _n, int _m, int *restrict _y, const int *_x, const int *_s)
- void pulse2comb (int _n, int _m, int *_x, int *_s, const int *_y)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.16.1 Function Documentation

6.16.1.1 void comb2pulse (int _n, int _m, int *restrict _y, const int * _x, const int * _s)

Definition at line 362 of file cwrs.c.BASE.9303.c.

References CELT_MEMSET, and k.

6.16.1.2 void cwrsi32 (int _n, int _m, celt_uint32_t _i, int * _x, int * _s, celt_uint32_t * _u)

Definition at line 261 of file cwrs.c.BASE.9303.c.

References k, and p.

6.16.1.3 void cwrsi64 (int _n, int _m, celt_uint64_t _i, int * _x, int * _s, celt_uint64_t * _u)

Definition at line 285 of file cwrs.c.BASE.9303.c.

References k, and p.

6.16.1.4 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 488 of file cwrs.c.BASE.9303.c.

References ALLOC, comb2pulse(), decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), fits_in64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.16.1.5 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 439 of file cwrs.c.BASE.9303.c.

References ALLOC, ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), fits_in64(), pulse2comb(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.16.1.6 int fits_in32 (int _n, int _m)

Definition at line 101 of file cwrs.c.BASE.9303.c.

6.16.1.7 int fits_in64 (int _n, int _m)

Definition at line 119 of file cwrs.c.BASE.9303.c.

6.16.1.8 int get_required_bits (int N, int K, int frac)

Definition at line 420 of file cwrs.c.BASE.9303.c.

References ALLOC, fits_in64(), get_required_bits(), log2_frac64(), ncwrs_u64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.16.1.9 celt_uint32_t icwrs32 (int _n, int _m, const int * _x, const int * _s, celt_uint32_t * _u)

Definition at line 315 of file cwrs.c.BASE.9303.c.

References k, and p.

6.16.1.10 celt_uint64_t icwrs64 (int _n, int _m, const int * _x, const int * _s, celt_uint64_t * _u)

Definition at line 336 of file cwrs.c.BASE.9303.c.

References k, and p.

6.16.1.11 int log2_frac64 (ec_uint64 val, int frac)

Definition at line 77 of file cwrs.c.BASE.9303.c.

References EC_ILOG64.

6.16.1.12 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 237 of file cwrs.c.BASE.9303.c.

References CELT_MEMSET, k, and ncwrs_unext32().

6.16.1.13 celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t * _u)

Definition at line 246 of file cwrs.c.BASE.9303.c.

References CELT_MEMSET, k, and ncwrs_unext64().

6.16.1.14 celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t * _ui)

Definition at line 200 of file cwrs.c.BASE.9303.c.

References celt_assert, and ret.

6.16.1.15 celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t * _ui)

Definition at line 217 of file cwrs.c.BASE.9303.c.

References celt_assert, and ret.

6.16.1.16 void pulse2comb (int _n, int _m, int * _x, int * _s, const int * _y)

Definition at line 376 of file cwrs.c.BASE.9303.c.

References k, and n.

6.17 libcelt/cwrs.c.LOCAL.9121.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
#include "arch.h"
```

Functions

- int log2_frac (ec_uint32 val, int frac)
- int fits_in32 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_x, int *_s, celt_uint32_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, const int *_x, const int *_s, celt_uint32_t *_u)
- void comb2pulse (int _n, int _m, int *restrict _y, const int *_x, const int *_s)
- void pulse2comb (int _n, int _m, int *_x, int *_s, const int *_y)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.17.1 Function Documentation

6.17.1.1 void comb2pulse (int _n, int _m, int *restrict _y, const int * _x, const int * _s)

Definition at line 220 of file cwrs.c.LOCAL.9121.c.

References CELT_MEMSET, and k.

6.17.1.2 void cwrsi32 (int _n, int _m, celt_uint32_t _i, int * _x, int * _s, celt_uint32_t * _u)

Definition at line 164 of file cwrs.c.LOCAL.9121.c.

References k, and p.

6.17.1.3 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 323 of file cwrs.c.LOCAL.9121.c.

References ALLOC, comb2pulse(), decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.17.1.4 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 284 of file cwrs.c.LOCAL.9121.c.

References ALLOC, ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), pulse2comb(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.17.1.5 int fits_in32 (int *_n*, int *_m*)

Definition at line 76 of file cwrs.c.LOCAL.9121.c.

6.17.1.6 int get_required_bits (int *N*, int *K*, int *frac*)

Definition at line 265 of file cwrs.c.LOCAL.9121.c.

References ALLOC, fits_in32(), get_required_bits(), log2_frac(), ncwrs_u32(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.17.1.7 celt_uint32_t icwrs32 (int *_n*, int *_m*, const int * *_x*, const int * *_s*, celt_uint32_t * *_u*)

Definition at line 194 of file cwrs.c.LOCAL.9121.c.

References k, and p.

6.17.1.8 int log2_frac (ec_uint32 *val*, int *frac*)

Definition at line 52 of file cwrs.c.LOCAL.9121.c.

References EC_ILOG.

6.17.1.9 celt_uint32_t ncwrs_u32 (int *_n*, int *_m*, celt_uint32_t * *_u*)

Definition at line 149 of file cwrs.c.LOCAL.9121.c.

References CELT_MEMSET, k, and ncwrs_unext32().

6.17.1.10 celt_uint32_t ncwrs_unext32 (int *_n*, celt_uint32_t * *_ui*)

Definition at line 129 of file cwrs.c.LOCAL.9121.c.

References celt_assert, and ret.

6.17.1.11 void pulse2comb (int *_n*, int *_m*, int * *_x*, int * *_s*, const int * *_y*)

Definition at line 234 of file cwrs.c.LOCAL.9121.c.

References k, and n.

6.18 libcelt/cwrs.c.LOCAL.9303.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
#include "arch.h"
```

Functions

- int log2_frac (ec_uint32 val, int frac)
- int fits_in32 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_x, int *_s, celt_uint32_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, const int *_x, const int *_s, celt_uint32_t *_u)
- void comb2pulse (int _n, int _m, int *restrict _y, const int *_x, const int *_s)
- void pulse2comb (int _n, int _m, int *_x, int *_s, const int *_y)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.18.1 Function Documentation

6.18.1.1 void comb2pulse (int _n, int _m, int *restrict _y, const int * _x, const int * _s)

Definition at line 220 of file cwrs.c.LOCAL.9303.c.

References CELT_MEMSET, and k.

6.18.1.2 void cwrsi32 (int _n, int _m, celt_uint32_t _i, int * _x, int * _s, celt_uint32_t * _u)

Definition at line 164 of file cwrs.c.LOCAL.9303.c.

References k, and p.

6.18.1.3 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 323 of file cwrs.c.LOCAL.9303.c.

References ALLOC, comb2pulse(), decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.18.1.4 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 284 of file cwrs.c.LOCAL.9303.c.

References ALLOC, ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), pulse2comb(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.18.1.5 int fits_in32 (int *_n*, int *_m*)

Definition at line 76 of file cwrs.c.LOCAL.9303.c.

6.18.1.6 int get_required_bits (int *N*, int *K*, int *frac*)

Definition at line 265 of file cwrs.c.LOCAL.9303.c.

References ALLOC, fits_in32(), get_required_bits(), log2_frac(), ncwrs_u32(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.18.1.7 celt_uint32_t icwrs32 (int *_n*, int *_m*, const int * *_x*, const int * *_s*, celt_uint32_t * *_u*)

Definition at line 194 of file cwrs.c.LOCAL.9303.c.

References k, and p.

6.18.1.8 int log2_frac (ec_uint32 *val*, int *frac*)

Definition at line 52 of file cwrs.c.LOCAL.9303.c.

References EC_ILOG.

6.18.1.9 celt_uint32_t ncwrs_u32 (int *_n*, int *_m*, celt_uint32_t * *_u*)

Definition at line 149 of file cwrs.c.LOCAL.9303.c.

References CELT_MEMSET, k, and ncwrs_unext32().

6.18.1.10 celt_uint32_t ncwrs_unext32 (int *_n*, celt_uint32_t * *_ui*)

Definition at line 129 of file cwrs.c.LOCAL.9303.c.

References celt_assert, and ret.

6.18.1.11 void pulse2comb (int *_n*, int *_m*, int * *_x*, int * *_s*, const int * *_y*)

Definition at line 234 of file cwrs.c.LOCAL.9303.c.

References k, and n.

6.19 libcelt/cwrs.c.REMOTE.9121.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
```

Defines

- #define MASK32 (0xFFFFFFFF)

Functions

- int log2_frac64 (ec_uint64 val, int frac)
- int fits_in32 (int _n, int _m)
- int fits_in64 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, celt_uint32_t _nc, int *_y, celt_uint32_t *_u)
- void cwrsi64 (int _n, int _m, celt_uint64_t _i, celt_uint64_t _nc, int *_y, celt_uint64_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t *_nc, const int *_y, celt_uint32_t *_u)
- celt_uint64_t icwrs64 (int _n, int _m, celt_uint64_t *_nc, const int *_y, celt_uint64_t *_u)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.19.1 Define Documentation

6.19.1.1 #define MASK32 (0xFFFFFFFF)

Definition at line 142 of file cwrs.c.REMOTE.9121.c.

6.19.2 Function Documentation

6.19.2.1 void cwrsi32 (int _n, int _m, celt_uint32_t _i, celt_uint32_t _nc, int * _y, celt_uint32_t * _u)

Definition at line 361 of file cwrs.c.REMOTE.9121.c.

References celt_assert, k, p, and yj.

6.19.2.2 void cwrsi64 (int _n, int _m, celt_uint64_t _i, celt_uint64_t _nc, int * _y, celt_uint64_t * _u)

Definition at line 398 of file cwrs.c.REMOTE.9121.c.

References celt_assert, k, p, and yj.

6.19.2.3 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 585 of file cwrs.c.REMOTE.9121.c.

References decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), and fits_in64().

6.19.2.4 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 541 of file cwrs.c.REMOTE.9121.c.

References ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), and fits_in64().

6.19.2.5 int fits_in32 (int _n, int _m)

Definition at line 101 of file cwrs.c.REMOTE.9121.c.

6.19.2.6 int fits_in64 (int _n, int _m)

Definition at line 119 of file cwrs.c.REMOTE.9121.c.

6.19.2.7 int get_required_bits (int N, int K, int frac)

Definition at line 522 of file cwrs.c.REMOTE.9121.c.

References ALLOC, fits_in64(), get_required_bits(), log2_frac64(), ncwrs_u64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.19.2.8 celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t * _nc, const int * _y, celt_uint32_t * _u)

Definition at line 434 of file cwrs.c.REMOTE.9121.c.

References celt_assert, and k.

6.19.2.9 celt_uint64_t icwrs64 (int _n, int _m, celt_uint64_t * _nc, const int * _y, celt_uint64_t * _u)

Definition at line 469 of file cwrs.c.REMOTE.9121.c.

6.19.2.10 int log2_frac64 (ec_uint64 val, int frac)

Definition at line 77 of file cwrs.c.REMOTE.9121.c.

References EC_ILOG64.

6.19.2.11 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 306 of file cwrs.c.REMOTE.9121.c.

References celt_assert, k, ret, and um2.

6.19.2.12 celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t * _u)

Definition at line 346 of file cwrs.c.REMOTE.9121.c.

References CELT_MEMSET, k, and ncwrs_unext64().

6.19.2.13 celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t * _ui)

Definition at line 269 of file cwrs.c.REMOTE.9121.c.

References celt_assert, and ret.

6.19.2.14 celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t * _ui)

Definition at line 286 of file cwrs.c.REMOTE.9121.c.

References celt_assert, and ret.

6.20 libcelt/cwrs.c.REMOTE.9303.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
```

Defines

- #define MASK32 (0xFFFFFFFF)

Functions

- int log2_frac64 (ec_uint64 val, int frac)
- int fits_in32 (int _n, int _m)
- int fits_in64 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, celt_uint32_t _nc, int *_y, celt_uint32_t *_u)
- void cwrsi64 (int _n, int _m, celt_uint64_t _i, celt_uint64_t _nc, int *_y, celt_uint64_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t *_nc, const int *_y, celt_uint32_t *_u)
- celt_uint64_t icwrs64 (int _n, int _m, celt_uint64_t *_nc, const int *_y, celt_uint64_t *_u)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.20.1 Define Documentation

6.20.1.1 #define MASK32 (0xFFFFFFFF)

Definition at line 142 of file cwrs.c.REMOTE.9303.c.

6.20.2 Function Documentation

6.20.2.1 void cwrsi32 (int *_n*, int *_m*, celt_uint32_t *_i*, celt_uint32_t *_nc*, int * *_y*, celt_uint32_t * *_u*)

Definition at line 361 of file cwrs.c.REMOTE.9303.c.

References celt_assert, k, p, and yj.

6.20.2.2 void cwrsi64 (int _n, int _m, celt_uint64_t _i, celt_uint64_t _nc, int * _y, celt_uint64_t * _u)

Definition at line 398 of file cwrs.c.REMOTE.9303.c.

References celt_assert, k, p, and yj.

6.20.2.3 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 585 of file cwrs.c.REMOTE.9303.c.

References decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), and fits_in64().

6.20.2.4 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 541 of file cwrs.c.REMOTE.9303.c.

References ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), and fits_in64().

6.20.2.5 int fits_in32 (int _n, int _m)

Definition at line 101 of file cwrs.c.REMOTE.9303.c.

6.20.2.6 int fits_in64 (int _n, int _m)

Definition at line 119 of file cwrs.c.REMOTE.9303.c.

6.20.2.7 int get_required_bits (int N, int K, int frac)

Definition at line 522 of file cwrs.c.REMOTE.9303.c.

References ALLOC, fits_in64(), get_required_bits(), log2_frac64(), ncwrs_u64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.20.2.8 celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t * _nc, const int * _y, celt_uint32_t * _u)

Definition at line 434 of file cwrs.c.REMOTE.9303.c.

References celt_assert, and k.

6.20.2.9 celt_uint64_t icwrs64 (int _n, int _m, celt_uint64_t * _nc, const int * _y, celt_uint64_t * _u)

Definition at line 469 of file cwrs.c.REMOTE.9303.c.

References celt_assert, and k.

6.20.2.10 int log2_frac64 (ec_uint64 val, int frac)

Definition at line 77 of file cwrs.c.REMOTE.9303.c.

References EC_ILOG64.

6.20.2.11 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 306 of file cwrs.c.REMOTE.9303.c.

References celt_assert, k, ret, and um2.

6.20.2.12 celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t * _u)

Definition at line 346 of file cwrs.c.REMOTE.9303.c.

References CELT_MEMSET, k, and ncwrs_unext64().

6.20.2.13 celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t * _ui)

Definition at line 269 of file cwrs.c.REMOTE.9303.c.

References celt_assert, and ret.

6.20.2.14 celt_uint64_t ncwrs_unext64 (int _n, celt_uint64_t * _ui)

Definition at line 286 of file cwrs.c.REMOTE.9303.c.

References celt_assert, and ret.

6.21 libcelt/cwrs.h File Reference

```
#include "arch.h"
#include "stack_alloc.h"
#include "entenc.h"
#include "entdec.h"
```

Functions

- int log2_frac64 (ec_uint64 val, int frac)
- int fits_in32 (int _n, int _m)
- int fits_in64 (int _n, int _m)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_y, celt_uint32_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t *_nc, const int *_y, celt_uint32_t *_u)
- celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t *_u)
- void cwrsi64 (int _n, int _m, celt_uint64_t _i, int *_y, celt_uint64_t *_u)
- celt_uint64_t icwrs64 (int _n, int _m, celt_uint64_t *_nc, const int *_y, celt_uint64_t *_u)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.21.1 Function Documentation

6.21.1.1 void cwrsi32 (int _n, int _m, celt_uint32_t _i, int * _y, celt_uint32_t * _u)

Definition at line 236 of file cwrs.c.

References celt_assert, k, p, and yj.

6.21.1.2 void cwrsi64 (int _n, int _m, celt_uint64_t _i, int * _y, celt_uint64_t * _u)

6.21.1.3 void decode_pulses (int * _y, int N, int K, ec_dec * dec)

Definition at line 368 of file cwrs-pc.c.

References ALLOC, celt_ilog2, comb2pulse(), decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), fits_in64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by alg_unquant(), and decode_pulses().

6.21.1.4 void encode_pulses (int * _y, int N, int K, ec_enc * enc)

Definition at line 332 of file cwrs-pc.c.

References ALLOC, celt_ilog2, ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), fits_in64(), pulse2comb(), RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by alg_quant(), and encode_pulses().

6.21.1.5 int fits_in32 (int _n, int _m)

Definition at line 76 of file cwrs.c.

Referenced by decode_pulses(), encode_pulses(), and get_required_bits().

6.21.1.6 int fits_in64 (int _n, int _m)

Definition at line 119 of file cwrs.c.BASE.9121.c.

Referenced by decode_pulses(), encode_pulses(), and get_required_bits().

6.21.1.7 int get_required_bits (int *N*, int *K*, int *frac*)

Definition at line 301 of file cwrs.c.

References ALLOC, fits_in32(), fits_in64(), get_required_bits(), log2_frac(), log2_frac64(), ncwrs_u32(), ncwrs_u64(), RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by compute_alloc_cache(), and get_required_bits().

6.21.1.8 celt_uint32_t icwrs32 (int _n, int _m, celt_uint32_t * _nc, const int * _y, celt_uint32_t * _u)

Definition at line 265 of file cwrs.c.

References celt_assert, and k.

6.21.1.9 celt_uint64_t icwrs64 (int _n, int _m, celt_uint64_t * _nc, const int * _y, celt_uint64_t * _u)

Definition at line 469 of file cwrs.c.REMOTE.9121.c.

References celt_assert, and k.

6.21.1.10 int log2_frac64 (ec_uint64 *val*, int *frac*)

Definition at line 77 of file cwrs.c.BASE.9121.c.

References EC_ILOG64.

Referenced by get_required_bits().

6.21.1.11 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 146 of file cwrs-pc.c.

References celt_assert, CELT_MEMSET, k, ncwrs_unext32(), ret, and um2.

Referenced by get_required_bits().

6.21.1.12 celt_uint64_t ncwrs_u64 (int _n, int _m, celt_uint64_t * _u)

Definition at line 155 of file cwrs-pc.c.

References CELT_MEMSET, k, and ncwrs_unext64().

Referenced by get_required_bits().

6.22 libcelt/cwrs32.c File Reference

```
#include "os_support.h"
#include <stdlib.h>
#include <string.h>
#include "cwrs.h"
#include "mathops.h"
```

Functions

- int log2_frac (ec_uint32 val, int frac)
- int fits_in32 (int _n, int _m)
- celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t *_ui)
- celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t *_u)
- void cwrsi32 (int _n, int _m, celt_uint32_t _i, int *_x, int *_s, celt_uint32_t *_u)
- celt_uint32_t icwrs32 (int _n, int _m, const int *_x, const int *_s, celt_uint32_t *_u)
- void comb2pulse (int _n, int _m, int *restrict _y, const int *_x, const int *_s)
- void pulse2comb (int _n, int _m, int *_x, int *_s, const int *_y)
- int get_required_bits (int N, int K, int frac)
- void encode_pulses (int *_y, int N, int K, ec_enc *enc)
- void decode_pulses (int *_y, int N, int K, ec_dec *dec)

6.22.1 Function Documentation

6.22.1.1 void comb2pulse (int *_n*, int *_m*, int *restrict _y*, const int * *_x*, const int * *_s*)

Definition at line 270 of file cwrs32.c.

References CELT_MEMSET, and k.

6.22.1.2 void cwrsi32 (int *_n*, int *_m*, celt_uint32_t *_i*, int * *_x*, int * *_s*, celt_uint32_t * *_u*)

Definition at line 212 of file cwrs32.c.

References k, and p.

6.22.1.3 void decode_pulses (int * *_y*, int *N*, int *K*, ec_dec * *dec*)

Definition at line 373 of file cwrs32.c.

References ALLOC, celt_ilog2, comb2pulse(), decode_pulses(), ec_dec_bits(), ec_dec_uint(), fits_in32(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.22.1.4 void encode_pulses (int * *_y*, int *N*, int *K*, ec_enc * *enc*)

Definition at line 334 of file cwrs32.c.

References ALLOC, celt_ilog2, ec_enc_bits(), ec_enc_uint(), encode_pulses(), fits_in32(), pulse2comb(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.22.1.5 int fits_in32 (int _n, int _m)

Definition at line 101 of file cwrs32.c.

6.22.1.6 int get_required_bits (int N, int K, int frac)

Definition at line 315 of file cwrs32.c.

References ALLOC, fits_in32(), get_required_bits(), log2_frac(), ncwrs_u32(), RESTORE_STACK, SAVE_STACK, and VARDECL.

6.22.1.7 celt_uint32_t icwrs32 (int _n, int _m, const int * _x, const int * _s, celt_uint32_t * _u)

Definition at line 243 of file cwrs32.c.

References k, and p.

6.22.1.8 int log2_frac (ec_uint32 val, int frac)

Definition at line 51 of file cwrs32.c.

References EC_ILOG.

6.22.1.9 celt_uint32_t ncwrs_u32 (int _n, int _m, celt_uint32_t * _u)

Definition at line 197 of file cwrs32.c.

References CELT_MEMSET, k, and ncwrs_unext32().

6.22.1.10 celt_uint32_t ncwrs_unext32 (int _n, celt_uint32_t * _ui)

Definition at line 177 of file cwrs32.c.

References celt_assert, and ret.

6.22.1.11 void pulse2comb (int _n, int _m, int * _x, int * _s, const int * _y)

Definition at line 284 of file cwrs32.c.

References k, and n.

6.23 libcelt/dist.m File Reference

Variables

- function p
- mi = round(min(v))
- ma = round(max(v))
- for k

6.23.1 Variable Documentation

6.23.1.1 for k

Initial value:

```
1:ma-mi+1
        p(k)=length(find(r==k+mi-1))
```

Definition at line 8 of file dist.m.

6.23.1.2 ma = round(max(v))

Definition at line 5 of file dist.m.

6.23.1.3 mi = round(min(v))

Definition at line 4 of file dist.m.

6.23.1.4 end p

Initial value:

```
dist(v)

    r = round(v)
```

Definition at line 1 of file dist.m.

Referenced by cwrsi32(), cwrsi64(), icwrs32(), and icwrs64().

6.24 libcelt/dump_modes.c File Reference

```
#include <stdio.h>
#include "modes.h"
#include "celt.h"
#include "rate.h"
```

Defines

- #define INT16 "%d"
- #define INT32 "%d"
- #define FLOAT "%f"
- #define WORD16 FLOAT
- #define WORD32 FLOAT

Functions

- void dump_modes (FILE *file, CELTMode **modes, int nb_modes)
- void dump_header (FILE *file, CELTMode **modes, int nb_modes)
- int main (int argc, char **argv)

6.24.1 Define Documentation

6.24.1.1 #define FLOAT "%f"

Definition at line 43 of file dump_modes.c.

6.24.1.2 #define INT16 "%d"

Definition at line 41 of file dump_modes.c.

6.24.1.3 #define INT32 "%d"

Definition at line 42 of file dump_modes.c.

Referenced by dump_modes().

6.24.1.4 #define WORD16 FLOAT

Definition at line 49 of file dump_modes.c.

Referenced by dump_modes().

6.24.1.5 #define WORD32 FLOAT

Definition at line 50 of file dump_modes.c.

6.24.2 Function Documentation

6.24.2.1 void dump_header (FILE * *file*, CELTMode ** *modes*, int *nb_modes*)

Definition at line 230 of file dump_modes.c.

References CELTMode::mdctSize, CELTMode::nbChannels, and CELTMode::overlap.

Referenced by main().

6.24.2.2 void dump_modes (FILE * *file*, CELTMode ** *modes*, int *nb_modes*)

Definition at line 54 of file dump_modes.c.

References CELTMode::allocVectors, CELTMode::bits, CELTMode::bits_stereo, PsyDecay::decayR, CELTMode::eBands, CELTMode::energy_alloc, CELTMode::ePredCoef, CELTMode::Fs, INT32, k, MAX_PERIOD, MAX_PULSES, CELTMode::mdctSize, CELTMode::nbAllocVectors, CELTMode::nbChannels, CELTMode::nbEBands, CELTMode::nbPBands, CELTMode::nbShortMdcts, CELTMode::overlap, CELTMode::pBands, CELTMode::pitchEnd, CELTMode::psy, CELTMode::shortMdctSize, CELTMode::window, and WORD16.

Referenced by main().

6.24.2.3 int main (int *argc*, char ** *argv*)

Definition at line 269 of file dump_modes.c.

References celt_mode_create(), celt_mode_destroy(), dump_header(), dump_modes(), and fclose().

6.25 libcelt/dump_out.c File Reference

```
#include "modes.h"
```

Variables

- const short eBands44100_256 [20]
- const short pBands44100_256 [10]
- const celt_word16_t window128 [128]
- const short allocVectors44100_256 [180]
- CELTMode mode44100_1_256_128

6.25.1 Variable Documentation

6.25.1.1 const short allocVectors44100_256[180]

Initial value:

```
 {
 5,  5,  3,  2,  1,  1,  1,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,
 6,  6,  4,  3,  3,  6,  6,  6,  6,  7,  5,  4,  0,  0,  0,  0,  0,
 7,  7,  4,  5,  3,  7,  7,  7,  7,  8,  6,  5,  4,  0,  0,  0,  0,
 8,  8,  5,  5,  5,  8,  8,  8,  9,  8,  7,  6,  5,  5,  5,  0,  0,
 9,  8,  7,  5,  6, 11,  9,  9, 10,  9,  9,  6,  5,  5,  5,  5,  0,
 9,  8,  7,  6,  6, 12, 11, 10, 12, 11, 10, 10, 10, 11, 10, 10,  5,
14, 14, 12,  9, 10, 17, 13, 12, 12, 12, 13, 15, 16, 17, 18, 20, 18,
23, 23, 19, 16, 16, 24, 18, 17, 18, 24, 22, 26, 29, 31, 34, 35, 35,
28, 28, 22, 22, 27, 30, 24, 20, 21, 27, 27, 36, 39, 43, 49, 60, 53,
36, 38, 30, 26, 27, 42, 35, 29, 29, 37, 37, 46, 49, 53, 59, 61, 59,
}
```

Definition at line 12 of file dump_out.c.

6.25.1.2 const short eBands44100_256[20]

Initial value:

```
 {
0, 4, 8, 12, 16, 20, 24, 28, 32, 37, 43, 51, 62, 74, 90, 110, 140, 180, 233, 256, }
```

Definition at line 3 of file dump_out.c.

6.25.1.3 CELTMode mode44100_1_256_128

Initial value:

```
 {
0xa110ca7e,
44100,
128,
256,
1,
1,
```

```
18,
8,
74,
eBands44100_256,
pBands44100_256,
0.800000,
10,
allocVectors44100_256,
0,
{512, 0, 0},
window128,
0xa110ca7e,
}
```

Definition at line 25 of file dump_out.c.

6.25.1.4 const short pBands44100_256[10]

Initial value:

```
 {
0, 4, 8, 12, 16, 24, 37, 62, 74, 256, }
```

Definition at line 6 of file dump_out.c.

6.25.1.5 const celt_word16_t window128[128]

Initial value:

```
 {
0.000059, 0.000532, 0.001478, 0.002896, 0.004785, 0.007145, 0.009973, 0.013269, 0.017029, 0.021251, 0.0259
```

Definition at line 9 of file dump_out.c.

6.26 libcelt/ecintrin.h File Reference

```
#include <math.h>
#include <limits.h>
```

Defines

- #define _ecintrin_H (1)
- #define EC_MAXI(_a, _b) ((_a)-((_a)-(_b)&-((_b)>(_a))))
- #define EC_MINI(_a, _b) ((_a)+((_b)-(_a)&-((_b)<(_a))))
- #define EC_SIGNI(_a) (((_a)>0)-((_a)<0))
- #define EC_SIGNMASK(_a) (-((_a)<0))
- #define EC_CLAMPI(_a, _b, _c) (EC_MAXI(_a,EC_MINI(_b,_c)))
- #define EC_ILOG(_x) (ec_ilog(_x))
- #define EC_ILOG64(_x) (ec_ilog64(_x))

6.26.1 Define Documentation

6.26.1.1 #define _ecintrin_H (1)

Definition at line 5 of file ecintrin.h.

6.26.1.2 #define EC_CLAMPI(_a, _b, _c) (EC_MAXI(_a,EC_MINI(_b,_c)))

Definition at line 49 of file ecintrin.h.

6.26.1.3 #define EC_ILOG(_x) (ec_ilog(_x))

Definition at line 77 of file ecintrin.h.

Referenced by ec_dec_tell(), ec_dec_uint(), ec_enc_tell(), ec_enc_uint(), and log2_frac().

6.26.1.4 #define EC_ILOG64(_x) (ec_ilog64(_x))

Definition at line 102 of file ecintrin.h.

Referenced by log2_frac64().

6.26.1.5 #define EC_MAXI(_a, _b) ((_a)-((_a)-(_b)&-((_b)>(_a))))

Definition at line 32 of file ecintrin.h.

6.26.1.6 #define EC_MINI(_a, _b) ((_a)+((_b)-(_a)&-((_b)<(_a))))

Definition at line 33 of file ecintrin.h.

Referenced by ec_decode(), and ec_decode_bin().

6.26.1.7 #define EC_SIGNI(_a) (((_a)>0)-((_a)<0))

Definition at line 38 of file ecintrin.h.

6.26.1.8 #define EC_SIGNMASK(_a) (-((_a)<0))

Definition at line 42 of file ecintrin.h.

6.27 libcelt/entcode.c File Reference

```
#include "entcode.h"
```

Functions

- int ec_ilog (ec_uint32 _v)

6.27.1 Function Documentation

6.27.1.1 int ec_ilog (ec_uint32 _v)

Definition at line 13 of file entcode.c.

References ret.

6.28 libcelt/entcode.h File Reference

```
#include "celt_types.h"
#include <limits.h>
#include "ecintrin.h"
```

Data Structures

- struct ec_byte_buffer

Defines

- #define _entcode_H (1)
- #define EC_UNIT_BITS (8)
- #define EC_UNIT_MASK ((1U<<EC_UNIT_BITS)-1)

Typedefs

- typedef celt_int32_t ec_int32
- typedef celt_uint32_t ec_uint32
- typedef celt_uint64_t ec_uint64
- typedef struct ec_byte_buffer ec_byte_buffer

Functions

- void ec_byte_writeinit (ec_byte_buffer *_b)
- void ec_byte_writetrunc (ec_byte_buffer *_b, long _bytes)
- void ec_byte_write1 (ec_byte_buffer *_b, unsigned _value)
- void ec_byte_write4 (ec_byte_buffer *_b, ec_uint32 _value)
- void ec_byte_writecopy (ec_byte_buffer *_b, void *_source, long _bytes)
- void ec_byte_writeclear (ec_byte_buffer *_b)
- void ec_byte_readinit (ec_byte_buffer *_b, unsigned char *_buf, long _bytes)
- int ec_byte_look1 (ec_byte_buffer *_b)
- int ec_byte_look4 (ec_byte_buffer *_b, ec_uint32 *_val)
- void ec_byte_adv1 (ec_byte_buffer *_b)
- void ec_byte_adv4 (ec_byte_buffer *_b)
- int ec_byte_read1 (ec_byte_buffer *_b)
- int ec_byte_read4 (ec_byte_buffer *_b, ec_uint32 *_val)
- int ec_ilog (ec_uint32 _v)
- int ec_ilog64 (ec_uint64 _v)

6.28.1 Define Documentation

6.28.1.1 #define _entcode_H (1)

Definition at line 4 of file entcode.h.

6.28.1.2 #define EC_UNIT_BITS (8)

Definition at line 18 of file entcode.h.

Referenced by ec_dec_bits(), ec_dec_uint(), ec_enc_bits(), and ec_enc_uint().

6.28.1.3 #define EC_UNIT_MASK ((1U<<EC_UNIT_BITS)-1)

Definition at line 20 of file entcode.h.

Referenced by ec_dec_bits(), and ec_enc_bits().

6.28.2 Typedef Documentation

6.28.2.1 typedef struct ec_byte_buffer ec_byte_buffer

Definition at line 13 of file entcode.h.

6.28.2.2 typedef celt_int32_t ec_int32

Definition at line 10 of file entcode.h.

6.28.2.3 typedef celt_uint32_t ec_uint32

Definition at line 11 of file entcode.h.

6.28.2.4 typedef celt_uint64_t ec_uint64

Definition at line 12 of file entcode.h.

6.28.3 Function Documentation

6.28.3.1 void ec_byte_adv1 (ec_byte_buffer * _b)

Definition at line 46 of file entdec.c.

References ec_byte_buffer::ptr.

6.28.3.2 void ec_byte_adv4 (ec_byte_buffer * _b)

Definition at line 50 of file entdec.c.

References ec_byte_buffer::ptr.

6.28.3.3 int ec_byte_look1 (ec_byte_buffer * _b)

Definition at line 15 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.28.3.4 int ec_byte_look4 (ec_byte_buffer * _b, ec_uint32 * _val)

Definition at line 22 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.28.3.5 int ec_byte_read1 (ec_byte_buffer * _b)

Definition at line 54 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.28.3.6 int ec_byte_read4 (ec_byte_buffer * _b, ec_uint32 * _val)

Definition at line 61 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.28.3.7 void ec_byte_readinit (ec_byte_buffer * _b, unsigned char * _buf, long _bytes)

Definition at line 10 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

Referenced by celt_decode(), and celt_decode_float().

6.28.3.8 void ec_byte_write1 (ec_byte_buffer * _b, unsigned _value)

Definition at line 21 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.28.3.9 void ec_byte_write4 (ec_byte_buffer * _b, ec_uint32 _value)

Definition at line 32 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.28.3.10 void ec_byte_writeclear (ec_byte_buffer * _b)

Definition at line 61 of file entenc.c.

References ec_byte_buffer::buf.

Referenced by celt_encoder_destroy().

6.28.3.11 void ec_byte_writecopy (ec_byte_buffer * _b, void * _source, long _bytes)

Definition at line 49 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.28.3.12 void ec_byte_writeinit (ec_byte_buffer * _b)

Definition at line 12 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

Referenced by celt_encoder_create().

6.28.3.13 void ec_byte_writetrunc (ec_byte_buffer * _b, long _bytes)

Definition at line 17 of file entenc.c.

References ec_byte_buffer::buf, and ec_byte_buffer::ptr.

6.28.3.14 int ec_ilog (ec_uint32 _v)

Definition at line 13 of file entcode.c.

References ret.

6.28.3.15 int ec_ilog64 (ec_uint64 _v)

6.29 libcelt/entdec.c File Reference

```
#include <stddef.h>
#include "entdec.h"
#include "os_support.h"
```

Functions

- void ec_byte_readinit (ec_byte_buffer *_b, unsigned char *_buf, long _bytes)
- int ec_byte_look1 (ec_byte_buffer *_b)
- int ec_byte_look4 (ec_byte_buffer *_b, ec_uint32 *_val)
- void ec_byte_adv1 (ec_byte_buffer *_b)
- void ec_byte_adv4 (ec_byte_buffer *_b)
- int ec_byte_read1 (ec_byte_buffer *_b)
- int ec_byte_read4 (ec_byte_buffer *_b, ec_uint32 *_val)
- ec_uint32 ec_dec_bits (ec_dec *_this, int _ftb)
- ec_uint32 ec_dec_uint (ec_dec *_this, ec_uint32 _ft)

6.29.1 Function Documentation

6.29.1.1 void ec_byte_adv1 (ec_byte_buffer * *_b*)

Definition at line 46 of file entdec.c.

References ec_byte_buffer::ptr.

6.29.1.2 void ec_byte_adv4 (ec_byte_buffer * *_b*)

Definition at line 50 of file entdec.c.

References ec_byte_buffer::ptr.

6.29.1.3 int ec_byte_look1 (ec_byte_buffer * *_b*)

Definition at line 15 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.29.1.4 int ec_byte_look4 (ec_byte_buffer * *_b*, ec_uint32 * *_val*)

Definition at line 22 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.29.1.5 int ec_byte_read1 (ec_byte_buffer * *_b*)

Definition at line 54 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.29.1.6 int ec_byte_read4 (ec_byte_buffer * _b, ec_uint32 * _val)

Definition at line 61 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.29.1.7 void ec_byte_readinit (ec_byte_buffer * _b, unsigned char * _buf, long _bytes)

Definition at line 10 of file entdec.c.

References ec_byte_buffer::buf, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

Referenced by celt_decode(), and celt_decode_float().

6.29.1.8 ec_uint32 ec_dec_bits (ec_dec * _this, int _ftb)

Definition at line 85 of file entdec.c.

References ec_dec_update(), ec_decode_bin(), EC_UNIT_BITS, and EC_UNIT_MASK.

Referenced by celt_decode_float(), decode_pulses(), ec_dec_uint(), intra_unquant(), and unquant_pitch().

6.29.1.9 ec_uint32 ec_dec_uint (ec_dec * _this, ec_uint32 _ft)

Definition at line 103 of file entdec.c.

References ec_dec_bits(), ec_dec_update(), ec_decode(), EC_ILOG, and EC_UNIT_BITS.

Referenced by celt_decode(), celt_decode_float(), and decode_pulses().

6.30 libcelt/entdec.h File Reference

```
#include "entcode.h"
```

Data Structures

- struct ec_dec

Defines

- #define _entdec_H (1)

Typedefs

- typedef struct ec_dec ec_dec

Functions

- void ec_dec_init (ec_dec *_this, ec_byte_buffer *_buf)
- unsigned ec_decode (ec_dec *_this, unsigned _ft)
- unsigned ec_decode_bin (ec_dec *_this, unsigned bits)
- void ec_dec_update (ec_dec *_this, unsigned _fl, unsigned _fh, unsigned _ft)
- ec_uint32 ec_dec_bits (ec_dec *_this, int _ftb)
- ec_uint64 ec_dec_bits64 (ec_dec *_this, int _ftb)
- ec_uint32 ec_dec_uint (ec_dec *_this, ec_uint32 _ft)
- ec_uint64 ec_dec_uint64 (ec_dec *_this, ec_uint64 _ft)
- long ec_dec_tell (ec_dec *_this, int _b)

6.30.1 Define Documentation

6.30.1.1 #define _entdec_H (1)

Definition at line 2 of file entdec.h.

6.30.2 Typedef Documentation

6.30.2.1 typedef struct ec_dec ec_dec

Definition at line 7 of file entdec.h.

6.30.3 Function Documentation

6.30.3.1 ec_uint32 ec_dec_bits (ec_dec * *_this*, int *_ftb*)

Definition at line 85 of file entdec.c.

References ec_dec_update(), ec_decode_bin(), EC_UNIT_BITS, and EC_UNIT_MASK.

Referenced by celt_decode_float(), decode_pulses(), ec_dec_uint(), intra_unquant(), and unquant_pitch().

6.30.3.2 ec_uint64 ec_dec_bits64 (ec_dec * _this, int _ftb)

6.30.3.3 void ec_dec_init (ec_dec * _this, ec_byte_buffer * _buf)

Definition at line 155 of file rangedec.c.

References ec_dec::buf, ec_dec::dif, EC_CODE_EXTRA, EC_SYM_BITS, ec_dec::rem, and ec_dec::rng.

Referenced by celt_decode(), and celt_decode_float().

6.30.3.4 long ec_dec_tell (ec_dec * _this, int _b)

Definition at line 189 of file rangedec.c.

References b, ec_dec::buf, EC_CODE_BITS, EC_ILOG, EC_SYM_BITS, and ec_dec::rng.

Referenced by celt_decode(), celt_decode_float(), and unquant_bands().

6.30.3.5 ec_uint32 ec_dec_uint (ec_dec * _this, ec_uint32 _ft)

Definition at line 103 of file entdec.c.

References ec_dec_bits(), ec_dec_update(), ec_decode(), EC_ILOG, and EC_UNIT_BITS.

Referenced by celt_decode(), celt_decode_float(), and decode_pulses().

6.30.3.6 ec_uint64 ec_dec_uint64 (ec_dec * _this, ec_uint64 _ft)

6.30.3.7 void ec_dec_update (ec_dec * _this, unsigned _fl, unsigned _fh, unsigned _ft)

Definition at line 181 of file rangedec.c.

References ec_dec::dif, IMUL32, ec_dec::nrm, and ec_dec::rng.

Referenced by ec_dec_bits(), ec_dec_uint(), and ec_laplace_decode_start().

6.30.3.8 unsigned ec_decode (ec_dec * _this, unsigned _ft)

Definition at line 165 of file rangedec.c.

References ec_dec::dif, EC_MINI, ec_dec::nrm, and ec_dec::rng.

Referenced by ec_dec_uint(), and ec_laplace_decode_start().

6.30.3.9 unsigned ec_decode_bin (ec_dec * _this, unsigned bits)

Definition at line 172 of file rangedec.c.

References ec_dec::dif, EC_MINI, ec_dec::nrm, and ec_dec::rng.

Referenced by ec_dec_bits().

6.31 libcelt/entenc.c File Reference

```
#include "os_support.h"
#include "entenc.h"
```

Defines

- #define EC_BUFFER_INCREMENT (256)

Functions

- void ec_byte_writeinit (ec_byte_buffer *_b)
- void ec_byte_writetrunc (ec_byte_buffer *_b, long _bytes)
- void ec_byte_write1 (ec_byte_buffer *_b, unsigned _value)
- void ec_byte_write4 (ec_byte_buffer *_b, ec_uint32 _value)
- void ec_byte_writecopy (ec_byte_buffer *_b, void *_source, long _bytes)
- void ec_byte_writeclear (ec_byte_buffer *_b)
- void ec_enc_bits (ec_enc *_this, ec_uint32 _fl, int _ftb)
- void ec_enc_uint (ec_enc *_this, ec_uint32 _fl, ec_uint32 _ft)

6.31.1 Define Documentation

6.31.1.1 #define EC_BUFFER_INCREMENT (256)

Definition at line 10 of file entenc.c.

Referenced by ec_byte_write1(), ec_byte_write4(), ec_byte_writecopy(), and ec_byte_writeinit().

6.31.2 Function Documentation

6.31.2.1 void ec_byte_write1 (ec_byte_buffer * _b, unsigned _value)

Definition at line 21 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.31.2.2 void ec_byte_write4 (ec_byte_buffer * _b, ec_uint32 _value)

Definition at line 32 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.31.2.3 void ec_byte_writeclear (ec_byte_buffer * _b)

Definition at line 61 of file entenc.c.

References ec_byte_buffer::buf.

Referenced by celt_encoder_destroy().

6.31.2.4 void ec_byte_writecopy (ec_byte_buffer * _b, void * _source, long _bytes)

Definition at line 49 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

6.31.2.5 void ec_byte_writeinit (ec_byte_buffer * _b)

Definition at line 12 of file entenc.c.

References ec_byte_buffer::buf, EC_BUFFER_INCREMENT, ec_byte_buffer::ptr, and ec_byte_buffer::storage.

Referenced by celt_encoder_create().

6.31.2.6 void ec_byte_writetrunc (ec_byte_buffer * _b, long _bytes)

Definition at line 17 of file entenc.c.

References ec_byte_buffer::buf, and ec_byte_buffer::ptr.

6.31.2.7 void ec_enc_bits (ec_enc * _this, ec_uint32 _fl, int _ftb)

Definition at line 67 of file entenc.c.

References ec_encode_bin(), EC_UNIT_BITS, and EC_UNIT_MASK.

Referenced by celt_encode(), celt_encode_float(), ec_enc_uint(), encode_pulses(), intra_prediction(), and quant_pitch().

6.31.2.8 void ec_enc_uint (ec_enc * _this, ec_uint32 _fl, ec_uint32 _ft)

Definition at line 80 of file entenc.c.

References ec_enc_bits(), ec_encode(), EC_ILOG, and EC_UNIT_BITS.

Referenced by celt_encode(), celt_encode_float(), and encode_pulses().

6.32 libcelt/entenc.h File Reference

```
#include <stddef.h>
#include "entcode.h"
```

Data Structures

- struct ec_enc

Defines

- #define _entenc_H (1)

Typedefs

- typedef struct ec_enc ec_enc

Functions

- void ec_enc_init (ec_enc *_this, ec_byte_buffer *_buf)
- void ec_encode (ec_enc *_this, unsigned _fl, unsigned _fh, unsigned _ft)
- void ec_encode_bin (ec_enc *_this, unsigned _fl, unsigned _fh, unsigned bits)
- void ec_enc_bits (ec_enc *_this, ec_uint32 _fl, int _ftb)
- void ec_enc_bits64 (ec_enc *_this, ec_uint64 _fl, int _ftb)
- void ec_enc_uint (ec_enc *_this, ec_uint32 _fl, ec_uint32 _ft)
- void ec_enc_uint64 (ec_enc *_this, ec_uint64 _fl, ec_uint64 _ft)
- long ec_enc_tell (ec_enc *_this, int _b)
- void ec_enc_done (ec_enc *_this)

6.32.1 Define Documentation

6.32.1.1 #define _entenc_H (1)

Definition at line 2 of file entenc.h.

6.32.2 Typedef Documentation

6.32.2.1 typedef struct ec_enc ec_enc

Definition at line 8 of file entenc.h.

6.32.3 Function Documentation

6.32.3.1 void ec_enc_bits (ec_enc * *_this*, ec_uint32 *_fl*, int *_ftb*)

Definition at line 67 of file entenc.c.

References ec_encode_bin(), EC_UNIT_BITS, and EC_UNIT_MASK.

Referenced by celt_encode(), celt_encode_float(), ec_enc_uint(), encode_pulses(), intra_prediction(), and quant_pitch().

6.32.3.2 void ec_enc_bits64 (ec_enc * *_this*, ec_uint64 *_fl*, int *_ftb*)

6.32.3.3 void ec_enc_done (ec_enc * *_this*)

Definition at line 133 of file rangeenc.c.

References EC_CODE_SHIFT, EC_CODE_TOP, ec_enc::ext, ec_enc::low, ec_enc::rem, and ec_enc::rng.

Referenced by celt_encode(), and celt_encode_float().

6.32.3.4 void ec_enc_init (ec_enc * *_this*, ec_byte_buffer * *_buf*)

Definition at line 77 of file rangeenc.c.

References ec_enc::buf, EC_CODE_TOP, ec_enc::ext, ec_enc::low, ec_enc::rem, and ec_enc::rng.

Referenced by celt_encode(), celt_encode_float(), and celt_encoder_create().

6.32.3.5 long ec_enc_tell (ec_enc * *_this*, int *_b*)

Definition at line 108 of file rangeenc.c.

References b, ec_enc::buf, EC_CODE_BITS, EC_ILOG, ec_enc::ext, ec_enc::rem, and ec_enc::rng.

Referenced by celt_encode(), celt_encode_float(), and quant_bands().

6.32.3.6 void ec_enc_uint (ec_enc * *_this*, ec_uint32 *_fl*, ec_uint32 *_ft*)

Definition at line 80 of file entenc.c.

References ec_enc_bits(), ec_encode(), EC_ILOG, and EC_UNIT_BITS.

Referenced by celt_encode(), celt_encode_float(), and encode_pulses().

6.32.3.7 void ec_enc_uint64 (ec_enc * *_this*, ec_uint64 *_fl*, ec_uint64 *_ft*)

6.32.3.8 void ec_encode (ec_enc * *_this*, unsigned *_fl*, unsigned *_fh*, unsigned *_ft*)

Definition at line 85 of file rangeenc.c.

References IMUL32, ec_enc::low, and ec_enc::rng.

Referenced by ec_enc_uint(), and ec_laplace_encode_start().

6.32.3.9 void ec_encode_bin (ec_enc * *_this*, unsigned *_fl*, unsigned *_fh*, unsigned *bits*)

Definition at line 96 of file rangeenc.c.

References IMUL32, ec_enc::low, and ec_enc::rng.

Referenced by ec_enc_bits().

6.33 libcelt/entropy.m File Reference

Functions

- plot ([0:length(p)-1]/(length(p)-1)∗(ma-mi)+mi, p)

Variables

- function b
- mi = round(min(v))
- ma = round(max(v))
- p = zeros(1,ma-mi+1)
- for k

6.33.1 Function Documentation

6.33.1.1 plot ([0:length(p)-1] /(length(p)-1)∗(ma-mi)+ *mi*, p)

6.33.2 Variable Documentation

6.33.2.1 b

Initial value:

```
entropy(v)

    r = round(v)
```

Definition at line 1 of file entropy.m.

Referenced by ec_dec_tell(), and ec_enc_tell().

6.33.2.2 for k

Initial value:

```
1:ma-mi+1
        p(k)=length(find(r==k+mi-1))
```

Definition at line 8 of file entropy.m.

6.33.2.3 ma = round(max(v))

Definition at line 5 of file entropy.m.

6.33.2.4 mi = round(min(v))

Definition at line 4 of file entropy.m.

6.33.2.5 end p = zeros(1,ma-mi+1)

Definition at line 7 of file entropy.m.

6.34 libcelt/entropy2.m File Reference

Variables

- function b
- mi = round(min(v))
- ma = round(max(v))
- p = zeros(1,ma-mi+1)
- for k

6.34.1 Variable Documentation

6.34.1.1 b

Initial value:

```
entropy2(v)

    r = round(v)
```

Definition at line 1 of file entropy2.m.

6.34.1.2 for k

Initial value:

```
1:ma-mi+1
        p(k)=length(find(r==k+mi-1))
```

Definition at line 8 of file entropy2.m.

6.34.1.3 ma = round(max(v))

Definition at line 5 of file entropy2.m.

6.34.1.4 mi = round(min(v))

Definition at line 4 of file entropy2.m.

6.34.1.5 end p = zeros(1,ma-mi+1)

Definition at line 7 of file entropy2.m.

6.35 libcelt/fixed_c5x.h File Reference

Fixed-point operations for the TI C5x DSP family.

```
#include "dsplib.h"
```

Defines

- #define MAX16(a, b) _max(a,b)
- #define MIN16(a, b) _min(a,b)
- #define MAX32(a, b) _lmax(a,b)
- #define MIN32(a, b) _lmin(a,b)
- #define VSHR32(a, shift) _lshl(a,-(shift))
- #define MULT16_16_Q15(a, b) (_smpy(a,b))
- #define MULT16_16SU(a, b) _lmpysu(a,b)
- #define MULT_16_16(a, b) _lmpy(a,b)
- #define MULT16_32_Q15(a, b) ADD32(SHL(MULT16_16((a),SHR((b),16)),1), SHR(MULT16_16SU((a),(b)),15))
- #define celt_ilog2(x) (30 - _lnorm(x))
- #define OVERRIDE_CELT_ILOG2
- #define celt_maxabs16(x, len) MAX16(maxval((DATA *)x, len),-minval((DATA *)x, len))
- #define OVERRIDE_CELT_MAXABS16
- #define OVERRIDE_FIND_MAX16

6.35.1 Detailed Description

Fixed-point operations for the TI C5x DSP family.

Definition in file fixed_c5x.h.

6.35.2 Define Documentation

6.35.2.1 #define celt_ilog2(x) (30 - _lnorm(x))

Definition at line 78 of file fixed_c5x.h.

Referenced by alg_quant(), decode_pulses(), encode_pulses(), and find_spectral_pitch().

6.35.2.2 #define celt_maxabs16(x, len) MAX16(maxval((DATA *)x, len),-minval((DATA *)x, len))

Definition at line 81 of file fixed_c5x.h.

6.35.2.3 #define MAX16(a, b) _max(a,b)

Definition at line 50 of file fixed_c5x.h.

6.35.2.4 #define MAX32(a, b) _lmax(a,b)

Definition at line 56 of file fixed_c5x.h.

6.35.2.5 #define MIN16(a, b) _min(a,b)

Definition at line 53 of file fixed_c5x.h.

6.35.2.6 #define MIN32(a, b) _lmin(a,b)

Definition at line 59 of file fixed_c5x.h.

6.35.2.7 #define MULT16_16_Q15(a, b) (_smpy(a,b))

Definition at line 65 of file fixed_c5x.h.

6.35.2.8 #define MULT16_16SU(a, b) _lmpysu(a,b)

Definition at line 68 of file fixed_c5x.h.

6.35.2.9 #define MULT16_32_Q15(a, b) ADD32(SHL(MULT16_16((a),SHR((b),16)),1), SHR(MULT16_16SU((a),(b)),15))

Definition at line 75 of file fixed_c5x.h.

6.35.2.10 #define MULT_16_16(a, b) _lmpy(a,b)

Definition at line 71 of file fixed_c5x.h.

6.35.2.11 #define OVERRIDE_CELT_ILOG2

Definition at line 79 of file fixed_c5x.h.

6.35.2.12 #define OVERRIDE_CELT_MAXABS16

Definition at line 82 of file fixed_c5x.h.

6.35.2.13 #define OVERRIDE_FIND_MAX16

Definition at line 84 of file fixed_c5x.h.

6.35.2.14 #define VSHR32(a, shift) _lshl(a,-(shift))

Definition at line 62 of file fixed_c5x.h.

6.36 libcelt/fixed_c6x.h File Reference

Fixed-point operations for the TI C6x DSP family.

Defines

- #define MULT16_16SU(a, b) _mpysu(a,b)
- #define MULT_16_16(a, b) _mpy(a,b)
- #define celt_ilog2(x) (30 - _norm(x))
- #define OVERRIDE_CELT_ILOG2
- #define MULT16_32_Q15(a, b) ADD32(SHL(_mpylh(a,b),1), SHR(_mpsu(a,b),15)

6.36.1 Detailed Description

Fixed-point operations for the TI C6x DSP family.

Definition in file fixed_c6x.h.

6.36.2 Define Documentation

6.36.2.1 #define celt_ilog2(x) (30 - _norm(x))

Definition at line 44 of file fixed_c6x.h.

6.36.2.2 #define MULT16_16SU(a, b) _mpysu(a,b)

Definition at line 39 of file fixed_c6x.h.

6.36.2.3 #define MULT16_32_Q15(a, b) ADD32(SHL(_mpylh(a,b),1), SHR(_mpsu(a,b),15)

Definition at line 48 of file fixed_c6x.h.

6.36.2.4 #define MULT_16_16(a, b) _mpy(a,b)

Definition at line 42 of file fixed_c6x.h.

6.36.2.5 #define OVERRIDE_CELT_ILOG2

Definition at line 45 of file fixed_c6x.h.

6.37 libcelt/fixed_debug.h File Reference

Fixed-point operations with debugging.

```
#include <stdio.h>
```

Defines

- #define MIPS_INC celt_mips++,
- #define MULT16_16SU(a, b) ((celt_word32_t)(celt_word16_t)(a)∗(celt_word32_t)(celt_uint16_t)(b))
- #define MULT32_32_Q31(a, b) ADD32(ADD32(SHL32(MULT16_16(SHR32((a),16),SHR((b),16)),1), SHR32(MULT16_16SU(SHR32((a),16),((b)&0x0000ffff)),15)), SHR32(MULT16_16SU(SHR32((b),16),((a)&0x0000ffff)),15))
- #define MULT16_32_Q16(a, b) ADD32(MULT16_16((a),SHR32((b),16)), SHR32(MULT16_16SU((a),((b)&0x0000ffff)),16))
- #define QCONST16(x, bits) ((celt_word16_t)(.5+(x)∗(((celt_word32_t)1)<<(bits))))
- #define QCONST32(x, bits) ((celt_word32_t)(.5+(x)∗(((celt_word32_t)1)<<(bits))))
- #define VERIFY_SHORT(x) ((x)<=32767&&(x)>=-32768)
- #define VERIFY_INT(x) ((x)<=2147483647LL&&(x)>=-2147483648LL)
- #define VERIFY_UINT(x) ((x)<=(2147483647LLU<<1))
- #define SHR(a, b) SHR32(a,b)
- #define PSHR(a, b) PSHR32(a,b)
- #define EXTRACT16(x) _EXTRACT16(x, __FILE__, __LINE__)
- #define EXTEND32(x) _EXTEND32(x, __FILE__, __LINE__)
- #define SHR16(a, shift) _SHR16(a, shift, __FILE__, __LINE__)
- #define SHL16(a, shift) _SHL16(a, shift, __FILE__, __LINE__)
- #define PSHR16(a, shift) (celt_mips–,SHR16(ADD16((a),((1<<((shift))>>1))),shift))
- #define PSHR32(a, shift) (celt_mips–,SHR32(ADD32((a),(((celt_word32_t)(1)<<((shift))>>1))),shift))
- #define VSHR32(a, shift) (((shift)>0) ? SHR32(a, shift) : SHL32(a, -(shift)))
- #define SATURATE16(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))
- #define SATURATE32(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))
- #define ROUND16(x, a) (celt_mips–,EXTRACT16(PSHR32((x),(a))))
- #define HALF32(x) (SHR32(x,1))
- #define ADD16(a, b) _ADD16(a, b, __FILE__, __LINE__)
- #define SUB16(a, b) _SUB16(a, b, __FILE__, __LINE__)
- #define ADD32(a, b) _ADD32(a, b, __FILE__, __LINE__)
- #define SUB32(a, b) _SUB32(a, b, __FILE__, __LINE__)
- #define UADD32(a, b) _UADD32(a, b, __FILE__, __LINE__)
- #define USUB32(a, b) _USUB32(a, b, __FILE__, __LINE__)
- #define MULT16_16(a, b) _MULT16_16(a, b, __FILE__, __LINE__)
- #define MAC16_16(c, a, b) (celt_mips-=2,ADD32((c),MULT16_16((a),(b))))
- #define MAC16_16_Q11(c, a, b) (ADD16((c),EXTRACT16(SHR32(MULT16_16((a),(b)),11))))
- #define MAC16_16_Q13(c, a, b) (ADD16((c),EXTRACT16(SHR32(MULT16_16((a),(b)),13))))
- #define MAC16_16_P13(c, a, b) (ADD16((c),SHR32(ADD32(4096,MULT16_16((a),(b))),13)))
- #define MULT16_32_QX(a, b, Q) _MULT16_32_QX(a, b, Q, __FILE__, __LINE__)
- #define MULT16_32_Q11(a, b) MULT16_32_QX(a,b,11)
- #define MAC16_32_Q11(c, a, b) ADD32((c),MULT16_32_Q11((a),(b)))
- #define MULT16_32_Q12(a, b) MULT16_32_QX(a,b,12)

- #define MULT16_32_Q13(a, b) MULT16_32_QX(a,b,13)
- #define MULT16_32_Q14(a, b) MULT16_32_QX(a,b,14)
- #define MULT16_32_Q15(a, b) MULT16_32_QX(a,b,15)
- #define MULT16_32_P15(a, b) MULT16_32_PX(a,b,15)
- #define MAC16_32_Q15(c, a, b) (celt_mips-=2,ADD32((c),MULT16_32_Q15((a),(b))))
- #define DIV32_16(a, b) _DIV32_16(a, b, __FILE__, __LINE__)
- #define DIV32(a, b) _DIV32(a, b, __FILE__, __LINE__)
- #define PDIV32(a, b) DIV32(ADD32((a),(b)>>1),b)
- #define PDIV32_16(a, b) DIV32_16(ADD32((a),(b)>>1),b)
- #define PRINT_MIPS(file) do {fprintf (file, "total complexity = %d MIPS\n", celt_mips);} while (0);

Variables

- long long celt_mips

6.37.1 Detailed Description

Fixed-point operations with debugging.

Definition in file fixed_debug.h.

6.37.2 Define Documentation

6.37.2.1 #define ADD16(a, b) _ADD16(a, b, __FILE__, __LINE__)

Definition at line 191 of file fixed_debug.h.

6.37.2.2 #define ADD32(a, b) _ADD32(a, b, __FILE__, __LINE__)

Definition at line 223 of file fixed_debug.h.

6.37.2.3 #define DIV32(a, b) _DIV32(a, b, __FILE__, __LINE__)

Definition at line 526 of file fixed_debug.h.

6.37.2.4 #define DIV32_16(a, b) _DIV32_16(a, b, __FILE__, __LINE__)

Definition at line 499 of file fixed_debug.h.

6.37.2.5 #define EXTEND32(x) _EXTEND32(x, __FILE__, __LINE__)

Definition at line 105 of file fixed_debug.h.

6.37.2.6 #define EXTRACT16(x) _EXTRACT16(x, __FILE__, __LINE__)

Definition at line 92 of file fixed_debug.h.

6.37.2.7 #define HALF32(x) (SHR32(x,1))

Definition at line 186 of file fixed_debug.h.

6.37.2.8 #define MAC16_16(c, a, b) (celt_mips-=2,ADD32((c),MULT16_16((a),(b))))

Definition at line 323 of file fixed_debug.h.

6.37.2.9 #define MAC16_16_P13(c, a, b) (ADD16((c),SHR32(ADD32(4096,MULT16_16((a),(b))),13)))

Definition at line 326 of file fixed_debug.h.

6.37.2.10 #define MAC16_16_Q11(c, a, b) (ADD16((c),EXTRACT16(SHR32(MULT16_16((a),(b)),11))))

Definition at line 324 of file fixed_debug.h.

6.37.2.11 #define MAC16_16_Q13(c, a, b) (ADD16((c),EXTRACT16(SHR32(MULT16_16((a),(b)),13))))

Definition at line 325 of file fixed_debug.h.

6.37.2.12 #define MAC16_32_Q11(c, a, b) ADD32((c),MULT16_32_Q11((a),(b)))

Definition at line 370 of file fixed_debug.h.

6.37.2.13 #define MAC16_32_Q15(c, a, b) (celt_mips-=2,ADD32((c),MULT16_32_Q15((a),(b))))

Definition at line 376 of file fixed_debug.h.

6.37.2.14 #define MIPS_INC celt_mips++,

Definition at line 46 of file fixed_debug.h.

6.37.2.15 #define MULT16_16(a, b) _MULT16_16(a, b, __FILE__, __LINE__)

Definition at line 308 of file fixed_debug.h.

6.37.2.16 #define MULT16_16SU(a, b) ((celt_word32_t)(celt_word16_t)(a)∗(celt_word32_t)(celt_uint16_t)(b))

Definition at line 48 of file fixed_debug.h.

6.37.2.17 #define MULT16_32_P15(a, b) MULT16_32_PX(a,b,15)

Definition at line 375 of file fixed_debug.h.

6.37.2.18 #define MULT16_32_Q11(a, b) MULT16_32_QX(a,b,11)

Definition at line 369 of file fixed_debug.h.

6.37.2.19 #define MULT16_32_Q12(a, b) MULT16_32_QX(a,b,12)

Definition at line 371 of file fixed_debug.h.

6.37.2.20 #define MULT16_32_Q13(a, b) MULT16_32_QX(a,b,13)

Definition at line 372 of file fixed_debug.h.

6.37.2.21 #define MULT16_32_Q14(a, b) MULT16_32_QX(a,b,14)

Definition at line 373 of file fixed_debug.h.

6.37.2.22 #define MULT16_32_Q15(a, b) MULT16_32_QX(a,b,15)

Definition at line 374 of file fixed_debug.h.

6.37.2.23 #define MULT16_32_Q16(a, b) ADD32(MULT16_16((a),SHR32((b),16)), SHR32(MULT16_16SU((a),((b)&0x0000ffff)),16))

16x32 multiplication, followed by a 16-bit shift right. Results fits in 32 bits

Definition at line 52 of file fixed_debug.h.

6.37.2.24 #define MULT16_32_QX(a, b, Q) _MULT16_32_QX(a, b, Q, __FILE__, __LINE__)

Definition at line 329 of file fixed_debug.h.

6.37.2.25 #define MULT32_32_Q31(a, b) ADD32(ADD32(SHL32(MULT16_16(SHR32((a),16),SHR((b),16)),1), SHR32(MULT16_16SU(SHR32((a),16),((b)&0x0000ffff)),15)), SHR32(MULT16_16SU(SHR32((b),16),((a)&0x0000ffff)),15))

Definition at line 49 of file fixed_debug.h.

6.37.2.26 #define PDIV32(a, b) DIV32(ADD32((a),(b)>>1),b)

Definition at line 546 of file fixed_debug.h.

6.37.2.27 #define PDIV32_16(a, b) DIV32_16(ADD32((a),(b)>>1),b)

Definition at line 547 of file fixed_debug.h.

6.37.2.28 #define PRINT_MIPS(file) do {fprintf (file, "total complexity = %d MIPS\n", celt_mips);} while (0);

Definition at line 550 of file fixed_debug.h.

6.37.2.29 #define PSHR(a, b) PSHR32(a,b)

Definition at line 63 of file fixed_debug.h.

6.37.2.30 #define PSHR16(a, shift) (celt_mips–,SHR16(ADD16((a),((1<<((shift))>>1))),shift))

Definition at line 178 of file fixed_debug.h.

6.37.2.31 #define PSHR32(a, shift) (celt_mips–,SHR32(ADD32((a),(((celt_word32_t)(1)<<((shift))>>1))),shift))

Definition at line 179 of file fixed_debug.h.

6.37.2.32 #define QCONST16(x, bits) ((celt_word16_t)(.5+(x)∗(((celt_word32_t)1)<<(bits))))

Definition at line 54 of file fixed_debug.h.

6.37.2.33 #define QCONST32(x, bits) ((celt_word32_t)(.5+(x)∗(((celt_word32_t)1)<<(bits))))

Definition at line 55 of file fixed_debug.h.

6.37.2.34 #define ROUND16(x, a) (celt_mips–,EXTRACT16(PSHR32((x),(a))))

Definition at line 185 of file fixed_debug.h.

6.37.2.35 #define SATURATE16(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))

Definition at line 182 of file fixed_debug.h.

6.37.2.36 #define SATURATE32(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))

Definition at line 183 of file fixed_debug.h.

6.37.2.37 #define SHL16(a, shift) _SHL16(a, shift, __FILE__, __LINE__)

Definition at line 132 of file fixed_debug.h.

6.37.2.38 #define SHR(a, b) SHR32(a,b)

Definition at line 62 of file fixed_debug.h.

6.37.2.39 #define SHR16(a, shift) _SHR16(a, shift, __FILE__, __LINE__)

Definition at line 118 of file fixed_debug.h.

6.37.2.40 #define SUB16(a, b) _SUB16(a, b, __FILE__, __LINE__)

Definition at line 208 of file fixed_debug.h.

6.37.2.41 #define SUB32(a, b) _SUB32(a, b, __FILE__, __LINE__)

Definition at line 240 of file fixed_debug.h.

6.37.2.42 #define UADD32(a, b) _UADD32(a, b, __FILE__, __LINE__)

Definition at line 256 of file fixed_debug.h.

6.37.2.43 #define USUB32(a, b) _USUB32(a, b, __FILE__, __LINE__)

Definition at line 274 of file fixed_debug.h.

6.37.2.44 #define VERIFY_INT(x) ((x)<=2147483647LL&&(x)>=-2147483648LL)

Definition at line 59 of file fixed_debug.h.

6.37.2.45 #define VERIFY_SHORT(x) ((x)<=32767&&(x)>=-32768)

Definition at line 58 of file fixed_debug.h.

6.37.2.46 #define VERIFY_UINT(x) ((x)<=(2147483647LLU<<1))

Definition at line 60 of file fixed_debug.h.

6.37.2.47 #define VSHR32(a, shift) (((shift)>0) ? SHR32(a, shift) : SHL32(a, -(shift)))

Definition at line 180 of file fixed_debug.h.

6.37.3 Variable Documentation

6.37.3.1 long long celt_mips

6.38 libcelt/fixed_generic.h File Reference

Generic fixed-point operations.

Defines

- #define MULT16_16SU(a, b) ((celt_word32_t)(celt_word16_t)(a)*(celt_word32_t)(celt_uint16_t)(b))
- #define MULT16_32_Q16(a, b) ADD32(MULT16_16((a),SHR((b),16)), SHR(MULT16_16SU((a),((b)&0x0000ffff)),16))
- #define MULT16_32_Q15(a, b) ADD32(SHL(MULT16_16((a),SHR((b),16)),1), SHR(MULT16_16SU((a),((b)&0x0000ffff)),15))
- #define MULT32_32_Q31(a, b) ADD32(ADD32(SHL(MULT16_16(SHR((a),16),SHR((b),16)),1), SHR(MULT16_16SU(SHR((a),16),((b)&0x0000ffff)),15)), SHR(MULT16_16SU(SHR((b),16),((a)&0x0000ffff)),15))
- #define MULT32_32_Q32(a, b) ADD32(ADD32(MULT16_16(SHR((a),16),SHR((b),16)), SHR(MULT16_16SU(SHR((a),16),((b)&0x0000ffff)),16)), SHR(MULT16_16SU(SHR((b),16),((a)&0x0000ffff)),16))
- #define QCONST16(x, bits) ((celt_word16_t)(.5+(x)*(((celt_word32_t)1)<<(bits))))
- #define QCONST32(x, bits) ((celt_word32_t)(.5+(x)*(((celt_word32_t)1)<<(bits))))
- #define NEG16(x) (-(x))
- #define NEG32(x) (-(x))
- #define EXTRACT16(x) ((celt_word16_t)(x))
- #define EXTEND32(x) ((celt_word32_t)(x))
- #define SHR16(a, shift) ((a) >> (shift))
- #define SHL16(a, shift) ((a) << (shift))
- #define SHR32(a, shift) ((a) >> (shift))
- #define SHL32(a, shift) ((celt_word32_t)(a) << (shift))
- #define PSHR16(a, shift) (SHR16((a)+((1<<((shift))>>1)),shift))
- #define PSHR32(a, shift) (SHR32((a)+((EXTEND32(1)<<((shift))>>1)),shift))
- #define VSHR32(a, shift) (((shift)>0) ? SHR32(a, shift) : SHL32(a, -(shift)))
- #define SATURATE16(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))
- #define SATURATE32(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))
- #define SHR(a, shift) ((a) >> (shift))
- #define SHL(a, shift) ((celt_word32_t)(a) << (shift))
- #define PSHR(a, shift) (SHR((a)+((EXTEND32(1)<<((shift))>>1)),shift))
- #define SATURATE(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))
- #define ROUND16(x, a) (EXTRACT16(PSHR32((x),(a))))
- #define HALF32(x) (SHR32(x,1))
- #define ADD16(a, b) ((celt_word16_t)((celt_word16_t)(a)+(celt_word16_t)(b)))
- #define SUB16(a, b) ((celt_word16_t)(a)-(celt_word16_t)(b))
- #define ADD32(a, b) ((celt_word32_t)(a)+(celt_word32_t)(b))
- #define SUB32(a, b) ((celt_word32_t)(a)-(celt_word32_t)(b))
- #define MULT16_16_16(a, b) ((((celt_word16_t)(a))*((celt_word16_t)(b))))
- #define MULT16_16(a, b) (((celt_word32_t)(celt_word16_t)(a))*((celt_word32_t)(celt_word16_t)(b)))
- #define MAC16_16(c, a, b) (ADD32((c),MULT16_16((a),(b))))
- #define MULT16_32_Q12(a, b) ADD32(MULT16_16((a),SHR((b),12)), SHR(MULT16_16((a),((b)&0x00000fff)),12))

- #define MULT16_32_Q13(a, b) ADD32(MULT16_16((a),SHR((b),13)), SHR(MULT16_16((a),((b)&0x00001fff)),13))
- #define MULT16_32_Q14(a, b) ADD32(MULT16_16((a),SHR((b),14)), SHR(MULT16_16((a),((b)&0x00003fff)),14))
- #define MULT16_32_Q11(a, b) ADD32(MULT16_16((a),SHR((b),11)), SHR(MULT16_16((a),((b)&0x000007ff)),11))
- #define MAC16_32_Q11(c, a, b) ADD32(c,ADD32(MULT16_16((a),SHR((b),11)), SHR(MULT16_16((a),((b)&0x000007ff)),11)))
- #define MULT16_32_P15(a, b) ADD32(MULT16_16((a),SHR((b),15)), PSHR(MULT16_16((a),((b)&0x00007fff)),15))
- #define MAC16_32_Q15(c, a, b) ADD32(c,ADD32(MULT16_16((a),SHR((b),15)), SHR(MULT16_16((a),((b)&0x00007fff)),15)))
- #define MAC16_16_Q11(c, a, b) (ADD32((c),SHR(MULT16_16((a),(b)),11)))
- #define MAC16_16_Q13(c, a, b) (ADD32((c),SHR(MULT16_16((a),(b)),13)))
- #define MAC16_16_P13(c, a, b) (ADD32((c),SHR(ADD32(4096,MULT16_16((a),(b))),13)))
- #define MULT16_16_Q11_32(a, b) (SHR(MULT16_16((a),(b)),11))
- #define MULT16_16_Q13(a, b) (SHR(MULT16_16((a),(b)),13))
- #define MULT16_16_Q14(a, b) (SHR(MULT16_16((a),(b)),14))
- #define MULT16_16_Q15(a, b) (SHR(MULT16_16((a),(b)),15))
- #define MULT16_16_P13(a, b) (SHR(ADD32(4096,MULT16_16((a),(b))),13))
- #define MULT16_16_P14(a, b) (SHR(ADD32(8192,MULT16_16((a),(b))),14))
- #define MULT16_16_P15(a, b) (SHR(ADD32(16384,MULT16_16((a),(b))),15))
- #define DIV32_16(a, b) ((celt_word16_t)(((celt_word32_t)(a))/((celt_word16_t)(b))))
- #define PDIV32_16(a, b) ((celt_word16_t)(((celt_word32_t)(a)+((celt_word16_t)(b)>>1))/((celt_word16_t)(b))))
- #define DIV32(a, b) (((celt_word32_t)(a))/((celt_word32_t)(b)))
- #define PDIV32(a, b) (((celt_word32_t)(a)+((celt_word16_t)(b)>>1))/((celt_word32_t)(b)))

6.38.1 Detailed Description

Generic fixed-point operations.

Definition in file fixed_generic.h.

6.38.2 Define Documentation

6.38.2.1 #define ADD16(a, b) ((celt_word16_t)((celt_word16_t)(a)+(celt_word16_t)(b)))

Add two 16-bit values

Definition at line 101 of file fixed_generic.h.

6.38.2.2 #define ADD32(a, b) ((celt_word32_t)(a)+(celt_word32_t)(b))

Add two 32-bit values

Definition at line 105 of file fixed_generic.h.

6.38.2.3 #define DIV32(a, b) (((celt_word32_t)(a))/((celt_word32_t)(b)))

Divide a 32-bit value by a 32-bit value. Result fits in 32 bits

Definition at line 155 of file fixed_generic.h.

6.38.2.4 #define DIV32_16(a, b) ((celt_word16_t)(((celt_word32_t)(a))/((celt_word16_t)(b))))

Divide a 32-bit value by a 16-bit value. Result fits in 16 bits

Definition at line 151 of file fixed_generic.h.

6.38.2.5 #define EXTEND32(x) ((celt_word32_t)(x))

Change a 16-bit value into a 32-bit value

Definition at line 66 of file fixed_generic.h.

6.38.2.6 #define EXTRACT16(x) ((celt_word16_t)(x))

Change a 32-bit value into a 16-bit value. The value is assumed to fit in 16-bit, otherwise the result is undefined

Definition at line 64 of file fixed_generic.h.

6.38.2.7 #define HALF32(x) (SHR32(x,1))

Divide by two

Definition at line 98 of file fixed_generic.h.

6.38.2.8 #define MAC16_16(c, a, b) (ADD32((c),MULT16_16((a),(b))))

16x16 multiply-add where the result fits in 32 bits

Definition at line 118 of file fixed_generic.h.

6.38.2.9 #define MAC16_16_P13(c, a, b) (ADD32((c),SHR(ADD32(4096,MULT16_16((a),(b))),13)))

Definition at line 139 of file fixed_generic.h.

6.38.2.10 #define MAC16_16_Q11(c, a, b) (ADD32((c),SHR(MULT16_16((a),(b)),11)))

Definition at line 137 of file fixed_generic.h.

6.38.2.11 #define MAC16_16_Q13(c, a, b) (ADD32((c),SHR(MULT16_16((a),(b)),13)))

Definition at line 138 of file fixed_generic.h.

6.38.2.12 #define MAC16_32_Q11(c, a, b) ADD32(c,ADD32(MULT16_16((a),SHR((b),11)), SHR(MULT16_16((a),((b)&0x000007ff)),11)))

16x32 multiply-add, followed by an 11-bit shift right. Results fits in 32 bits

Definition at line 129 of file fixed_generic.h.

6.38.2.13 #define MAC16_32_Q15(c, a, b) ADD32(c,ADD32(MULT16_16((a),SHR((b),15)), SHR(MULT16_16((a),((b)&0x00007fff)),15)))

16x32 multiply-add, followed by a 15-bit shift right. Results fits in 32 bits

Definition at line 134 of file fixed_generic.h.

6.38.2.14 #define MULT16_16(a, b) (((celt_word32_t)(celt_word16_t)(a))∗((celt_word32_t)(celt_word16_t)(b)))

16x16 multiplication where the result fits in 32 bits

Definition at line 115 of file fixed_generic.h.

6.38.2.15 #define MULT16_16_16(a, b) ((((celt_word16_t)(a))∗((celt_word16_t)(b))))

16x16 multiplication where the result fits in 16 bits

Definition at line 111 of file fixed_generic.h.

6.38.2.16 #define MULT16_16_P13(a, b) (SHR(ADD32(4096,MULT16_16((a),(b))),13))

Definition at line 146 of file fixed_generic.h.

6.38.2.17 #define MULT16_16_P14(a, b) (SHR(ADD32(8192,MULT16_16((a),(b))),14))

Definition at line 147 of file fixed_generic.h.

6.38.2.18 #define MULT16_16_P15(a, b) (SHR(ADD32(16384,MULT16_16((a),(b))),15))

Definition at line 148 of file fixed_generic.h.

6.38.2.19 #define MULT16_16_Q11_32(a, b) (SHR(MULT16_16((a),(b)),11))

Definition at line 141 of file fixed_generic.h.

6.38.2.20 #define MULT16_16_Q13(a, b) (SHR(MULT16_16((a),(b)),13))

Definition at line 142 of file fixed_generic.h.

6.38.2.21 #define MULT16_16_Q14(a, b) (SHR(MULT16_16((a),(b)),14))

Definition at line 143 of file fixed_generic.h.

6.38.2.22 #define MULT16_16_Q15(a, b) (SHR(MULT16_16((a),(b)),15))

Definition at line 144 of file fixed_generic.h.

6.38.2.23 #define MULT16_16SU(a, b) ((celt_word32_t)(celt_word16_t)(a)∗(celt_word32_t)(celt_-uint16_t)(b))

Multiply a 16-bit signed value by a 16-bit unsigned value. The result is a 32-bit signed value

Definition at line 39 of file fixed_generic.h.

6.38.2.24 #define MULT16_32_P15(a, b) ADD32(MULT16_16((a),SHR((b),15)), PSHR(MULT16_16((a),((b)&0x00007fff)),15))

16x32 multiplication, followed by a 15-bit shift right (round-to-nearest). Results fits in 32 bits

Definition at line 132 of file fixed_generic.h.

6.38.2.25 #define MULT16_32_Q11(a, b) ADD32(MULT16_16((a),SHR((b),11)), SHR(MULT16_16((a),((b)&0x000007ff)),11))

16x32 multiplication, followed by an 11-bit shift right. Results fits in 32 bits

Definition at line 127 of file fixed_generic.h.

6.38.2.26 #define MULT16_32_Q12(a, b) ADD32(MULT16_16((a),SHR((b),12)), SHR(MULT16_16((a),((b)&0x00000fff)),12))

16x32 multiplication, followed by a 12-bit shift right. Results fits in 32 bits

Definition at line 120 of file fixed_generic.h.

6.38.2.27 #define MULT16_32_Q13(a, b) ADD32(MULT16_16((a),SHR((b),13)), SHR(MULT16_16((a),((b)&0x00001fff)),13))

16x32 multiplication, followed by a 13-bit shift right. Results fits in 32 bits

Definition at line 122 of file fixed_generic.h.

6.38.2.28 #define MULT16_32_Q14(a, b) ADD32(MULT16_16((a),SHR((b),14)), SHR(MULT16_16((a),((b)&0x00003fff)),14))

16x32 multiplication, followed by a 14-bit shift right. Results fits in 32 bits

Definition at line 124 of file fixed_generic.h.

6.38.2.29 #define MULT16_32_Q15(a, b) ADD32(SHL(MULT16_16((a),SHR((b),16)),1), SHR(MULT16_16SU((a),((b)&0x0000ffff)),15))

16x32 multiplication, followed by a 15-bit shift right. Results fits in 32 bits

Definition at line 45 of file fixed_generic.h.

6.38.2.30 #define MULT16_32_Q16(a, b) ADD32(MULT16_16((a),SHR((b),16)), SHR(MULT16_16SU((a),((b)&0x0000ffff)),16))

16x32 multiplication, followed by a 16-bit shift right. Results fits in 32 bits

Definition at line 42 of file fixed_generic.h.

6.38.2.31 #define MULT32_32_Q31(a, b) ADD32(ADD32(SHL(MULT16_16(SHR((a),16),SHR((b),16)),1), SHR(MULT16_16SU(SHR((a),16),((b)&0x0000ffff)),15)), SHR(MULT16_16SU(SHR((b),16),((a)&0x0000ffff)),15))

32x32 multiplication, followed by a 31-bit shift right. Results fits in 32 bits

Definition at line 48 of file fixed_generic.h.

6.38.2.32 #define MULT32_32_Q32(a, b) ADD32(ADD32(MULT16_16(SHR((a),16),SHR((b),16)), SHR(MULT16_16SU(SHR((a),16),((b)&0x0000ffff)),16)), SHR(MULT16_16SU(SHR((b),16),((a)&0x0000ffff)),16))

32x32 multiplication, followed by a 32-bit shift right. Results fits in 32 bits

Definition at line 51 of file fixed_generic.h.

6.38.2.33 #define NEG16(x) (-(x))

Negate a 16-bit value

Definition at line 59 of file fixed_generic.h.

6.38.2.34 #define NEG32(x) (-(x))

Negate a 32-bit value

Definition at line 61 of file fixed_generic.h.

6.38.2.35 #define PDIV32(a, b) (((celt_word32_t)(a)+((celt_word16_t)(b)>>1))/((celt_word32_t)(b)))

Divide a 32-bit value by a 32-bit value and round to nearest. Result fits in 32 bits

Definition at line 157 of file fixed_generic.h.

6.38.2.36 #define PDIV32_16(a, b) ((celt_word16_t)(((celt_word32_t)(a)+((celt_word16_t)(b)>>1))/((celt_word16_t)(b))))

Divide a 32-bit value by a 16-bit value and round to nearest. Result fits in 16 bits

Definition at line 153 of file fixed_generic.h.

6.38.2.37 #define PSHR(a, shift) (SHR((a)+((EXTEND32(1)<<((shift))>>1)),shift))

Definition at line 92 of file fixed_generic.h.

6.38.2.38 #define PSHR16(a, shift) (SHR16((a)+((1<<((shift))>>1)),shift))

16-bit arithmetic shift right with rounding-to-nearest instead of rounding down

Definition at line 78 of file fixed_generic.h.

6.38.2.39 #define PSHR32(a, shift) (SHR32((a)+((EXTEND32(1)<<((shift))>>1)),shift))

32-bit arithmetic shift right with rounding-to-nearest instead of rounding down

Definition at line 80 of file fixed_generic.h.

6.38.2.40 #define QCONST16(x, bits) ((celt_word16_t)(.5+(x)∗(((celt_word32_t)1)<<(bits))))

Compile-time conversion of float constant to 16-bit value

Definition at line 54 of file fixed_generic.h.

6.38.2.41 #define QCONST32(x, bits) ((celt_word32_t)(.5+(x)∗(((celt_word32_t)1)<<(bits))))

Compile-time conversion of float constant to 32-bit value

Definition at line 56 of file fixed_generic.h.

6.38.2.42 #define ROUND16(x, a) (EXTRACT16(PSHR32((x),(a))))

Shift by a and round-to-neareast 32-bit value. Result is a 16-bit value

Definition at line 96 of file fixed_generic.h.

6.38.2.43 #define SATURATE(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))

Definition at line 93 of file fixed_generic.h.

6.38.2.44 #define SATURATE16(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))

Saturates 16-bit value to +/- a

Definition at line 85 of file fixed_generic.h.

6.38.2.45 #define SATURATE32(x, a) (((x)>(a) ? (a) : (x)<-(a) ? -(a) : (x)))

Saturates 32-bit value to +/- a

Definition at line 87 of file fixed_generic.h.

6.38.2.46 #define SHL(a, shift) ((celt_word32_t)(a) << (shift))

Definition at line 91 of file fixed_generic.h.

6.38.2.47 #define SHL16(a, shift) ((a) << (shift))

Arithmetic shift-left of a 16-bit value

Definition at line 71 of file fixed_generic.h.

6.38.2.48 #define SHL32(a, shift) ((celt_word32_t)(a) << (shift))

Arithmetic shift-left of a 32-bit value

Definition at line 75 of file fixed_generic.h.

6.38.2.49 #define SHR(a, shift) ((a) >> (shift))

"RAW" macros, should not be used outside of this header file

Definition at line 90 of file fixed_generic.h.

6.38.2.50 #define SHR16(a, shift) ((a) >> (shift))

Arithmetic shift-right of a 16-bit value

Definition at line 69 of file fixed_generic.h.

6.38.2.51 #define SHR32(a, shift) ((a) >> (shift))

Arithmetic shift-right of a 32-bit value

Definition at line 73 of file fixed_generic.h.

6.38.2.52 #define SUB16(a, b) ((celt_word16_t)(a)-(celt_word16_t)(b))

Subtract two 16-bit values

Definition at line 103 of file fixed_generic.h.

6.38.2.53 #define SUB32(a, b) ((celt_word32_t)(a)-(celt_word32_t)(b))

Subtract two 32-bit values

Definition at line 107 of file fixed_generic.h.

6.38.2.54 #define VSHR32(a, shift) (((shift)>0) ? SHR32(a, shift) : SHL32(a, -(shift)))

32-bit arithmetic shift right where the argument can be negative

Definition at line 82 of file fixed_generic.h.

6.39 libcelt/float_cast.h File Reference

```
#include <math.h>
```

Defines

- #define float2int(flt) ((int)(floor(.5+flt)))

6.39.1 Define Documentation

6.39.1.1 #define float2int(flt) ((int)(floor(.5+flt)))

Definition at line 102 of file float_cast.h.

6.40 libcelt/gcc_ice.c File Reference

Functions

- void foo (unsigned *_this, short _sz, short _sym, short _val)

6.40.1 Function Documentation

6.40.1.1 void foo (unsigned * *_this*, short *_sz*, short *_sym*, short *_val*)

Definition at line 1 of file gcc_ice.c.

6.41 libcelt/header.c File Reference

```
#include "celt_header.h"
#include "os_support.h"
#include "modes.h"
```

Functions

- void celt_header_init (CELTHeader *header, const CELTMode *m)
- int celt_header_to_packet (const CELTHeader *header, unsigned char *packet, celt_uint32_t size)
- int celt_header_from_packet (const unsigned char *packet, celt_uint32_t size, CELTHeader *header)

6.41.1 Function Documentation

6.41.1.1 int celt_header_from_packet (const unsigned char * *packet*, celt_uint32_t *size*, CELTHeader * *header*)

Definition at line 106 of file header.c.

References CELT_COPY.

6.41.1.2 void celt_header_init (CELTHeader * *header*, const CELTMode * *m*)

Creates a basic header struct

Definition at line 65 of file header.c.

References CELTHeader::bytes_per_packet, CELT_COPY, CELT_GET_BITSTREAM_VERSION, celt_mode_info(), CELTHeader::codec_id, CELTHeader::codec_version, CELTHeader::extra_headers, CELTHeader::frame_size, CELTMode::Fs, CELTHeader::header_size, CELTMode::mdctSize, CELTHeader::nb_channels, CELTMode::nbChannels, CELTMode::overlap, CELTHeader::overlap, CELTHeader::sample_rate, and CELTHeader::version_id.

6.41.1.3 int celt_header_to_packet (const CELTHeader * *header*, unsigned char * *packet*, celt_uint32_t *size*)

Definition at line 80 of file header.c.

References CELT_BAD_ARG, CELT_COPY, and CELT_MEMSET.

6.42 libcelt/kfft_double.h File Reference

```
#include "kiss_fft.h"
#include "_kiss_fft_guts.h"
```

Defines

- #define cpx32_fft_alloc(length) kiss_fft_alloc(length, 0, 0);
- #define cpx32_fft_free(state) kiss_fft_free(state)
- #define cpx32_fft(state, X, Y, nx) kiss_fft(state,(const kiss_fft_cpx *)(X), (kiss_fft_cpx *)(Y))
- #define cpx32_ifft(state, X, Y, nx) kiss_ifft(state,(const kiss_fft_cpx *)(X), (kiss_fft_cpx *)(Y))

6.42.1 Define Documentation

6.42.1.1 #define cpx32_fft(state, X, Y, nx) kiss_fft(state,(const kiss_fft_cpx *)(X), (kiss_fft_cpx *)(Y))

Definition at line 63 of file kfft_double.h.

Referenced by mdct_forward().

6.42.1.2 #define cpx32_fft_alloc(length) kiss_fft_alloc(length, 0, 0);

Definition at line 61 of file kfft_double.h.

Referenced by mdct_init().

6.42.1.3 #define cpx32_fft_free(state) kiss_fft_free(state)

Definition at line 62 of file kfft_double.h.

Referenced by mdct_clear().

6.42.1.4 #define cpx32_ifft(state, X, Y, nx) kiss_ifft(state,(const kiss_fft_cpx *)(X), (kiss_fft_cpx *)(Y))

Definition at line 64 of file kfft_double.h.

Referenced by mdct_backward().

6.43 libcelt/kfft_single.c File Reference

6.44 libcelt/kfft_single.h File Reference

```
#include "kiss_fft.h"
#include "kiss_fftr.h"
#include "_kiss_fft_guts.h"
```

Defines

- #define real16_fft_alloc(length) kiss_fftr_alloc_celt_single(length, 0, 0);
- #define real16_fft_free(state) kiss_fft_free(state)
- #define real16_fft_inplace(state, X, nx) kiss_fftr_inplace(state,X)
- #define BITREV(state, i) ((state) → substate → bitrev[i])
- #define real16_ifft(state, X, Y, nx) kiss_fftri(state,X, Y)

6.44.1 Define Documentation

6.44.1.1 #define BITREV(state, i) ((state) → substate → bitrev[i])

Definition at line 79 of file kfft_single.h.

Referenced by find_spectral_pitch().

6.44.1.2 #define real16_fft_alloc(length) kiss_fftr_alloc_celt_single(length, 0, 0);

Definition at line 76 of file kfft_single.h.

Referenced by pitch_state_alloc().

6.44.1.3 #define real16_fft_free(state) kiss_fft_free(state)

Definition at line 77 of file kfft_single.h.

Referenced by pitch_state_free().

6.44.1.4 #define real16_fft_inplace(state, X, nx) kiss_fftr_inplace(state,X)

Definition at line 78 of file kfft_single.h.

Referenced by find_spectral_pitch().

6.44.1.5 #define real16_ifft(state, X, Y, nx) kiss_fftri(state,X, Y)

Definition at line 80 of file kfft_single.h.

Referenced by find_spectral_pitch().

6.45 libcelt/kiss_fft.c File Reference

```
#include "_kiss_fft_guts.h"
#include "arch.h"
#include "os_support.h"
#include "mathops.h"
#include "stack_alloc.h"
```

Functions

- void kf_work (kiss_fft_cpx *Fout, const kiss_fft_cpx *f, const size_t fstride, int in_stride, int *factors, const kiss_fft_cfg st, int N, int s2, int m2)
- void ki_work (kiss_fft_cpx *Fout, const kiss_fft_cpx *f, const size_t fstride, int in_stride, int *factors, const kiss_fft_cfg st, int N, int s2, int m2)
- kiss_fft_cfg kiss_fft_alloc (int nfft, void *mem, size_t *lenmem)
- void kiss_fft_stride (kiss_fft_cfg st, const kiss_fft_cpx *fin, kiss_fft_cpx *fout, int in_stride)
- void kiss_fft (kiss_fft_cfg cfg, const kiss_fft_cpx *fin, kiss_fft_cpx *fout)
- void kiss_ifft_stride (kiss_fft_cfg st, const kiss_fft_cpx *fin, kiss_fft_cpx *fout, int in_stride)
- void kiss_ifft (kiss_fft_cfg cfg, const kiss_fft_cpx *fin, kiss_fft_cpx *fout)

6.45.1 Function Documentation

6.45.1.1 void kf_work (kiss_fft_cpx * *Fout*, const kiss_fft_cpx * *f*, const size_t *fstride*, int *in_stride*, int * *factors*, const kiss_fft_cfg *st*, int *N*, int *s2*, int *m2*)

Definition at line 512 of file kiss_fft.c.

References celt_fatal, and kf_work.

6.45.1.2 void ki_work (kiss_fft_cpx * *Fout*, const kiss_fft_cpx * *f*, const size_t *fstride*, int *in_stride*, int * *factors*, const kiss_fft_cfg *st*, int *N*, int *s2*, int *m2*)

Internal function. Can be useful when you want to do the bit-reversing yourself

Definition at line 548 of file kiss_fft.c.

References celt_fatal, and ki_work.

6.45.1.3 void kiss_fft (kiss_fft_cfg *cfg*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*)

kiss_fft(cfg,in_out_buf)

Perform an FFT on a complex input buffer. for a forward FFT, fin should be f[0] , f[1] , ... ,f[nfft-1] fout will be F[0] , F[1] , ... ,F[nfft-1] Note that each element is complex and can be accessed like f[k].r and f[k].i

Definition at line 678 of file kiss_fft.c.

References kiss_fft_stride.

6.45.1.4 kiss_fft_cfg kiss_fft_alloc (int *nfft*, void * *mem*, size_t * *lenmem*)

kiss_fft_alloc

Initialize a FFT (or IFFT) algorithm's cfg/state buffer.

typical usage: kiss_fft_cfg mycfg=kiss_fft_alloc(1024,0,NULL,NULL);

The return value from fft_alloc is a cfg buffer used internally by the fft routine or NULL.

If lenmem is NULL, then kiss_fft_alloc will allocate a cfg buffer using malloc. The returned value should be free()d when done to avoid memory leaks.

The state can be placed in a user supplied buffer 'mem': If lenmem is not NULL and mem is not NULL and *lenmem is large enough, then the function places the cfg in mem and the size used in *lenmem and returns mem.

If lenmem is not NULL and (mem is NULL or *lenmem is not large enough), then the function returns NULL and places the minimum cfg buffer size in *lenmem.

Definition at line 615 of file kiss_fft.c.

References kiss_fft_state::bitrev, DIV32, kiss_fft_state::factors, kf_cexp, kf_cexp2, KISS_FFT_MALLOC, kiss_fft_state::nfft, kiss_fft_state::scale, SHL32, and kiss_fft_state::twiddles.

6.45.1.5 void kiss_fft_stride (kiss_fft_cfg *cfg*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*, int *fin_stride*)

A more generic version of the above function. It reads its input from every Nth sample.

Definition at line 658 of file kiss_fft.c.

References kiss_fft_state::bitrev, celt_fatal, kiss_fft_state::factors, kiss_fft_cpx::i, kf_work, kiss_fft_state::nfft, kiss_fft_cpx::r, and kiss_fft_state::scale.

6.45.1.6 void kiss_ifft (kiss_fft_cfg *cfg*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*)

Definition at line 697 of file kiss_fft.c.

References kiss_ifft_stride.

6.45.1.7 void kiss_ifft_stride (kiss_fft_cfg *st*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*, int *in_stride*)

Definition at line 683 of file kiss_fft.c.

References kiss_fft_state::bitrev, celt_fatal, kiss_fft_state::factors, ki_work, and kiss_fft_state::nfft.

6.46 libcelt/kiss_fft.h File Reference

```
#include <stdlib.h>
#include <math.h>
#include "arch.h"
```

Data Structures

- struct kiss_fft_cpx
- struct kiss_twiddle_cpx

Defines

- #define KISS_FFT_MALLOC celt_alloc
- #define kiss_fft_scalar float
- #define kiss_twiddle_scalar float
- #define KF_SUFFIX _celt_single
- #define CAT_SUFFIX(a, b) a ## b
- #define SUF(a, b) CAT_SUFFIX(a, b)
- #define kiss_fft_alloc SUF(kiss_fft_alloc,KF_SUFFIX)
- #define kf_work SUF(kf_work,KF_SUFFIX)
- #define ki_work SUF(ki_work,KF_SUFFIX)
- #define kiss_fft SUF(kiss_fft,KF_SUFFIX)
- #define kiss_ifft SUF(kiss_ifft,KF_SUFFIX)
- #define kiss_fft_stride SUF(kiss_fft_stride,KF_SUFFIX)
- #define kiss_ifft_stride SUF(kiss_ifft_stride,KF_SUFFIX)
- #define kiss_fft_free celt_free

Typedefs

- typedef struct kiss_fft_state * kiss_fft_cfg

Functions

- kiss_fft_cfg kiss_fft_alloc (int nfft, void *mem, size_t *lenmem)
- void kf_work (kiss_fft_cpx *Fout, const kiss_fft_cpx *f, const size_t fstride, int in_stride, int *factors, const kiss_fft_cfg st, int N, int s2, int m2)
- void ki_work (kiss_fft_cpx *Fout, const kiss_fft_cpx *f, const size_t fstride, int in_stride, int *factors, const kiss_fft_cfg st, int N, int s2, int m2)
- void kiss_fft (kiss_fft_cfg cfg, const kiss_fft_cpx *fin, kiss_fft_cpx *fout)
- void kiss_ifft (kiss_fft_cfg cfg, const kiss_fft_cpx *fin, kiss_fft_cpx *fout)
- void kiss_fft_stride (kiss_fft_cfg cfg, const kiss_fft_cpx *fin, kiss_fft_cpx *fout, int fin_stride)
- void kiss_ifft_stride (kiss_fft_cfg cfg, const kiss_fft_cpx *fin, kiss_fft_cpx *fout, int fin_stride)

6.46.1 Define Documentation

6.46.1.1 #define CAT_SUFFIX(a, b) a ## b

Definition at line 57 of file kiss_fft.h.

6.46.1.2 #define KF_SUFFIX _celt_single

Definition at line 50 of file kiss_fft.h.

6.46.1.3 #define kf_work SUF(kf_work,KF_SUFFIX)

Definition at line 61 of file kiss_fft.h.

Referenced by kf_work(), kiss_fft_stride(), and kiss_fftr_inplace().

6.46.1.4 #define ki_work SUF(ki_work,KF_SUFFIX)

Definition at line 62 of file kiss_fft.h.

Referenced by ki_work(), kiss_fftri(), and kiss_ifft_stride().

6.46.1.5 #define kiss_fft SUF(kiss_fft,KF_SUFFIX)

Definition at line 63 of file kiss_fft.h.

Referenced by kiss_fftr().

6.46.1.6 #define kiss_fft_alloc SUF(kiss_fft_alloc,KF_SUFFIX)

Definition at line 60 of file kiss_fft.h.

Referenced by kiss_fftr_alloc().

6.46.1.7 #define kiss_fft_free celt_free

If kiss_fft_alloc allocated a buffer, it is one contiguous buffer and can be simply free()d when no longer needed

Definition at line 134 of file kiss_fft.h.

6.46.1.8 #define KISS_FFT_MALLOC celt_alloc

Definition at line 30 of file kiss_fft.h.

Referenced by kiss_fft_alloc(), and kiss_fftr_alloc().

6.46.1.9 #define kiss_fft_scalar float

Definition at line 48 of file kiss_fft.h.

Referenced by mdct_backward(), and mdct_forward().

6.46.1.10 #define kiss_fft_stride SUF(kiss_fft_stride,KF_SUFFIX)

Definition at line 65 of file kiss_fft.h.

Referenced by kiss_fft().

6.46.1.11 #define kiss_ifft SUF(kiss_ifft,KF_SUFFIX)

Definition at line 64 of file kiss_fft.h.

6.46.1.12 #define kiss_ifft_stride SUF(kiss_ifft_stride,KF_SUFFIX)

Definition at line 66 of file kiss_fft.h.

Referenced by kiss_ifft().

6.46.1.13 #define kiss_twiddle_scalar float

Definition at line 49 of file kiss_fft.h.

Referenced by mdct_init().

6.46.1.14 #define SUF(a, b) CAT_SUFFIX(a, b)

Definition at line 58 of file kiss_fft.h.

6.46.2 Typedef Documentation

6.46.2.1 typedef struct kiss_fft_state* kiss_fft_cfg

Definition at line 79 of file kiss_fft.h.

6.46.3 Function Documentation

6.46.3.1 void kf_work (kiss_fft_cpx * *Fout*, const kiss_fft_cpx * *f*, const size_t *fstride*, int *in_stride*, int * *factors*, const kiss_fft_cfg *st*, int *N*, int *s2*, int *m2*)

Definition at line 512 of file kiss_fft.c.

References celt_fatal, and kf_work.

6.46.3.2 void ki_work (kiss_fft_cpx * *Fout*, const kiss_fft_cpx * *f*, const size_t *fstride*, int *in_stride*, int * *factors*, const kiss_fft_cfg *st*, int *N*, int *s2*, int *m2*)

Internal function. Can be useful when you want to do the bit-reversing yourself

Definition at line 548 of file kiss_fft.c.

References celt_fatal, and ki_work.

6.46.3.3 void kiss_fft (kiss_fft_cfg *cfg*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*)

kiss_fft(cfg,in_out_buf)

Perform an FFT on a complex input buffer. for a forward FFT, fin should be f[0] , f[1] , ... ,f[nfft-1] fout will be F[0] , F[1] , ... ,F[nfft-1] Note that each element is complex and can be accessed like f[k].r and f[k].i

Definition at line 678 of file kiss_fft.c.

References kiss_fft_stride.

6.46.3.4 kiss_fft_cfg kiss_fft_alloc (int *nfft*, void * *mem*, size_t * *lenmem*)

kiss_fft_alloc

Initialize a FFT (or IFFT) algorithm's cfg/state buffer.

typical usage: kiss_fft_cfg mycfg=kiss_fft_alloc(1024,0,NULL,NULL);

The return value from fft_alloc is a cfg buffer used internally by the fft routine or NULL.

If lenmem is NULL, then kiss_fft_alloc will allocate a cfg buffer using malloc. The returned value should be free()d when done to avoid memory leaks.

The state can be placed in a user supplied buffer 'mem': If lenmem is not NULL and mem is not NULL and *lenmem is large enough, then the function places the cfg in mem and the size used in *lenmem and returns mem.

If lenmem is not NULL and (mem is NULL or *lenmem is not large enough), then the function returns NULL and places the minimum cfg buffer size in *lenmem.

Definition at line 615 of file kiss_fft.c.

References kiss_fft_state::bitrev, DIV32, kiss_fft_state::factors, kf_cexp, kf_cexp2, KISS_FFT_MALLOC, kiss_fft_state::nfft, kiss_fft_state::scale, SHL32, and kiss_fft_state::twiddles.

6.46.3.5 void kiss_fft_stride (kiss_fft_cfg *cfg*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*, int *fin_stride*)

A more generic version of the above function. It reads its input from every Nth sample.

Definition at line 658 of file kiss_fft.c.

References kiss_fft_state::bitrev, celt_fatal, kiss_fft_state::factors, kiss_fft_cpx::i, kf_work, kiss_fft_state::nfft, kiss_fft_cpx::r, and kiss_fft_state::scale.

6.46.3.6 void kiss_ifft (kiss_fft_cfg *cfg*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*)

Definition at line 697 of file kiss_fft.c.

References kiss_ifft_stride.

6.46.3.7 void kiss_ifft_stride (kiss_fft_cfg *cfg*, const kiss_fft_cpx * *fin*, kiss_fft_cpx * *fout*, int *fin_stride*)

Definition at line 683 of file kiss_fft.c.

References kiss_fft_state::bitrev, celt_fatal, kiss_fft_state::factors, ki_work, and kiss_fft_state::nfft.

6.47 libcelt/kiss_fftr.c File Reference

```
#include "os_support.h"
#include "mathops.h"
#include "kiss_fftr.h"
#include "_kiss_fft_guts.h"
```

Functions

- kiss_fftr_cfg kiss_fftr_alloc (int nfft, void *mem, size_t *lenmem)
- void kiss_fftr_twiddles (kiss_fftr_cfg st, kiss_fft_scalar *freqdata)
- void kiss_fftr (kiss_fftr_cfg st, const kiss_fft_scalar *timedata, kiss_fft_scalar *freqdata)
- void kiss_fftr_inplace (kiss_fftr_cfg st, kiss_fft_scalar *X)
- void kiss_fftri (kiss_fftr_cfg st, const kiss_fft_scalar *freqdata, kiss_fft_scalar *timedata)

6.47.1 Function Documentation

6.47.1.1 void kiss_fftr (kiss_fftr_cfg *st*, const kiss_fft_scalar * *timedata*, kiss_fft_scalar * *freqdata*)

Definition at line 123 of file kiss_fftr.c.

References kiss_fft, kiss_fftr_twiddles, and kiss_fftr_state::substate.

6.47.1.2 kiss_fftr_cfg kiss_fftr_alloc (int *nfft*, void * *mem*, size_t * *lenmem*)

Definition at line 31 of file kiss_fftr.c.

References DIV32, kf_cexp, kf_cexp2, kiss_fft_alloc, KISS_FFT_MALLOC, kiss_fft_state::scale, SHL32, kiss_fftr_state::substate, and kiss_fftr_state::super_twiddles.

6.47.1.3 void kiss_fftr_inplace (kiss_fftr_cfg *st*, kiss_fft_scalar * *X*)

Definition at line 131 of file kiss_fftr.c.

References kiss_fft_state::factors, kf_work, kiss_fftr_twiddles, and kiss_fftr_state::substate.

6.47.1.4 void kiss_fftr_twiddles (kiss_fftr_cfg *st*, kiss_fft_scalar * *freqdata*)

Definition at line 79 of file kiss_fftr.c.

References ADD32, C_FIXDIV, C_MULC, CHECK_OVERFLOW_OP, EXT32, HALF_OF, kiss_fft_cpx::i, k, kiss_fft_state::nfft, PSHR32, kiss_fft_cpx::r, SHR32, SUB32, kiss_fftr_state::substate, and kiss_fftr_state::super_twiddles.

6.47.1.5 void kiss_fftri (kiss_fftr_cfg *st*, const kiss_fft_scalar * *freqdata*, kiss_fft_scalar * *timedata*)

Definition at line 137 of file kiss_fftr.c.

References kiss_fft_state::bitrev, C_ADD, C_MUL, C_SUB, kiss_fft_state::factors, kiss_fft_cpx::i, k, ki_work, kiss_fft_state::nfft, kiss_fft_cpx::r, kiss_fftr_state::substate, and kiss_fftr_state::super_twiddles.

6.48 libcelt/kiss_fftr.h File Reference

```
#include "kiss_fft.h"
```

Data Structures

- struct kiss_fftr_state

Defines

- #define kiss_fftr_alloc SUF(kiss_fftr_alloc,KF_SUFFIX)
- #define kiss_fftr_inplace SUF(kiss_fftr_inplace,KF_SUFFIX)
- #define kiss_fftr_alloc SUF(kiss_fftr_alloc,KF_SUFFIX)
- #define kiss_fftr_twiddles SUF(kiss_fftr_twiddles,KF_SUFFIX)
- #define kiss_fftr SUF(kiss_fftr,KF_SUFFIX)
- #define kiss_fftri SUF(kiss_fftri,KF_SUFFIX)
- #define kiss_fftr_free speex_free

Typedefs

- typedef struct kiss_fftr_state * kiss_fftr_cfg

Functions

- kiss_fftr_cfg kiss_fftr_alloc (int nfft, void *mem, size_t *lenmem)
- void kiss_fftr_twiddles (kiss_fftr_cfg st, kiss_fft_scalar *freqdata)
- void kiss_fftr (kiss_fftr_cfg st, const kiss_fft_scalar *timedata, kiss_fft_scalar *freqdata)
- void kiss_fftr_inplace (kiss_fftr_cfg st, kiss_fft_scalar *X)
- void kiss_fftri (kiss_fftr_cfg st, const kiss_fft_scalar *freqdata, kiss_fft_scalar *timedata)

6.48.1 Define Documentation

6.48.1.1 #define kiss_fftr SUF(kiss_fftr,KF_SUFFIX)

Definition at line 13 of file kiss_fftr.h.

6.48.1.2 #define kiss_fftr_alloc SUF(kiss_fftr_alloc,KF_SUFFIX)

Definition at line 11 of file kiss_fftr.h.

6.48.1.3 #define kiss_fftr_alloc SUF(kiss_fftr_alloc,KF_SUFFIX)

Definition at line 11 of file kiss_fftr.h.

6.48.1.4 #define kiss_fftr_free speex_free

Definition at line 59 of file kiss_fftr.h.

6.48.1.5 #define kiss_fftr_inplace SUF(kiss_fftr_inplace,KF_SUFFIX)

Definition at line 10 of file kiss_fftr.h.

6.48.1.6 #define kiss_fftr_twiddles SUF(kiss_fftr_twiddles,KF_SUFFIX)

Definition at line 12 of file kiss_fftr.h.

Referenced by kiss_fftr(), and kiss_fftr_inplace().

6.48.1.7 #define kiss_fftri SUF(kiss_fftri,KF_SUFFIX)

Definition at line 14 of file kiss_fftr.h.

6.48.2 Typedef Documentation

6.48.2.1 typedef struct kiss_fftr_state* kiss_fftr_cfg

Definition at line 32 of file kiss_fftr.h.

6.48.3 Function Documentation

6.48.3.1 void kiss_fftr (kiss_fftr_cfg *st*, const kiss_fft_scalar * *timedata*, kiss_fft_scalar * *freqdata*)

Definition at line 123 of file kiss_fftr.c.

References kiss_fft, kiss_fftr_twiddles, and kiss_fftr_state::substate.

6.48.3.2 kiss_fftr_cfg kiss_fftr_alloc (int *nfft*, void * *mem*, size_t * *lenmem*)

Definition at line 31 of file kiss_fftr.c.

References DIV32, kf_cexp, kf_cexp2, kiss_fft_alloc, KISS_FFT_MALLOC, kiss_fft_state::scale, SHL32, kiss_fftr_state::substate, and kiss_fftr_state::super_twiddles.

6.48.3.3 void kiss_fftr_inplace (kiss_fftr_cfg *st*, kiss_fft_scalar * *X*)

Definition at line 131 of file kiss_fftr.c.

References kiss_fft_state::factors, kf_work, kiss_fftr_twiddles, and kiss_fftr_state::substate.

6.48.3.4 void kiss_fftr_twiddles (kiss_fftr_cfg *st*, kiss_fft_scalar * *freqdata*)

Definition at line 79 of file kiss_fftr.c.

References ADD32, C_FIXDIV, C_MULC, CHECK_OVERFLOW_OP, EXT32, HALF_OF, kiss_fft_cpx::i, k, kiss_fft_state::nfft, PSHR32, kiss_fft_cpx::r, SHR32, SUB32, kiss_fftr_state::substate, and kiss_fftr_state::super_twiddles.

6.48.3.5 void kiss_fftri (kiss_fftr_cfg *st*, const kiss_fft_scalar * *freqdata*, kiss_fft_scalar * *timedata*)

Definition at line 137 of file kiss_fftr.c.

References kiss_fft_state::bitrev, C_ADD, C_MUL, C_SUB, kiss_fft_state::factors, kiss_fft_cpx::i, k, ki_work, kiss_fft_state::nfft, kiss_fft_cpx::r, kiss_fftr_state::substate, and kiss_fftr_state::super_twiddles.

6.49 libcelt/laplace.c File Reference

```
#include "laplace.h"
```

Functions

- int ec_laplace_get_start_freq (int decay)
- void ec_laplace_encode_start (ec_enc *enc, int *value, int decay, int fs)
- void ec_laplace_encode (ec_enc *enc, int *value, int decay)
- int ec_laplace_decode_start (ec_dec *dec, int decay, int fs)
- int ec_laplace_decode (ec_dec *dec, int decay)

6.49.1 Function Documentation

6.49.1.1 int ec_laplace_decode (ec_dec * *dec*, int *decay*)

Decode a value that is assumed to be the realisation of a Laplace-distributed random process

Parameters:

dec Entropy decoder state

decay Probability of the value +/- 1, multiplied by 16384

Returns:

Value decoded

Definition at line 126 of file laplace.c.

References ec_laplace_decode_start(), and ec_laplace_get_start_freq().

6.49.1.2 int ec_laplace_decode_start (ec_dec * *dec*, int *decay*, int *fs*)

Definition at line 90 of file laplace.c.

References ec_dec_update(), and ec_decode().

Referenced by ec_laplace_decode().

6.49.1.3 void ec_laplace_encode (ec_enc * *enc*, int * *value*, int *decay*)

Encode a value that is assumed to be the realisation of a Laplace-distributed random process

Parameters:

enc Entropy encoder state

value Value to encode

decay Probability of the value +/- 1, multiplied by 16384

Definition at line 83 of file laplace.c.

References ec_laplace_encode_start(), and ec_laplace_get_start_freq().

6.49.1.4 void ec_laplace_encode_start (ec_enc * *enc*, int * *value*, int *decay*, int *fs*)

Definition at line 43 of file laplace.c.

References ec_encode().

Referenced by ec_laplace_encode().

6.49.1.5 int ec_laplace_get_start_freq (int *decay*)

Definition at line 38 of file laplace.c.

Referenced by ec_laplace_decode(), ec_laplace_encode(), and quant_prob_alloc().

6.50 libcelt/laplace.h File Reference

```
#include "entenc.h"
#include "entdec.h"
```

Functions

- int ec_laplace_get_start_freq (int decay)
- void ec_laplace_encode (ec_enc *enc, int *value, int decay)
- void ec_laplace_encode_start (ec_enc *enc, int *value, int decay, int fs)
- int ec_laplace_decode (ec_dec *dec, int decay)
- int ec_laplace_decode_start (ec_dec *dec, int decay, int fs)

6.50.1 Function Documentation

6.50.1.1 int ec_laplace_decode (ec_dec * *dec*, int *decay*)

Decode a value that is assumed to be the realisation of a Laplace-distributed random process

Parameters:

dec Entropy decoder state

decay Probability of the value +/- 1, multiplied by 16384

Returns:

Value decoded

Definition at line 126 of file laplace.c.

References ec_laplace_decode_start(), and ec_laplace_get_start_freq().

6.50.1.2 int ec_laplace_decode_start (ec_dec * *dec*, int *decay*, int *fs*)

Definition at line 90 of file laplace.c.

References ec_dec_update(), and ec_decode().

Referenced by ec_laplace_decode().

6.50.1.3 void ec_laplace_encode (ec_enc * *enc*, int * *value*, int *decay*)

Encode a value that is assumed to be the realisation of a Laplace-distributed random process

Parameters:

enc Entropy encoder state

value Value to encode

decay Probability of the value +/- 1, multiplied by 16384

Definition at line 83 of file laplace.c.

References ec_laplace_encode_start(), and ec_laplace_get_start_freq().

6.50.1.4 void ec_laplace_encode_start (ec_enc * *enc*, int * *value*, int *decay*, int *fs*)

Definition at line 43 of file laplace.c.

References ec_encode().

Referenced by ec_laplace_encode().

6.50.1.5 int ec_laplace_get_start_freq (int *decay*)

Definition at line 38 of file laplace.c.

Referenced by ec_laplace_decode(), ec_laplace_encode(), and quant_prob_alloc().

6.51 libcelt/mathops.h File Reference

Various math functions.

```
#include "arch.h"
#include "entcode.h"
#include "os_support.h"
```

Defines

- #define celt_sqrt(x) ((float)sqrt(x))
- #define celt_psqrt(x) ((float)sqrt(x))
- #define celt_rsqrt(x) (1.f/celt_sqrt(x))
- #define celt_acos acos
- #define celt_exp exp
- #define celt_cos_norm(x) (cos((.5f∗M_PI)∗(x)))
- #define celt_atan atan
- #define celt_rcp(x) (1.f/(x))
- #define celt_div(a, b) ((a)/(b))

6.51.1 Detailed Description

Various math functions.

Definition in file mathops.h.

6.51.2 Define Documentation

6.51.2.1 #define celt_acos acos

Definition at line 91 of file mathops.h.

6.51.2.2 #define celt_atan atan

Definition at line 94 of file mathops.h.

6.51.2.3 #define celt_cos_norm(x) (cos((.5f∗M_PI)∗(x)))

Definition at line 93 of file mathops.h.

Referenced by mdct_init().

6.51.2.4 #define celt_div(a, b) ((a)/(b))

Definition at line 96 of file mathops.h.

Referenced by compute_pitch_gain(), intra_prediction(), and intra_unquant().

6.51.2.5 #define celt_exp exp

Definition at line 92 of file mathops.h.

6.51.2.6 #define celt_psqrt(x) ((float)sqrt(x))

Definition at line 89 of file mathops.h.

Referenced by alg_quant().

6.51.2.7 #define celt_rcp(x) (1.f/(x))

Definition at line 95 of file mathops.h.

Referenced by celt_encode_float(), and renormalise_vector().

6.51.2.8 #define celt_rsqrt(x) (1.f/celt_sqrt(x))

Definition at line 90 of file mathops.h.

Referenced by find_spectral_pitch().

6.51.2.9 #define celt_sqrt(x) ((float)sqrt(x))

Definition at line 88 of file mathops.h.

Referenced by quant_bands(), quant_pitch(), renormalise_vector(), stereo_mix(), and unquant_bands().

6.52 libcelt/mdct-pc.c File Reference

```
#include "mdct.h"
#include "kfft_double.h"
#include <math.h>
#include "os_support.h"
#include "mathops.h"
#include "stack_alloc.h"
```

Defines

- #define M_PI 3.141592653

Functions

- void mdct_init (mdct_lookup *l, int N)
- void mdct_clear (mdct_lookup *l)
- void mdct_forward (const mdct_lookup *l, kiss_fft_scalar *in, kiss_fft_scalar *restrict out)
- void mdct_backward (const mdct_lookup *l, kiss_fft_scalar *in, kiss_fft_scalar *restrict out)

6.52.1 Define Documentation

6.52.1.1 #define M_PI 3.141592653

Definition at line 57 of file mdct-pc.c.

Referenced by celt_mode_create(), and mdct_init().

6.52.2 Function Documentation

6.52.2.1 void mdct_backward (const mdct_lookup * *l*, kiss_fft_scalar * *in*, kiss_fft_scalar *restrict *out*)

Definition at line 163 of file mdct-pc.c.

References ALLOC, cpx32_ifft, mdct_lookup::kfft, kiss_fft_scalar, mdct_lookup::n, RESTORE_STACK, S_MUL, SAVE_STACK, mdct_lookup::trig, and VARDECL.

6.52.2.2 void mdct_clear (mdct_lookup * *l*)

Definition at line 83 of file mdct-pc.c.

Referenced by celt_mode_destroy().

6.52.2.3 void mdct_forward (const mdct_lookup * *l*, kiss_fft_scalar * *in*, kiss_fft_scalar *restrict *out*)

Definition at line 89 of file mdct-pc.c.

References ALLOC, cpx32_fft, HALF32, mdct_lookup::kfft, kiss_fft_scalar, mdct_lookup::n, RESTORE_STACK, S_MUL, SAVE_STACK, mdct_lookup::trig, and VARDECL.

6.52.2.4 void mdct_init (mdct_lookup * *l*, int *N*)

Definition at line 60 of file mdct-pc.c.

Referenced by celt_mode_create().

6.53 libcelt/mdct.c File Reference

```
#include "mdct.h"
#include "kfft_double.h"
#include <math.h>
#include "os_support.h"
#include "mathops.h"
#include "stack_alloc.h"
```

Defines

- #define M_PI 3.141592653

Functions

- void mdct_init (mdct_lookup *l, int N)
- void mdct_clear (mdct_lookup *l)
- void mdct_forward (const mdct_lookup *l, kiss_fft_scalar *in, kiss_fft_scalar *restrict out, const celt_word16_t *window, int overlap)
- void mdct_backward (const mdct_lookup *l, kiss_fft_scalar *in, kiss_fft_scalar *restrict out, const celt_word16_t *restrict window, int overlap)

6.53.1 Define Documentation

6.53.1.1 #define M_PI 3.141592653

Definition at line 57 of file mdct.c.

6.53.2 Function Documentation

6.53.2.1 void mdct_backward (const mdct_lookup * *l*, kiss_fft_scalar * *in*, kiss_fft_scalar *restrict *out*, const celt_word16_t *restrict *window*, int *overlap*)

Definition at line 180 of file mdct.c.

References ALLOC, cpx32_ifft, mdct_lookup::kfft, kiss_fft_scalar, MULT16_32_Q15, mdct_lookup::n, RESTORE_STACK, S_MUL, SAVE_STACK, mdct_lookup::trig, and VARDECL.

6.53.2.2 void mdct_clear (mdct_lookup * *l*)

Definition at line 83 of file mdct.c.

References cpx32_fft_free, mdct_lookup::kfft, and mdct_lookup::trig.

6.53.2.3 void mdct_forward (const mdct_lookup * *l*, kiss_fft_scalar * *in*, kiss_fft_scalar *restrict *out*, const celt_word16_t * *window*, int *overlap*)

Definition at line 89 of file mdct.c.

References ALLOC, cpx32_fft, mdct_lookup::kfft, kiss_fft_scalar, MULT16_32_Q15, mdct_lookup::n, RESTORE_STACK, S_MUL, SAVE_STACK, mdct_lookup::trig, and VARDECL.

6.53.2.4 void mdct_init (mdct_lookup * *l*, int *N*)

Definition at line 60 of file mdct.c.

References ADD32, celt_cos_norm, cpx32_fft_alloc, DIV32, EXTEND32, mdct_lookup::kfft, kiss_twiddle_scalar, M_PI, mdct_lookup::n, SHL32, and mdct_lookup::trig.

6.54 libcelt/mdct.h File Reference

```
#include "kiss_fft.h"
#include "arch.h"
```

Data Structures

- struct mdct_lookup

Functions

- void mdct_init (mdct_lookup *l, int N)
- void mdct_clear (mdct_lookup *l)
- void mdct_forward (const mdct_lookup *l, kiss_fft_scalar *in, kiss_fft_scalar *out, const celt_word16_t *window, int overlap)
- void mdct_backward (const mdct_lookup *l, kiss_fft_scalar *in, kiss_fft_scalar *out, const celt_word16_t *restrict window, int overlap)

6.54.1 Function Documentation

6.54.1.1 void mdct_backward (const mdct_lookup * *l*, kiss_fft_scalar * *in*, kiss_fft_scalar * *out*, const celt_word16_t *restrict *window*, int *overlap*)

Compute a backward MDCT (no scaling) and performs weighted overlap-add (scales implicitly by 1/2)

6.54.1.2 void mdct_clear (mdct_lookup * *l*)

Definition at line 83 of file mdct-pc.c.

References cpx32_fft_free, mdct_lookup::kfft, and mdct_lookup::trig.

Referenced by celt_mode_destroy().

6.54.1.3 void mdct_forward (const mdct_lookup * *l*, kiss_fft_scalar * *in*, kiss_fft_scalar * *out*, const celt_word16_t * *window*, int *overlap*)

Compute a forward MDCT and scale by 4/N

6.54.1.4 void mdct_init (mdct_lookup * *l*, int *N*)

Definition at line 60 of file mdct-pc.c.

References ADD32, celt_cos_norm, cpx32_fft_alloc, DIV32, EXTEND32, mdct_lookup::kfft, kiss_twiddle_scalar, M_PI, mdct_lookup::n, SHL32, and mdct_lookup::trig.

Referenced by celt_mode_create().

6.55 libcelt/mfrngcod.h File Reference

```
#include "entcode.h"
```

Defines

- #define _mfrngcode_H (1)
- #define EC_SYM_BITS (8)
- #define EC_CODE_BITS (32)
- #define EC_SYM_MAX ((1U<<EC_SYM_BITS)-1)
- #define EC_CODE_SHIFT (EC_CODE_BITS-EC_SYM_BITS-1)
- #define EC_CODE_TOP (((ec_uint32)1U)<<EC_CODE_BITS-1)
- #define EC_CODE_BOT (EC_CODE_TOP>>EC_SYM_BITS)
- #define EC_CODE_CARRY (((ec_uint32)EC_SYM_MAX)<<EC_CODE_SHIFT)
- #define EC_CODE_EXTRA ((EC_CODE_BITS-2)%EC_SYM_BITS+1)
- #define EC_CODE_MASK ((((ec_uint32)1U)<<EC_CODE_BITS-1)-1<<1|1)
- #define EC_FOF_RSV1 (1<<EC_SYM_BITS-1)
- #define EC_FOF_RSV1_MASK (EC_FOF_RSV1-1)

6.55.1 Define Documentation

6.55.1.1 #define _mfrngcode_H (1)

Definition at line 2 of file mfrngcod.h.

6.55.1.2 #define EC_CODE_BITS (32)

Definition at line 10 of file mfrngcod.h.

Referenced by ec_dec_tell(), and ec_enc_tell().

6.55.1.3 #define EC_CODE_BOT (EC_CODE_TOP>>EC_SYM_BITS)

Definition at line 18 of file mfrngcod.h.

6.55.1.4 #define EC_CODE_CARRY (((ec_uint32)EC_SYM_MAX)<<EC_CODE_SHIFT)

Definition at line 20 of file mfrngcod.h.

6.55.1.5 #define EC_CODE_EXTRA ((EC_CODE_BITS-2)%EC_SYM_BITS+1)

Definition at line 22 of file mfrngcod.h.

Referenced by ec_dec_init().

6.55.1.6 #define EC_CODE_MASK ((((ec_uint32)1U)<<EC_CODE_BITS-1)-1<<1|1)

Definition at line 27 of file mfrngcod.h.

6.55.1.7 #define EC_CODE_SHIFT (EC_CODE_BITS-EC_SYM_BITS-1)

Definition at line 14 of file mfrngcod.h.

Referenced by ec_enc_done().

6.55.1.8 #define EC_CODE_TOP (((ec_uint32)1U)<<EC_CODE_BITS-1)

Definition at line 16 of file mfrngcod.h.

Referenced by ec_enc_done(), and ec_enc_init().

6.55.1.9 #define EC_FOF_RSV1 (1<<EC_SYM_BITS-1)

Definition at line 32 of file mfrngcod.h.

6.55.1.10 #define EC_FOF_RSV1_MASK (EC_FOF_RSV1-1)

Definition at line 34 of file mfrngcod.h.

6.55.1.11 #define EC_SYM_BITS (8)

Definition at line 8 of file mfrngcod.h.

Referenced by ec_dec_init(), and ec_dec_tell().

6.55.1.12 #define EC_SYM_MAX ((1U<<EC_SYM_BITS)-1)

Definition at line 12 of file mfrngcod.h.

6.56 libcelt/modes.c File Reference

```
#include "celt.h"
#include "modes.h"
#include "rate.h"
#include "os_support.h"
#include "stack_alloc.h"
#include "quant_bands.h"
```

Defines

- #define MODEVALID 0xa110ca7e
- #define MODEFREED 0xb10cf8ee
- #define M_PI 3.141592653
- #define PBANDS 8
- #define MIN_BINS 3
- #define BARK_BANDS 25
- #define BITALLOC_SIZE 12

Functions

- int celt_mode_info (const CELTMode *mode, int request, celt_int32_t *value)
- CELTMode * celt_mode_create (celt_int32_t Fs, int channels, int frame_size, int *error)
- void celt_mode_destroy (CELTMode *mode)
- int check_mode (const CELTMode *mode)

6.56.1 Define Documentation

6.56.1.1 #define BARK_BANDS 25

Definition at line 89 of file modes.c.

6.56.1.2 #define BITALLOC_SIZE 12

Definition at line 107 of file modes.c.

6.56.1.3 #define M_PI 3.141592653

Definition at line 51 of file modes.c.

6.56.1.4 #define MIN_BINS 3

Definition at line 84 of file modes.c.

Referenced by celt_mode_create().

6.56.1.5 #define MODEFREED 0xb10cf8ee

Definition at line 48 of file modes.c.

Referenced by celt_mode_destroy(), and check_mode().

6.56.1.6 #define MODEVALID 0xa110ca7e

Definition at line 47 of file modes.c.

Referenced by celt_mode_create(), and check_mode().

6.56.1.7 #define PBANDS 8

Definition at line 79 of file modes.c.

6.56.2 Function Documentation

6.56.2.1 int check_mode (const CELTMode * *mode*)

Definition at line 466 of file modes.c.

References CELT_INVALID_MODE, CELT_OK, CELTMode::marker_end, CELTMode::marker_start, MODEFREED, and MODEVALID.

Referenced by celt_decode(), celt_decode_float(), celt_decoder_create(), celt_decoder_destroy(), celt_encode(), celt_encode_float(), celt_encoder_create(), celt_encoder_destroy(), and celt_mode_destroy().

6.57 libcelt/modes.h File Reference

```
#include "celt_types.h"
#include "celt.h"
#include "arch.h"
#include "mdct.h"
#include "psy.h"
#include "pitch.h"
```

Data Structures

- struct CELTMode
  Mode definition.

Defines

- #define CELT_BITSTREAM_VERSION 0x80000005
- #define MAX_PERIOD 1024
- #define CHANNELS(mode) ((mode) → nbChannels)
- #define MDCT(mode) (&(mode) → mdct)
- #define OVERLAP(mode) ((mode) → overlap)
- #define FRAMESIZE(mode) ((mode) → mdctSize)

Functions

- int check_mode (const CELTMode *mode)

6.57.1 Define Documentation

6.57.1.1 #define CELT_BITSTREAM_VERSION 0x80000005

Definition at line 42 of file modes.h.

Referenced by celt_mode_info().

6.57.1.2 #define CHANNELS(mode) ((mode) → nbChannels)

Definition at line 54 of file modes.h.

Referenced by celt_decode(), celt_decode_float(), celt_decoder_create(), celt_encode(), celt_encode_float(), compute_allocation(), compute_band_energies(), compute_pitch_gain(), denormalise_bands(), find_spectral_pitch(), intra_fold(), intra_prediction(), intra_unquant(), normalise_bands(), pitch_quant_bands(), quant_bands(), renormalise_bands(), stereo_mix(), and unquant_bands().

6.57.1.3 #define FRAMESIZE(mode) ((mode) → mdctSize)

Definition at line 65 of file modes.h.

6.57.1.4 #define MAX_PERIOD 1024

Definition at line 48 of file modes.h.

Referenced by celt_decode(), celt_decode_float(), celt_decoder_create(), celt_encode(), celt_encode_float(), celt_encoder_create(), celt_mode_create(), do_plc(), dump_modes(), and find_spectral_pitch().

6.57.1.5 #define MDCT(mode) (&(mode) → mdct)

Definition at line 58 of file modes.h.

6.57.1.6 #define OVERLAP(mode) ((mode) → overlap)

Definition at line 61 of file modes.h.

Referenced by find_spectral_pitch().

6.57.2 Function Documentation

6.57.2.1 int check_mode (const CELTMode ∗ *mode*)

Definition at line 466 of file modes.c.

References CELT_INVALID_MODE, CELT_OK, CELTMode::marker_end, CELTMode::marker_start, MODEFREED, and MODEVALID.

Referenced by celt_decode(), celt_decode_float(), celt_decoder_create(), celt_decoder_destroy(), celt_encode(), celt_encode_float(), celt_encoder_create(), celt_encoder_destroy(), and celt_mode_destroy().

6.58 libcelt/os_support.h File Reference

```
#include <string.h>
#include <stdio.h>
#include <stdlib.h>
```

Defines

- #define CELT_COPY(dst, src, n) (memcpy((dst), (src), (n)∗sizeof(∗(dst)) + 0∗((dst)-(src))))
- #define CELT_MOVE(dst, src, n) (memmove((dst), (src), (n)∗sizeof(∗(dst)) + 0∗((dst)-(src))))
- #define CELT_MEMSET(dst, c, n) (memset((dst), (c), (n)∗sizeof(∗(dst))))

6.58.1 Define Documentation

6.58.1.1 #define CELT_COPY(dst, src, n) (memcpy((dst), (src), (n)∗sizeof(∗(dst)) + 0∗((dst)-(src))))

Copy n bytes of memory from src to dst. The 0∗ term provides compile-time type checking

Definition at line 92 of file os_support.h.

Referenced by celt_encode(), celt_encode_float(), celt_header_from_packet(), celt_header_init(), celt_header_to_packet(), and celt_mode_create().

6.58.1.2 #define CELT_MEMSET(dst, c, n) (memset((dst), (c), (n)∗sizeof(∗(dst))))

Set n bytes of memory to value of c, starting at address s

Definition at line 103 of file os_support.h.

Referenced by celt_header_to_packet(), comb2pulse(), find_spectral_pitch(), ncwrs_u32(), and ncwrs_u64().

6.58.1.3 #define CELT_MOVE(dst, src, n) (memmove((dst), (src), (n)∗sizeof(∗(dst)) + 0∗((dst)-(src))))

Copy n bytes of memory from src to dst, allowing overlapping regions. The 0∗ term provides compile-time type checking

Definition at line 98 of file os_support.h.

Referenced by celt_decode(), celt_decode_float(), celt_encode(), and celt_encode_float().

6.59 libcelt/pgain_table.h File Reference

Variables

- celt_uint16_t pgain_table [512]

6.59.1 Variable Documentation

6.59.1.1 celt_uint16_t pgain_table[512]

Definition at line 3 of file pgain_table.h.

Referenced by quant_pitch().

6.60 libcelt/pitch-pc.c File Reference

```
#include "kfft_single.h"
#include "pitch.h"
#include "psy.h"
#include "os_support.h"
#include "mathops.h"
#include "modes.h"
#include "stack_alloc.h"
```

Defines

- #define normalise16(x, len, val)
- #define INPUT_SHIFT 15

Functions

- kiss_fftr_cfg pitch_state_alloc (int max_lag)
- void pitch_state_free (kiss_fftr_cfg st)
- void find_spectral_pitch (const CELTMode *m, kiss_fftr_cfg fft, const struct PsyDecay *decay, const celt_sig_t *restrict x, const celt_sig_t *restrict y, const celt_word16_t *restrict window, int len, int *pitch)

6.60.1 Define Documentation

6.60.1.1 #define INPUT_SHIFT 15

Definition at line 105 of file pitch-pc.c.

Referenced by find_spectral_pitch().

6.60.1.2 #define normalise16(x, len, val)

Definition at line 102 of file pitch-pc.c.

Referenced by find_spectral_pitch().

6.60.2 Function Documentation

6.60.2.1 void find_spectral_pitch (const CELTMode * *m*, kiss_fftr_cfg *fft*, const struct PsyDecay * *decay*, const celt_sig_t *restrict *x*, const celt_sig_t *restrict *y*, const celt_word16_t *restrict *window*, int *len*, int * *pitch*)

Definition at line 107 of file pitch-pc.c.

References ADD16, ALLOC, BITREV, celt_ilog2, CELT_MEMSET, celt_rsqrt, CHANNELS, compute_masking(), EPSILON, EXTRACT16, INPUT_SHIFT, MAX_PERIOD, MULT16_16, MULT16_16_Q15,

n, normalise16, OVERLAP, real16_fft_inplace, real16_ifft, RESTORE_STACK, SAVE_STACK, SHR32, SUB16, and VARDECL.

Referenced by celt_encode(), celt_encode_float(), and do_plc().

6.60.2.2 kiss_fftr_cfg pitch_state_alloc (int *max_lag*)

Definition at line 53 of file pitch-pc.c.

Referenced by celt_mode_create().

6.60.2.3 void pitch_state_free (kiss_fftr_cfg *st*)

Definition at line 58 of file pitch-pc.c.

Referenced by celt_mode_destroy().

6.61 libcelt/pitch.c File Reference

Pitch analysis.

```
#include "kfft_single.h"
#include "pitch.h"
#include "psy.h"
#include "os_support.h"
#include "mathops.h"
#include "modes.h"
#include "stack_alloc.h"
```

Defines

- #define normalise16(x, len, val)
- #define INPUT_SHIFT 15

Functions

- kiss_fftr_cfg pitch_state_alloc (int max_lag)
- void pitch_state_free (kiss_fftr_cfg st)
- void find_spectral_pitch (const CELTMode *m, kiss_fftr_cfg fft, const struct PsyDecay *decay, const celt_sig_t *restrict x, const celt_sig_t *restrict y, const celt_word16_t *restrict window, int len, int max_pitch, int *pitch)

6.61.1 Detailed Description

Pitch analysis.

Definition in file pitch.c.

6.61.2 Define Documentation

6.61.2.1 #define INPUT_SHIFT 15

Definition at line 105 of file pitch.c.

6.61.2.2 #define normalise16(x, len, val)

Definition at line 102 of file pitch.c.

6.61.3 Function Documentation

6.61.3.1 void find_spectral_pitch (const CELTMode * *m*, kiss_fftr_cfg *fft*, const struct PsyDecay * *decay*, const celt_sig_t *restrict *x*, const celt_sig_t *restrict *y*, const celt_word16_t *restrict *window*, int *len*, int *max_pitch*, int * *pitch*)

Definition at line 107 of file pitch.c.

References ADD16, ALLOC, BITREV, celt_ilog2, CELT_MEMSET, celt_rsqrt, CHANNELS, compute_masking(), EPSILON, EXTRACT16, INPUT_SHIFT, MAX_PERIOD, MULT16_16, MULT16_16_16, MULT16_16_Q15, n, normalise16, OVERLAP, real16_fft_inplace, real16_ifft, RESTORE_STACK, SAVE_STACK, SHR32, SUB16, and VARDECL.

6.61.3.2 kiss_fftr_cfg pitch_state_alloc (int *max_lag*)

Definition at line 53 of file pitch.c.

References real16_fft_alloc.

6.61.3.3 void pitch_state_free (kiss_fftr_cfg *st*)

Definition at line 58 of file pitch.c.

References real16_fft_free.

6.62 libcelt/pitch.h File Reference

Pitch analysis.

```
#include "kiss_fftr.h"
#include "psy.h"
#include "modes.h"
```

Functions

- kiss_fftr_cfg pitch_state_alloc (int max_lag)
- void pitch_state_free (kiss_fftr_cfg st)
- void find_spectral_pitch (const CELTMode *m, kiss_fftr_cfg fft, const struct PsyDecay *decay, const celt_sig_t *x, const celt_sig_t *y, const celt_word16_t *window, int len, int max_pitch, int *pitch)

6.62.1 Detailed Description

Pitch analysis.

Definition in file pitch.h.

6.62.2 Function Documentation

6.62.2.1 void find_spectral_pitch (const CELTMode * *m*, kiss_fftr_cfg *fft*, const struct PsyDecay * *decay*, const celt_sig_t * *x*, const celt_sig_t * *y*, const celt_word16_t * *window*, int *len*, int *max_pitch*, int * *pitch*)

Find the optimal delay for the pitch prediction. Computation is done in the frequency domain, both to save time and to make it easier to apply psychoacoustic weighting

6.62.2.2 kiss_fftr_cfg pitch_state_alloc (int *max_lag*)

Definition at line 53 of file pitch-pc.c.

References real16_fft_alloc.

Referenced by celt_mode_create().

6.62.2.3 void pitch_state_free (kiss_fftr_cfg *st*)

Definition at line 58 of file pitch-pc.c.

References real16_fft_free.

Referenced by celt_mode_destroy().

6.63 libcelt/plc.c File Reference

Defines

- #define LPC_ORDER 12

Functions

- celt_word32_t _celt_lpc (celt_word16_t *lpc, const celt_word16_t *ac, int p)
- void fir_mem16 (const celt_word16_t *x, const celt_word16_t *num, celt_word16_t *y, int N, int ord, celt_word32_t *mem)
- void iir_mem16 (const celt_word16_t *x, const celt_word16_t *den, celt_word16_t *y, int N, int ord, celt_word32_t *mem)
- void _celt_autocorr (const celt_word16_t *x, float *ac, const float *window, int overlap, int lag, int n)
- void do_plc (const CELTMode *m, kiss_fftr_cfg fft, const struct PsyDecay *decay, float *out_mem, float *out, float *time, int len)

6.63.1 Define Documentation

6.63.1.1 #define LPC_ORDER 12

Definition at line 115 of file plc.c.

Referenced by do_plc().

6.63.2 Function Documentation

6.63.2.1 void _celt_autocorr (const celt_word16_t * *x*, float * *ac*, const float * *window*, int *overlap*, int *lag*, int *n*)

Definition at line 85 of file plc.c.

Referenced by do_plc().

6.63.2.2 celt_word32_t _celt_lpc (celt_word16_t * *lpc*, const celt_word16_t * *ac*, int *p*)

Definition at line 5 of file plc.c.

References ADD16, DIV32_16, EXTEND32, MAC16_16_P13, MULT16_16, MULT16_16_Q13, NEG32, PSHR32, SHL32, SUB16, and SUB32.

Referenced by do_plc().

6.63.2.3 void do_plc (const CELTMode * *m*, kiss_fftr_cfg *fft*, const struct PsyDecay * *decay*, float * *out_mem*, float * *out*, float * *time*, int *len*)

Definition at line 116 of file plc.c.

References _celt_autocorr(), _celt_lpc(), find_spectral_pitch(), LPC_ORDER, MAX_PERIOD, CELTMode::overlap, and CELTMode::window.

6.63.2.4 void fir_mem16 (const celt_word16_t * *x*, const celt_word16_t * *num*, celt_word16_t * *y*, int *N*, int *ord*, celt_word32_t * *mem*)

Definition at line 49 of file plc.c.

References ADD32, EXTEND32, EXTRACT16, MAC16_16, MULT16_16, PSHR32, and SATURATE.

6.63.2.5 void iir_mem16 (const celt_word16_t * *x*, const celt_word16_t * *den*, celt_word16_t * *y*, int *N*, int *ord*, celt_word32_t * *mem*)

Definition at line 67 of file plc.c.

References ADD32, EXTEND32, EXTRACT16, MAC16_16, MULT16_16, NEG16, PSHR32, and SATURATE.

6.64 libcelt/plot_BER.m File Reference

Functions

- gset xrange[1e-6:1e-2] load BER semilogx (BER(:, 1), BER(:, 2),'x-', BER(:, 1), BER(:, 3),'*-b') xlabel('Bit error rate')
- ylabel ('PESQ MOS')
- legend ('32-bit FEC', 'No FEC')

6.64.1 Function Documentation

6.64.1.1 legend ('32-bit FEC', 'No FEC')

6.64.1.2 gset xrange [1e-6:1e-2] load BER semilogx (BER(:, 1), BER(:, 2), 'x-', BER(:, 1), BER(:, 3), '*-b')

6.64.1.3 ylabel ('PESQ MOS')

6.65 libcelt/plot_PLC.m File Reference

Functions

- axis([0, 15, 1.5, 4.1]) load PLC plot(PLC(xlabel ('Packet loss(%)')
- ylabel ('PESQ MOS')

Variables

- legend

6.65.1 Function Documentation

6.65.1.1 axis ([0,15,1.5,4.1]) load PLC plot (PLC(xlabel ('Packet loss(%)')

6.65.1.2 ylabel ('PESQ MOS')

6.65.2 Variable Documentation

6.65.2.1 legend

Definition at line 6 of file plot_PLC.m.

6.66 libcelt/printeps2.m File Reference

Functions

- sleep (1)
- rename ("printout.eps", file)
- system (["mv printout.eps ", file])

Variables

- function printeps2 automatic_replot = 0
- filestring

6.66.1 Function Documentation

6.66.1.1 rename ("printout.eps", file)

6.66.1.2 sleep (1)

6.66.1.3 system ()

6.66.2 Variable Documentation

6.66.2.1 function printeps2 automatic_replot = 0

Definition at line 3 of file printeps2.m.

6.66.2.2 filestring

Initial value:

```
 concat(concat('\"', file),'\"')
gset output "printout.eps"
%gset term postscript eps color
%gset term postscript eps monochrome
%gset term postscript eps monochrome lw 2 "Times-Roman" 20
gset term postscript eps color lw 2 "Times-Roman" 20
replot
gset term x11
automatic_replot=1
```

Definition at line 4 of file printeps2.m.

6.67 libcelt/psy.c File Reference

```
#include "psy.h"
#include <math.h>
#include "os_support.h"
#include "arch.h"
#include "stack_alloc.h"
```

Defines

- #define toBARK(n) (13.1f∗atan(.00074f∗(n))+2.24f∗atan((n)∗(n)∗1.85e-8f)+1e-4f∗(n))
- #define fromBARK(z) (102.f∗(z)-2.f∗pow(z,2.f)+.4f∗pow(z,3.f)+pow(1.46f,z)-1.f)

Functions

- void psydecay_init (struct PsyDecay ∗decay, int len, celt_int32_t Fs)
- void psydecay_clear (struct PsyDecay ∗decay)
- void compute_masking (const struct PsyDecay ∗decay, celt_word16_t ∗X, celt_mask_t ∗restrict mask, int len)

6.67.1 Define Documentation

6.67.1.1 #define fromBARK(z) (102.f∗(z)-2.f∗pow(z,2.f)+.4f∗pow(z,3.f)+pow(1.46f,z)-1.f)

Definition at line 44 of file psy.c.

6.67.1.2 #define toBARK(n) (13.1f∗atan(.00074f∗(n))+2.24f∗atan((n)∗(n)∗1.85e-8f)+1e-4f∗(n))

Definition at line 43 of file psy.c.

6.67.2 Function Documentation

6.67.2.1 void compute_masking (const struct PsyDecay ∗ *decay*, celt_word16_t ∗ *X*, celt_mask_t ∗restrict *mask*, int *len*)

Definition at line 133 of file psy.c.

References ADD32, and MULT16_16.

Referenced by find_spectral_pitch().

6.67.2.2 void psydecay_clear (struct PsyDecay ∗ *decay*)

Free the memory allocated for the spreading function

Definition at line 75 of file psy.c.

References PsyDecay::decayR.

Referenced by celt_encoder_destroy(), and celt_mode_destroy().

6.67.2.3 void psydecay_init (struct PsyDecay * *decay*, int *len*, celt_int32_t *Fs*)

Pre-compute the decay of the psycho-acoustic spreading function

Definition at line 51 of file psy.c.

References PsyDecay::decayR, and Q15ONE.

Referenced by celt_encoder_create(), and celt_mode_create().

6.68 libcelt/psy.h File Reference

```
#include "arch.h"
```

Data Structures

- struct PsyDecay

Functions

- void psydecay_init (struct PsyDecay *decay, int len, celt_int32_t Fs)
- void psydecay_clear (struct PsyDecay *decay)
- void compute_masking (const struct PsyDecay *decay, celt_word16_t *X, celt_mask_t *mask, int len)
- void compute_mdct_masking (const struct PsyDecay *decay, celt_word32_t *X, celt_word16_t *long_window, celt_mask_t *mask, int len)

6.68.1 Function Documentation

6.68.1.1 void compute_masking (const struct PsyDecay * *decay*, celt_word16_t * *X*, celt_mask_t * *mask*, int *len*)

Compute the masking curve for an input (DFT) spectrum X

6.68.1.2 void compute_mdct_masking (const struct PsyDecay * *decay*, celt_word32_t * *X*, celt_word16_t * *long_window*, celt_mask_t * *mask*, int *len*)

Compute the masking curve for an input (MDCT) spectrum X

Referenced by celt_encode(), and celt_encode_float().

6.68.1.3 void psydecay_clear (struct PsyDecay * *decay*)

Free the memory allocated for the spreading function

Definition at line 75 of file psy.c.

References PsyDecay::decayR.

Referenced by celt_encoder_destroy(), and celt_mode_destroy().

6.68.1.4 void psydecay_init (struct PsyDecay * *decay*, int *len*, celt_int32_t *Fs*)

Pre-compute the decay of the psycho-acoustic spreading function

Definition at line 51 of file psy.c.

References PsyDecay::decayR, and Q15ONE.

Referenced by celt_encoder_create(), and celt_mode_create().

6.69 libcelt/quant_bands.c File Reference

```
#include "quant_bands.h"
#include "laplace.h"
#include <math.h>
#include "os_support.h"
#include "arch.h"
#include "mathops.h"
#include "stack_alloc.h"
```

Functions

- int * quant_prob_alloc (const CELTMode *m)
- void quant_prob_free (int *freq)
- void quant_coarse_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, int budget, int *prob, celt_word16_t *error, ec_enc *enc)
- void quant_fine_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, celt_word16_t *error, int *fine_quant, ec_enc *enc)
- void unquant_coarse_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, int budget, int *prob, ec_dec *dec)
- void unquant_fine_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, int *fine_quant, ec_dec *dec)

Variables

- const celt_word16_t eMeans [24] = {45.f, -8.f, -12.f, -2.5f, 1.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f}

6.69.1 Function Documentation

6.69.1.1 void quant_coarse_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, int *budget*, int * *prob*, celt_word16_t * *error*, ec_enc * *enc*)

Definition at line 252 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_encode_float().

6.69.1.2 void quant_fine_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, celt_word16_t * *error*, int * *fine_quant*, ec_enc * *enc*)

Definition at line 277 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_encode_float().

6.69.1.3 int* quant_prob_alloc (const CELTMode * *m*)

Definition at line 90 of file quant_bands.c.

References ec_laplace_get_start_freq(), and CELTMode::nbEBands.

Referenced by celt_mode_create().

6.69.1.4 void quant_prob_free (int * *freq*)

Definition at line 103 of file quant_bands.c.

Referenced by celt_mode_destroy().

6.69.1.5 void unquant_coarse_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, int *budget*, int * *prob*, ec_dec * *dec*)

Definition at line 303 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_-STACK, and VARDECL.

Referenced by celt_decode_float().

6.69.1.6 void unquant_fine_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, int * *fine_quant*, ec_dec * *dec*)

Definition at line 325 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_-STACK, and VARDECL.

Referenced by celt_decode_float().

6.69.2 Variable Documentation

6.69.2.1 const celt_word16_t eMeans[24] = {45.f, -8.f, -12.f, -2.5f, 1.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f, 0.f}

Definition at line 47 of file quant_bands.c.

6.70 libcelt/quant_bands.h File Reference

```
#include "arch.h"
#include "modes.h"
#include "entenc.h"
#include "entdec.h"
```

Functions

- int * quant_prob_alloc (const CELTMode *m)
- void quant_prob_free (int *freq)
- void compute_fine_allocation (const CELTMode *m, int *bits, int budget)
- void quant_coarse_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, int budget, int *prob, celt_word16_t *error, ec_enc *enc)
- void quant_fine_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, celt_word16_t *error, int *fine_quant, ec_enc *enc)
- void unquant_coarse_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, int budget, int *prob, ec_dec *dec)
- void unquant_fine_energy (const CELTMode *m, celt_ener_t *eBands, celt_word16_t *oldEBands, int *fine_quant, ec_dec *dec)

6.70.1 Function Documentation

6.70.1.1 void compute_fine_allocation (const CELTMode * *m*, int * *bits*, int *budget*)

6.70.1.2 void quant_coarse_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, int *budget*, int * *prob*, celt_word16_t * *error*, ec_enc * *enc*)

Definition at line 252 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_encode_float().

6.70.1.3 void quant_fine_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, celt_word16_t * *error*, int * *fine_quant*, ec_enc * *enc*)

Definition at line 277 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_encode_float().

6.70.1.4 int* quant_prob_alloc (const CELTMode * *m*)

Definition at line 90 of file quant_bands.c.

References ec_laplace_get_start_freq(), and CELTMode::nbEBands.

Referenced by celt_mode_create().

6.70.1.5 void quant_prob_free (int * *freq*)

Definition at line 103 of file quant_bands.c.

Referenced by celt_mode_destroy().

6.70.1.6 void unquant_coarse_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, int *budget*, int * *prob*, ec_dec * *dec*)

Definition at line 303 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_decode_float().

6.70.1.7 void unquant_fine_energy (const CELTMode * *m*, celt_ener_t * *eBands*, celt_word16_t * *oldEBands*, int * *fine_quant*, ec_dec * *dec*)

Definition at line 325 of file quant_bands.c.

References ALLOC, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_decode_float().

6.71 libcelt/quant_pitch.c File Reference

```
#include "quant_pitch.h"
#include <math.h>
#include "pgain_table.h"
#include "arch.h"
#include "mathops.h"
```

Defines

- #define PGAIN_ODD(codebook, i) ((1.f/32768.f)*(celt_word16_-t)(((codebook)[(i)]&0x00ffU)<<7))
- #define PGAIN_EVEN(codebook, i) ((1.f/32768.f)*(celt_word16_-t)(((codebook)[(i)]&0xff00U)>>1))
- #define PGAIN_ODD14(codebook, i) PGAIN_ODD(codebook, i)
- #define PGAIN_EVEN14(codebook, i) PGAIN_EVEN(codebook, i)
- #define PGAIN(codebook, i) ((i)&1 ? PGAIN_ODD(codebook, (i)>>1) : PGAIN_-EVEN(codebook, (i)>>1))
- #define Q1515ONE MULT16_16(Q15ONE,Q15ONE)

Functions

- int vq_index (const celt_pgain_t *in, const celt_uint16_t *codebook, int len, int entries)
- int quant_pitch (celt_pgain_t *gains, int len, ec_enc *enc)
- int unquant_pitch (celt_pgain_t *gains, int len, ec_dec *dec)

6.71.1 Define Documentation

6.71.1.1 #define PGAIN(codebook, i) ((i)&1 ? PGAIN_ODD(codebook, (i)>>1) : PGAIN_EVEN(codebook, (i)>>1))

Definition at line 55 of file quant_pitch.c.

6.71.1.2 #define PGAIN_EVEN(codebook, i) ((1.f/32768.f)*(celt_word16_-t)(((codebook)[(i)]&0xff00U)>>1))

Definition at line 50 of file quant_pitch.c.

6.71.1.3 #define PGAIN_EVEN14(codebook, i) PGAIN_EVEN(codebook, i)

Definition at line 52 of file quant_pitch.c.

Referenced by vq_index().

6.71.1.4 #define PGAIN_ODD(codebook, i) ((1.f/32768.f)*(celt_word16_-t)(((codebook)[(i)]&0x00ffU)<<7))

Definition at line 49 of file quant_pitch.c.

6.71.1.5 #define PGAIN_ODD14(codebook, i) PGAIN_ODD(codebook, i)

Definition at line 51 of file quant_pitch.c.

Referenced by vq_index().

6.71.1.6 #define Q1515ONE MULT16_16(Q15ONE,Q15ONE)

Definition at line 58 of file quant_pitch.c.

Referenced by quant_pitch().

6.71.2 Function Documentation

6.71.2.1 int quant_pitch (celt_pgain_t * *gains*, int *len*, ec_enc * *enc*)

If this returns 0, the gain is zero (don't encode the pitch index)

Definition at line 95 of file quant_pitch.c.

References ADD32, celt_sqrt, ec_enc_bits(), EXTEND32, MULT16_16, pgain_table, Q1515ONE, Q15ONE, QCONST32, SHR16, and vq_index().

Referenced by celt_encode(), and celt_encode_float().

6.71.2.2 int unquant_pitch (celt_pgain_t * *gains*, int *len*, ec_dec * *dec*)

If this returns 0, the gain is zero (don't decode the pitch index)

Definition at line 126 of file quant_pitch.c.

References ec_dec_bits().

Referenced by celt_decode(), and celt_decode_float().

6.71.2.3 int vq_index (const celt_pgain_t * *in*, const celt_uint16_t * *codebook*, int *len*, int *entries*)

Taken from Speex.Finds the index of the entry in a codebook that best matches the input

Definition at line 61 of file quant_pitch.c.

References MAC16_16, PGAIN_EVEN14, PGAIN_ODD14, SUB16, and VERY_LARGE32.

Referenced by quant_pitch().

6.72 libcelt/quant_pitch.h File Reference

```
#include "arch.h"
#include "entenc.h"
#include "entdec.h"
```

Functions

- int quant_pitch (celt_pgain_t *gains, int len, ec_enc *enc)
- int unquant_pitch (celt_pgain_t *gains, int len, ec_dec *dec)

6.72.1 Function Documentation

6.72.1.1 int quant_pitch (celt_pgain_t * *gains*, int *len*, ec_enc * *enc*)

If this returns 0, the gain is zero (don't encode the pitch index)

Definition at line 95 of file quant_pitch.c.

References ADD32, celt_sqrt, ec_enc_bits(), EXTEND32, MULT16_16, pgain_table, Q1515ONE, Q15ONE, QCONST32, SHR16, and vq_index().

Referenced by celt_encode(), and celt_encode_float().

6.72.1.2 int unquant_pitch (celt_pgain_t * *gains*, int *len*, ec_dec * *dec*)

If this returns 0, the gain is zero (don't decode the pitch index)

Definition at line 126 of file quant_pitch.c.

References ec_dec_bits().

Referenced by celt_decode(), and celt_decode_float().

6.73 libcelt/rangedec.c File Reference

```
#include "arch.h"
#include "entdec.h"
#include "mfrngcod.h"
```

Functions

- void ec_dec_init (ec_dec *_this, ec_byte_buffer *_buf)
- unsigned ec_decode (ec_dec *_this, unsigned _ft)
- unsigned ec_decode_bin (ec_dec *_this, unsigned bits)
- void ec_dec_update (ec_dec *_this, unsigned _fl, unsigned _fh, unsigned _ft)
- long ec_dec_tell (ec_dec *_this, int _b)

6.73.1 Function Documentation

6.73.1.1 void ec_dec_init (ec_dec * *_this*, ec_byte_buffer * *_buf*)

Definition at line 155 of file rangedec.c.

References ec_dec::buf, ec_dec::dif, EC_CODE_EXTRA, EC_SYM_BITS, ec_dec::rem, and ec_dec::rng.

Referenced by celt_decode(), and celt_decode_float().

6.73.1.2 long ec_dec_tell (ec_dec * *_this*, int *_b*)

Definition at line 189 of file rangedec.c.

References b, ec_dec::buf, EC_CODE_BITS, EC_ILOG, EC_SYM_BITS, and ec_dec::rng.

Referenced by celt_decode(), celt_decode_float(), and unquant_bands().

6.73.1.3 void ec_dec_update (ec_dec * *_this*, unsigned *_fl*, unsigned *_fh*, unsigned *_ft*)

Definition at line 181 of file rangedec.c.

References ec_dec::dif, IMUL32, ec_dec::nrm, and ec_dec::rng.

Referenced by ec_dec_bits(), ec_dec_uint(), and ec_laplace_decode_start().

6.73.1.4 unsigned ec_decode (ec_dec * *_this*, unsigned *_ft*)

Definition at line 165 of file rangedec.c.

References ec_dec::dif, EC_MINI, ec_dec::nrm, and ec_dec::rng.

Referenced by ec_dec_uint(), and ec_laplace_decode_start().

6.73.1.5 unsigned ec_decode_bin (ec_dec * *_this*, unsigned *bits*)

Definition at line 172 of file rangedec.c.

References ec_dec::dif, EC_MINI, ec_dec::nrm, and ec_dec::rng.

Referenced by ec_dec_bits().

6.74 libcelt/rangeenc.c File Reference

```
#include "arch.h"
#include "entenc.h"
#include "mfrngcod.h"
```

Functions

- void ec_enc_init (ec_enc *_this, ec_byte_buffer *_buf)
- void ec_encode (ec_enc *_this, unsigned _fl, unsigned _fh, unsigned _ft)
- void ec_encode_bin (ec_enc *_this, unsigned _fl, unsigned _fh, unsigned bits)
- long ec_enc_tell (ec_enc *_this, int _b)
- void ec_enc_done (ec_enc *_this)

6.74.1 Function Documentation

6.74.1.1 void ec_enc_done (ec_enc * *_this*)

Definition at line 133 of file rangeenc.c.

References EC_CODE_SHIFT, EC_CODE_TOP, ec_enc::ext, ec_enc::low, ec_enc::rem, and ec_enc::rng.

Referenced by celt_encode(), and celt_encode_float().

6.74.1.2 void ec_enc_init (ec_enc * *_this*, ec_byte_buffer * *_buf*)

Definition at line 77 of file rangeenc.c.

References ec_enc::buf, EC_CODE_TOP, ec_enc::ext, ec_enc::low, ec_enc::rem, and ec_enc::rng.

Referenced by celt_encode(), celt_encode_float(), and celt_encoder_create().

6.74.1.3 long ec_enc_tell (ec_enc * *_this*, int *_b*)

Definition at line 108 of file rangeenc.c.

References b, ec_enc::buf, EC_CODE_BITS, EC_ILOG, ec_enc::ext, ec_enc::rem, and ec_enc::rng.

Referenced by celt_encode(), celt_encode_float(), and quant_bands().

6.74.1.4 void ec_encode (ec_enc * *_this*, unsigned *_fl*, unsigned *_fh*, unsigned *_ft*)

Definition at line 85 of file rangeenc.c.

References IMUL32, ec_enc::low, and ec_enc::rng.

Referenced by ec_enc_uint(), and ec_laplace_encode_start().

6.74.1.5 void ec_encode_bin (ec_enc * *_this*, unsigned *_fl*, unsigned *_fh*, unsigned *bits*)

Definition at line 96 of file rangeenc.c.

References IMUL32, ec_enc::low, and ec_enc::rng.

Referenced by ec_enc_bits().

6.75 libcelt/rate.c File Reference

```
#include <math.h>
#include "modes.h"
#include "cwrs.h"
#include "arch.h"
#include "os_support.h"
#include "entcode.h"
#include "rate.h"
```

Functions

- celt_int16_t ** compute_alloc_cache (CELTMode *m, int C)
- void compute_allocation (const CELTMode *m, int *offsets, const int *stereo_mode, int total, int *pulses, int *ebits)

6.75.1 Function Documentation

6.75.1.1 celt_int16_t** compute_alloc_cache (CELTMode * *m*, int *C*)

Computes a cache of the pulses->bits mapping in each band

Definition at line 48 of file rate.c.

References BITOVERFLOW, BITRES, CELTMode::eBands, get_required_bits(), MAX_PULSES, CELTMode::nbEBands, and CELTMode::pitchEnd.

Referenced by celt_mode_create().

6.75.1.2 void compute_allocation (const CELTMode * *m*, int * *offsets*, const int * *stereo_mode*, int *total*, int * *pulses*, int * *ebits*)

Compute the pulse allocation, i.e. how many pulses will go in each band.

Parameters:

m mode

offsets Requested increase or decrease in the number of bits for each band

total Number of bands

pulses Number of pulses per band (returned)

Returns:

Total number of bits allocated

Definition at line 147 of file rate.c.

References ALLOC, CELTMode::allocVectors, BITRES, CELTMode::bits, CELTMode::bits_stereo, CHANNELS, CELTMode::energy_alloc, CELTMode::nbAllocVectors, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_decode_float(), celt_encode_float(), quant_bands(), and unquant_bands().

6.76 libcelt/rate.h File Reference

Defines

- #define MAX_PULSES 128
- #define LOG_MAX_PULSES 7
- #define BITRES 4
- #define BITROUND 8
- #define BITOVERFLOW 30000

Functions

- celt_int16_t ** compute_alloc_cache (CELTMode *m, int C)
- void compute_allocation (const CELTMode *m, int *offsets, const int *stereo_mode, int total, int *pulses, int *ebits)

6.76.1 Define Documentation

6.76.1.1 #define BITOVERFLOW 30000

Definition at line 40 of file rate.h.

Referenced by compute_alloc_cache().

6.76.1.2 #define BITRES 4

Definition at line 38 of file rate.h.

Referenced by compute_alloc_cache(), compute_allocation(), quant_bands(), and unquant_bands().

6.76.1.3 #define BITROUND 8

Definition at line 39 of file rate.h.

6.76.1.4 #define LOG_MAX_PULSES 7

Definition at line 36 of file rate.h.

6.76.1.5 #define MAX_PULSES 128

Definition at line 35 of file rate.h.

Referenced by compute_alloc_cache(), and dump_modes().

6.76.2 Function Documentation

6.76.2.1 celt_int16_t** compute_alloc_cache (CELTMode * *m*, int *C*)

Computes a cache of the pulses->bits mapping in each band

Definition at line 48 of file rate.c.

References BITOVERFLOW, BITRES, CELTMode::eBands, get_required_bits(), MAX_PULSES, CELTMode::nbEBands, and CELTMode::pitchEnd.

Referenced by celt_mode_create().

6.76.2.2 void compute_allocation (const CELTMode * *m*, int * *offsets*, const int * *stereo_mode*, int *total*, int * *pulses*, int * *ebits*)

Compute the pulse allocation, i.e. how many pulses will go in each band.

Parameters:

m mode

offsets Requested increase or decrease in the number of bits for each band

total Number of bands

pulses Number of pulses per band (returned)

Returns:

Total number of bits allocated

Definition at line 147 of file rate.c.

References ALLOC, CELTMode::allocVectors, BITRES, CELTMode::bits, CELTMode::bits_stereo, CHANNELS, CELTMode::energy_alloc, CELTMode::nbAllocVectors, CELTMode::nbChannels, CELTMode::nbEBands, RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by celt_decode_float(), celt_encode_float(), quant_bands(), and unquant_bands().

6.77 libcelt/read_audio.m File Reference

Functions

- fclose (F)

Variables

- function x

6.77.1 Function Documentation

6.77.1.1 fclose (F)

Referenced by main().

6.77.2 Variable Documentation

6.77.2.1 x

Initial value:

```
read_audio(file)
F=fopen(file,'rb')
```

Definition at line 1 of file read_audio.m.

6.78 libcelt/stack_alloc.h File Reference

Temporary memory allocation on stack.

```
#include "os_support.h"
```

Defines

- #define ALIGN(stack, size) ((stack) += ((size) - (long)(stack)) & ((size) - 1))
- #define PUSH(stack, size, type) (ALIGN((stack),sizeof(type)/sizeof(char)),(stack)+=(size)∗(sizeof(type)/sizeof(char)),(ty (size)∗(sizeof(type)/sizeof(char))))
- #define RESTORE_STACK (global_stack = _saved_stack)
- #define VARDECL(type, var) type ∗var
- #define ALLOC(var, size, type) var = PUSH(global_stack, size, type)
- #define SAVE_STACK char ∗_saved_stack = global_stack;
- #define ALLOC_STACK (global_stack = (global_stack==0) ? celt_alloc_scratch(25000) : global_- stack)

Variables

- char ∗ global_stack

6.78.1 Detailed Description

Temporary memory allocation on stack.

Definition in file stack_alloc.h.

6.78.2 Define Documentation

6.78.2.1 #define ALIGN(stack, size) ((stack) += ((size) - (long)(stack)) & ((size) - 1))

Aligns the stack to a 'size' boundary

Parameters:

stack Stack

size New size boundary

Definition at line 118 of file stack_alloc.h.

6.78.2.2 #define ALLOC(var, size, type) var = PUSH(global_stack, size, type)

Allocate 'size' elements of 'type' on stack

Parameters:

var Name of variable to allocate

size Number of elements

type Type of element

Definition at line 132 of file stack_alloc.h.

Referenced by alg_quant(), alg_unquant(), celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), compute_allocation(), decode_pulses(), encode_pulses(), find_spectral_pitch(), get_required_bits(), mdct_backward(), mdct_forward(), quant_bands(), quant_coarse_energy(), quant_fine_energy(), renormalise_bands(), unquant_bands(), unquant_coarse_energy(), and unquant_fine_energy().

6.78.2.3 #define ALLOC_STACK (global_stack = (global_stack==0) ? celt_alloc_scratch(25000) : global_stack)

Definition at line 134 of file stack_alloc.h.

Referenced by celt_mode_create().

6.78.2.4 #define PUSH(stack, size, type) (ALIGN((stack),sizeof(type)/sizeof(char)),(stack)+=(size)∗(sizeof(type)/sizeof(char)),(type∗)((stack)-(size)∗(sizeof(type)/sizeof(char))))

Allocates 'size' elements of type 'type' on the stack

Parameters:

stack Stack

size Number of elements

type Type of element

Definition at line 119 of file stack_alloc.h.

6.78.2.5 #define RESTORE_STACK (global_stack = _saved_stack)

Definition at line 120 of file stack_alloc.h.

Referenced by alg_quant(), alg_unquant(), celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), compute_allocation(), decode_pulses(), encode_pulses(), find_spectral_pitch(), get_required_bits(), mdct_backward(), mdct_forward(), quant_bands(), quant_coarse_energy(), quant_fine_energy(), renormalise_bands(), unquant_bands(), unquant_coarse_energy(), and unquant_fine_energy().

6.78.2.6 #define SAVE_STACK char ∗_saved_stack = global_stack;

Definition at line 133 of file stack_alloc.h.

Referenced by alg_quant(), alg_unquant(), celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), compute_allocation(), decode_pulses(), encode_pulses(), find_spectral_pitch(), get_required_bits(), mdct_backward(), mdct_forward(), quant_bands(), quant_coarse_energy(), quant_fine_energy(), renormalise_bands(), unquant_bands(), unquant_coarse_energy(), and unquant_fine_energy().

6.78.2.7 #define VARDECL(type, var) type ∗var

Declare variable on stack

Parameters:

var Variable to declare

Definition at line 131 of file stack_alloc.h.

Referenced by alg_quant(), alg_unquant(), celt_decode(), celt_decode_float(), celt_encode(), celt_encode_float(), compute_allocation(), decode_pulses(), encode_pulses(), find_spectral_pitch(), get_required_bits(), mdct_backward(), mdct_forward(), quant_bands(), quant_coarse_energy(), quant_fine_energy(), renormalise_bands(), unquant_bands(), unquant_coarse_energy(), and unquant_fine_energy().

6.78.3 Variable Documentation

6.78.3.1 char* global_stack

6.79 libcelt/static_modes.c File Reference

```
#include "modes.h"
#include "rate.h"
```

Defines

- #define allocCache_band1_44100_256_1 allocCache_band0_44100_256_1
- #define allocCache_band2_44100_256_1 allocCache_band1_44100_256_1
- #define allocCache_band3_44100_256_1 allocCache_band2_44100_256_1
- #define allocCache_band4_44100_256_1 allocCache_band3_44100_256_1
- #define allocCache_band5_44100_256_1 allocCache_band4_44100_256_1
- #define allocCache_band6_44100_256_1 allocCache_band5_44100_256_1
- #define allocCache_band7_44100_256_1 allocCache_band6_44100_256_1
- #define allocCache_band8_44100_256_1 allocCache_band7_44100_256_1
- #define allocCache_band11_44100_256_1 allocCache_band10_44100_256_1
- #define allocCache_band1_44100_256_2 allocCache_band0_44100_256_2
- #define allocCache_band2_44100_256_2 allocCache_band1_44100_256_2
- #define allocCache_band3_44100_256_2 allocCache_band2_44100_256_2
- #define allocCache_band4_44100_256_2 allocCache_band3_44100_256_2
- #define allocCache_band5_44100_256_2 allocCache_band4_44100_256_2
- #define allocCache_band6_44100_256_2 allocCache_band5_44100_256_2
- #define allocCache_band7_44100_256_2 allocCache_band6_44100_256_2
- #define allocCache_band8_44100_256_2 allocCache_band7_44100_256_2
- #define allocCache_band11_44100_256_2 allocCache_band10_44100_256_2
- #define TOTAL_MODES 2

6.79.1 Define Documentation

6.79.1.1 #define allocCache_band11_44100_256_1 allocCache_band10_44100_256_1

Definition at line 82 of file static_modes.c.

6.79.1.2 #define allocCache_band11_44100_256_2 allocCache_band10_44100_256_2

Definition at line 207 of file static_modes.c.

6.79.1.3 #define allocCache_band1_44100_256_1 allocCache_band0_44100_256_1

Definition at line 70 of file static_modes.c.

6.79.1.4 #define allocCache_band1_44100_256_2 allocCache_band0_44100_256_2

Definition at line 195 of file static_modes.c.

6.79.1.5 #define allocCache_band2_44100_256_1 allocCache_band1_44100_256_1

Definition at line 71 of file static_modes.c.

6.79.1.6 #define allocCache_band2_44100_256_2 allocCache_band1_44100_256_2

Definition at line 196 of file static_modes.c.

6.79.1.7 #define allocCache_band3_44100_256_1 allocCache_band2_44100_256_1

Definition at line 72 of file static_modes.c.

6.79.1.8 #define allocCache_band3_44100_256_2 allocCache_band2_44100_256_2

Definition at line 197 of file static_modes.c.

6.79.1.9 #define allocCache_band4_44100_256_1 allocCache_band3_44100_256_1

Definition at line 73 of file static_modes.c.

6.79.1.10 #define allocCache_band4_44100_256_2 allocCache_band3_44100_256_2

Definition at line 198 of file static_modes.c.

6.79.1.11 #define allocCache_band5_44100_256_1 allocCache_band4_44100_256_1

Definition at line 74 of file static_modes.c.

6.79.1.12 #define allocCache_band5_44100_256_2 allocCache_band4_44100_256_2

Definition at line 199 of file static_modes.c.

6.79.1.13 #define allocCache_band6_44100_256_1 allocCache_band5_44100_256_1

Definition at line 75 of file static_modes.c.

6.79.1.14 #define allocCache_band6_44100_256_2 allocCache_band5_44100_256_2

Definition at line 200 of file static_modes.c.

6.79.1.15 #define allocCache_band7_44100_256_1 allocCache_band6_44100_256_1

Definition at line 76 of file static_modes.c.

6.79.1.16 #define allocCache_band7_44100_256_2 allocCache_band6_44100_256_2

Definition at line 201 of file static_modes.c.

6.79.1.17 #define allocCache_band8_44100_256_1 allocCache_band7_44100_256_1

Definition at line 77 of file static_modes.c.

6.79.1.18 #define allocCache_band8_44100_256_2 allocCache_band7_44100_256_2

Definition at line 202 of file static_modes.c.

6.79.1.19 #define TOTAL_MODES 2

Definition at line 304 of file static_modes.c.

Referenced by celt_mode_create().

6.80 libcelt/static_modes.h File Reference

Defines

- #define FRAMESIZE(mode) 256
- #define OVERLAP(mode) 128

6.80.1 Define Documentation

6.80.1.1 #define FRAMESIZE(mode) 256

Definition at line 2 of file static_modes.h.

6.80.1.2 #define OVERLAP(mode) 128

Definition at line 3 of file static_modes.h.

6.81 libcelt/static_modes_def.h File Reference

```
#include "modes.h"
#include "rate.h"
```

Defines

- #define allocCache_band1_51200_256_1 allocCache_band0_51200_256_1
- #define allocCache_band2_51200_256_1 allocCache_band1_51200_256_1
- #define allocCache_band3_51200_256_1 allocCache_band2_51200_256_1
- #define allocCache_band4_51200_256_1 allocCache_band3_51200_256_1
- #define allocCache_band5_51200_256_1 allocCache_band4_51200_256_1
- #define allocCache_band6_51200_256_1 allocCache_band5_51200_256_1
- #define allocCache_band7_51200_256_1 allocCache_band6_51200_256_1
- #define TOTAL_MODES 1

6.81.1 Define Documentation

6.81.1.1 #define allocCache_band1_51200_256_1 allocCache_band0_51200_256_1

Definition at line 50 of file static_modes_def.h.

6.81.1.2 #define allocCache_band2_51200_256_1 allocCache_band1_51200_256_1

Definition at line 51 of file static_modes_def.h.

6.81.1.3 #define allocCache_band3_51200_256_1 allocCache_band2_51200_256_1

Definition at line 52 of file static_modes_def.h.

6.81.1.4 #define allocCache_band4_51200_256_1 allocCache_band3_51200_256_1

Definition at line 53 of file static_modes_def.h.

6.81.1.5 #define allocCache_band5_51200_256_1 allocCache_band4_51200_256_1

Definition at line 54 of file static_modes_def.h.

6.81.1.6 #define allocCache_band6_51200_256_1 allocCache_band5_51200_256_1

Definition at line 55 of file static_modes_def.h.

6.81.1.7 #define allocCache_band7_51200_256_1 allocCache_band6_51200_256_1

Definition at line 56 of file static_modes_def.h.

6.81.1.8 #define TOTAL_MODES 1

Definition at line 102 of file static_modes_def.h.

6.82 libcelt/testcelt.c File Reference

```
#include "celt.h"
#include "arch.h"
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <string.h>
```

Functions

- int main (int argc, char *argv[ ])

6.82.1 Function Documentation

6.82.1.1 int main (int *argc*, char * *argv*[ ])

Definition at line 43 of file testcelt.c.

References celt_decode(), celt_decoder_create(), celt_decoder_destroy(), celt_encode(), celt_encoder_create(), celt_encoder_ctl(), celt_encoder_destroy(), CELT_GET_FRAME_SIZE, CELT_GET_LOOKAHEAD, CELT_GET_NB_CHANNELS, celt_mode_create(), celt_mode_destroy(), celt_mode_info(), CELT_SET_COMPLEXITY, fclose(), fwrite(), and PRINT_MIPS.

6.83 libcelt/tunecelt.c File Reference

```
#include "celt.h"
#include "arch.h"
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <string.h>
```

Functions

- int main (int argc, char *argv[ ])

6.83.1 Function Documentation

6.83.1.1 int main (int *argc*, char * *argv*[ ])

Definition at line 43 of file tunecelt.c.

References celt_encode(), celt_encoder_create(), celt_encoder_destroy(), CELT_GET_FRAME_SIZE, CELT_GET_LOOKAHEAD, CELT_GET_NB_CHANNELS, celt_mode_create(), celt_mode_destroy(), celt_mode_info(), fclose(), and fwrite().

6.84 libcelt/ventropy.m File Reference

Variables

- function v

6.84.1 Variable Documentation

6.84.1.1 function v

Initial value:

```
 ventropy(M)

for k=1:size(M,2)
   v(k) = entropy2(M(:,k))
```

Definition at line 1 of file ventropy.m.

6.85 libcelt/vpred.m File Reference

Variables

- function p

6.85.1 Variable Documentation

6.85.1.1 function p

Initial value:

```
 vpred(M)

for k=1:size(M,2)
   p(k) = M(1:end-1,k)'*M(2:end,k)/(M(1:end-1,k)'*M(1:end-1,k))
```

Definition at line 1 of file vpred.m.

6.86 libcelt/vpred2.m File Reference

Functions

- D (:, k+1)
- end D (:, 1)

Variables

- function [p, D]

6.86.1 Function Documentation

6.86.1.1 end D (:, 1)

6.86.1.2 D (:, k+1)

6.86.2 Variable Documentation

6.86.2.1 function[p, D]

Initial value:

```
 vpred2(M)

for k=1:size(M,2)-1
   p(k) = M(1:end,k)'*M(1:end,k+1)/(M(1:end,k)'*M(1:end,k))
```

Definition at line 1 of file vpred2.m.

6.87 libcelt/vq.c File Reference

```
#include "mathops.h"
#include "cwrs.h"
#include "vq.h"
#include "arch.h"
#include "os_support.h"
```

Defines

- #define KGAIN 6

Functions

- void alg_quant (celt_norm_t *X, celt_mask_t *W, int N, int K, const celt_norm_t *P, ec_enc *enc)
- void alg_unquant (celt_norm_t *X, int N, int K, celt_norm_t *P, ec_dec *dec)
- void renormalise_vector (celt_norm_t *X, celt_word16_t value, int N, int stride)
- void intra_prediction (const CELTMode *m, celt_norm_t *restrict x, celt_mask_t *W, int N, int K, celt_norm_t *Y, celt_norm_t *restrict P, int N0, int B, ec_enc *enc)
- void intra_unquant (const CELTMode *m, celt_norm_t *x, int N, int K, celt_norm_t *Y, celt_norm_t *restrict P, int N0, int B, ec_dec *dec)
- void intra_fold (const CELTMode *m, celt_norm_t *x, int N, celt_norm_t *Y, celt_norm_t *restrict P, int N0, int B)

6.87.1 Define Documentation

6.87.1.1 #define KGAIN 6

Definition at line 305 of file vq.c.

Referenced by intra_prediction(), and intra_unquant().

6.87.2 Function Documentation

6.87.2.1 void alg_quant (celt_norm_t * *X*, celt_mask_t * *W*, int *N*, int *K*, const celt_norm_t * *P*, ec_enc * *enc*)

Algebraic pulse-vector quantiser. The signal x is replaced by the sum of the pitch and a combination of pulses such that its norm is still equal to 1. This is the function that will typically require the most CPU.

Parameters:

x Residual signal to quantise/encode (returns quantised version)

W Perceptual weight to use when optimising (currently unused)

N Number of samples to encode

K Number of pulses to use

p Pitch vector (it is assumed that p+x is a unit vector)

enc Entropy encoder state

Definition at line 94 of file vq.c.

References ADD16, ALLOC, celt_assert2, celt_ilog2, celt_psqrt, encode_pulses(), EXTRACT16, MAC16_16, MULT16_16, MULT16_16_Q15, NORM_SCALING, QCONST16, RESTORE_STACK, ROUND16, SAVE_STACK, SHL16, SHR32, SUB16, VARDECL, VERY_LARGE16, and y.

Referenced by quant_bands().

6.87.2.2 void alg_unquant (celt_norm_t * *X*, int *N*, int *K*, celt_norm_t * *P*, ec_dec * *dec*)

Decode pulse vector and combine the result with the pitch vector to produce the final normalised signal in the current band.

Definition at line 258 of file vq.c.

References ALLOC, decode_pulses(), RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by unquant_bands().

6.87.2.3 void intra_fold (const CELTMode * *m*, celt_norm_t * *x*, int *N*, celt_norm_t * *Y*, celt_norm_t *restrict *P*, int *N0*, int *B*)

Definition at line 353 of file vq.c.

References CHANNELS, Q15ONE, and renormalise_vector().

Referenced by quant_bands(), and unquant_bands().

6.87.2.4 void intra_prediction (const CELTMode * *m*, celt_norm_t *restrict *x*, celt_mask_t * *W*, int *N*, int *K*, celt_norm_t * *Y*, celt_norm_t *restrict *P*, int *N0*, int *B*, ec_enc * *enc*)

Intra-frame predictor that matches a section of the current frame (at lower frequencies) to encode the current band.

Parameters:

x Residual signal to quantise/encode (returns quantised version)

W Perceptual weight

N Number of samples to encode

K Number of pulses to use

Y Lower frequency spectrum to use, normalised to the same standard deviation

P Pitch vector (it is assumed that p+x is a unit vector)

B Stride (number of channels multiplied by the number of MDCTs per frame)

N0 Number of valid offsets

enc Entropy encoder state

Definition at line 307 of file vq.c.

References celt_div, CHANNELS, ec_enc_bits(), KGAIN, MAC16_16, MULT16_16, Q15_ONE, and renormalise_vector().

Referenced by quant_bands().

6.87.2.5 void intra_unquant (const CELTMode * *m*, celt_norm_t * *x*, int *N*, int *K*, celt_norm_t * *Y*, celt_norm_t *restrict *P*, int *N0*, int *B*, ec_dec * *dec*)

Definition at line 335 of file vq.c.

References celt_div, CHANNELS, ec_dec_bits(), KGAIN, MULT16_16, Q15_ONE, and renormalise_vector().

Referenced by unquant_bands().

6.87.2.6 void renormalise_vector (celt_norm_t * *X*, celt_word16_t *value*, int *N*, int *stride*)

Definition at line 268 of file vq.c.

References celt_rcp, celt_sqrt, EPSILON, MAC16_16, MULT16_16, MULT16_16_Q15, PSHR32, and SHL32.

Referenced by intra_fold(), intra_prediction(), intra_unquant(), and renormalise_bands().

6.88 libcelt/vq.h File Reference

Vector quantisation of the residual.

```
#include "entenc.h"
#include "entdec.h"
#include "modes.h"
```

Functions

- void alg_quant (celt_norm_t *X, celt_mask_t *W, int N, int K, const celt_norm_t *P, ec_enc *enc)
- void alg_unquant (celt_norm_t *X, int N, int K, celt_norm_t *P, ec_dec *dec)
- void renormalise_vector (celt_norm_t *X, celt_word16_t value, int N, int stride)
- void intra_prediction (const CELTMode *m, celt_norm_t *restrict x, celt_mask_t *W, int N, int K, celt_norm_t *Y, celt_norm_t *restrict P, int N0, int B, ec_enc *enc)
- void intra_unquant (const CELTMode *m, celt_norm_t *x, int N, int K, celt_norm_t *Y, celt_norm_t *P, int N0, int B, ec_dec *dec)
- void intra_fold (const CELTMode *m, celt_norm_t *x, int N, celt_norm_t *Y, celt_norm_t *P, int N0, int B)

6.88.1 Detailed Description

Vector quantisation of the residual.

Definition in file vq.h.

6.88.2 Function Documentation

6.88.2.1 void alg_quant (celt_norm_t * *X*, celt_mask_t * *W*, int *N*, int *K*, const celt_norm_t * *P*, ec_enc * *enc*)

Algebraic pulse-vector quantiser. The signal x is replaced by the sum of the pitch and a combination of pulses such that its norm is still equal to 1. This is the function that will typically require the most CPU.

Parameters:

x Residual signal to quantise/encode (returns quantised version)

W Perceptual weight to use when optimising (currently unused)

N Number of samples to encode

K Number of pulses to use

p Pitch vector (it is assumed that p+x is a unit vector)

enc Entropy encoder state

Definition at line 94 of file vq.c.

References ADD16, ALLOC, celt_assert2, celt_ilog2, celt_psqrt, encode_pulses(), EXTRACT16, MAC16_16, MULT16_16, MULT16_16_Q15, NORM_SCALING, QCONST16, RESTORE_STACK, ROUND16, SAVE_STACK, SHL16, SHR32, SUB16, VARDECL, VERY_LARGE16, and y.

Referenced by quant_bands().

6.88.2.2 void alg_unquant (celt_norm_t * X, int N, int K, celt_norm_t * P, ec_dec * dec)

Algebraic pulse decoder

Parameters:

x Decoded normalised spectrum (returned)

N Number of samples to decode

K Number of pulses to use

p Pitch vector (automatically added to x)

dec Entropy decoder state

Decode pulse vector and combine the result with the pitch vector to produce the final normalised signal in the current band.

Definition at line 258 of file vq.c.

References ALLOC, decode_pulses(), RESTORE_STACK, SAVE_STACK, and VARDECL.

Referenced by unquant_bands().

6.88.2.3 void intra_fold (const CELTMode * m, celt_norm_t * x, int N, celt_norm_t * Y, celt_norm_t * P, int N0, int B)

Encode the entire band as a "fold" from other parts of the spectrum. No bits required (only use is case of an emergency!)

6.88.2.4 void intra_prediction (const CELTMode * m, celt_norm_t *restrict x, celt_mask_t * W, int N, int K, celt_norm_t * Y, celt_norm_t *restrict P, int N0, int B, ec_enc * enc)

Intra-frame predictor that matches a section of the current frame (at lower frequencies) to encode the current band.

Parameters:

x Residual signal to quantise/encode (returns quantised version)

W Perceptual weight

N Number of samples to encode

K Number of pulses to use

Y Lower frequency spectrum to use, normalised to the same standard deviation

P Pitch vector (it is assumed that p+x is a unit vector)

B Stride (number of channels multiplied by the number of MDCTs per frame)

N0 Number of valid offsets

enc Entropy encoder state

Definition at line 307 of file vq.c.

References celt_div, CHANNELS, ec_enc_bits(), KGAIN, MAC16_16, MULT16_16, Q15_ONE, and renormalise_vector().

Referenced by quant_bands().

6.88.2.5 void intra_unquant (const CELTMode * *m*, celt_norm_t * *x*, int *N*, int *K*, celt_norm_t * *Y*, celt_norm_t * *P*, int *N0*, int *B*, ec_dec * *dec*)

6.88.2.6 void renormalise_vector (celt_norm_t * *X*, celt_word16_t *value*, int *N*, int *stride*)

Definition at line 268 of file vq.c.

References celt_rcp, celt_sqrt, EPSILON, MAC16_16, MULT16_16, MULT16_16_Q15, PSHR32, and SHL32.

Referenced by intra_fold(), intra_prediction(), intra_unquant(), and renormalise_bands().

6.89 libcelt/window_fig.m File Reference

Functions

- plot (xw, W, xw+256, W, xw+512, W) xlabel('Time[samples]')
- ylabel ('Amplitude')

Variables

- n = 0:127
- w = sin(.5∗pi∗ sin(.5∗pi∗(n+.5)/128).^2)
- W = [zeros(1,64),w,ones(1,128),w(end:-1:1),zeros(1,64)]
- xw = [0:511]
- legend

6.89.1 Function Documentation

6.89.1.1 plot (xw, W, xw+ *256*, W, xw+ *512*, W)

6.89.1.2 ylabel ('Amplitude')

6.89.2 Variable Documentation

6.89.2.1 legend

Definition at line 8 of file window_fig.m.

6.89.2.2 n = 0:127

Definition at line 1 of file window_fig.m.

Referenced by find_spectral_pitch(), pulse2comb(), quant_bands(), and unquant_bands().

6.89.2.3 W = [zeros(1,64),w,ones(1,128),w(end:-1:1),zeros(1,64)]

Definition at line 3 of file window_fig.m.

6.89.2.4 w = sin(.5∗pi∗ sin(.5∗pi∗(n+.5)/128).^2)

Definition at line 2 of file window_fig.m.

6.89.2.5 xw = [0:511]

Definition at line 4 of file window_fig.m.

6.90 libcelt/write_audio.m File Reference

Functions

- function write_audio (x, file) F
- fwrite (F, x,'short')
- fclose (F)

6.90.1 Function Documentation

6.90.1.1 fclose (F)

6.90.1.2 fwrite (F, x, 'short')

Referenced by main().

6.90.1.3 function write_audio (x, file)

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